Same Evidence, Different Futures: Inductive Reasoning in Language Model Forecasting

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Abstract

Forecasting requires calibrated probabilities grounded in how evidence maps to outcomes: the same observation implies different futures under different relationships. We ask whether language models (LMs) recover those relationships or rely on surface regularities. Across real-world and controlled settings, models distinguish probative evidence from topical controls, use context to choose among competing relationships, and revise those choices when direct observations arrive. Episode-specific conventions and controlled transfer extend this pattern beyond familiar cues and settings. Outcome-based training improves held-out market forecasts, and several checkpoints reach the crowd-level floor. These results establish context-dependent inductive reasoning as an observable forecasting behavior within the tested tasks. Evaluation should measure aggregate accuracy and whether forecasts change for the right structural reason.

1 Introduction

A forecast is a compact theory of how a world works. News of a policy proposal, an endorsement, or an economic shock matters only through a relationship: which actors can respond, which institutions transmit the effect, and whether the local setting amplifies or dampens it. The same observation therefore supports different predictions in different contexts. National political news carries more predictive weight in a media environment organized around national politics than where local issues dominate. Forecasting requires selecting the relationship that applies before extrapolating from the evidence. Figure 1 illustrates the identification problem: identical observations support different forecasts under different assumptions about the generator.

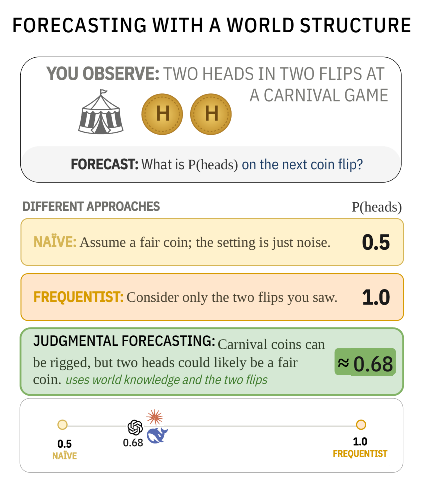

Figure 1: **A descriptive probe.** Three forecasts after two heads from an unfamiliar coin illustrate why two observations do not identify the operative relationship (Appendix A).

LMs now approach crowd accuracy, a practical performance floor; ensembles match it on selected question sets [Halawi et al., 2024, Schoenegger et al., 2024]. This makes the process behind a probability increasingly important. High average scores can coexist with reactions to the wrong cue or the transfer of a relationship into a setting where it no longer applies.

But accuracy alone does not resolve this problem. The same probability can come from a base rate, a retrieved

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precedent, a market price, a lexical shortcut, or a relationship inferred from the current context. Generated rationales do not resolve the ambiguity because explanations need not track model behavior [Jacovi and Goldberg, 2020, Turpin et al., 2023]. We therefore ask a process question: *when an LM forecasts an outcome, does it behave as if it selects, revises, and transfers predictive structure?* The problem is central to **judgmental forecasting**, which combines base rates, sparse observations, qualitative evidence, and domain knowledge to estimate future events [Mellers et al., 2014, Tetlock and Gardner, 2016].

We call this behavioral process *inductive reasoning*: selecting a predictive relationship from context and sparse examples, revising that selection as local evidence accumulates, and applying it to a new case [Tenenbaum et al., 2006, Griffiths et al., 2010, Lake et al., 2017]. We use *world structure* for the predictive relationships among evidence, context, and outcomes. These definitions presume neither an explicit causal graph nor a general internal world model.

1.1 Empirical Progression

We ask three questions:

- **Do LMs react selectively to evidence? (Experiment 1).** Across unresolved markets, every tested deployment distinguishes outcome-bearing packets from topical controls and moves in the withheld direction at high rates. A lexical baseline is directionally weaker than every deployment, establishing relevance-sensitive updating.
- **Can an LM select a relationship based on context? (Experiment 2).** In Coin City, a controlled task, models forecast a target city's polls from response regimes displayed in two reference cities. Changing only the contextual cue changes which regime forecasts follow; direct target observations then weaken the cue. Episode-randomized labels test induction of a local convention, and open-model decoding and activation patching connect that distinction to behavior in the tested checkpoints.
- **Can finetuning transfer contextual relationships? (Experiments 3 and 4).** Episode-matched training improves joint-shift transfer at two model scales. Separately, outcome-based training improves the tested bases on held-out historical markets. Scaling is nonmonotonic; several trained checkpoints reach the crowd-level floor.

Our contribution is a behavioral account of relationship use. Across these settings, forecasts select, revise, and transfer context-dependent relationships. Evaluation should pair aggregate realism with contrasts that change the relationship while holding easier cues fixed.

2 Forecasting, induction, and model validity

Judgmental forecasting. Strong human forecasters combine base rates, decomposition, diverse evidence, and repeated revision [Tetlock, 2005, Mellers et al., 2014, Tetlock and Gardner, 2016]. Forecasting benchmarks increasingly evaluate LMs on time-indexed or leakage-controlled questions [Zou et al., 2022, Karger et al., 2025]; retrieval and ensembling bring selected systems near crowd accuracy [Halawi et al., 2024, Schoenegger et al., 2024]. These results show what probabilities models produce, but not what drives them. Our focus is the relationship behind the update: whether evidence is treated as probative, which contextual analogue is selected, and whether that choice survives a change in domain or mechanism.

Induction and behavioral evidence. Transformers infer unseen functions from input–output examples within their training distribution [Garg et al., 2022, Chan et al., 2022]; LMs also solve selected abstract analogy tasks [Webb et al., 2023, Yasunaga et al., 2024]. Our selection tests ask a different question. The target displays no examples of its own, so a relationship must be carried from a demonstrated analogue under a qualitative cue, and an arbitrary-label variant reverses the cue's mapping across episodes. World-model accounts emphasize representations of environmental state or dynamics [Ha and Schmidhuber, 2018, LeCun, 2022, Li et al., 2023]; shortcut-learning results document failures under structural shift [Geirhos et al., 2020, Mitchell, 2023, Vafa et al., 2025]. We infer process from predicted changes and invariances in paired behavioral tests, not from accuracy alone.

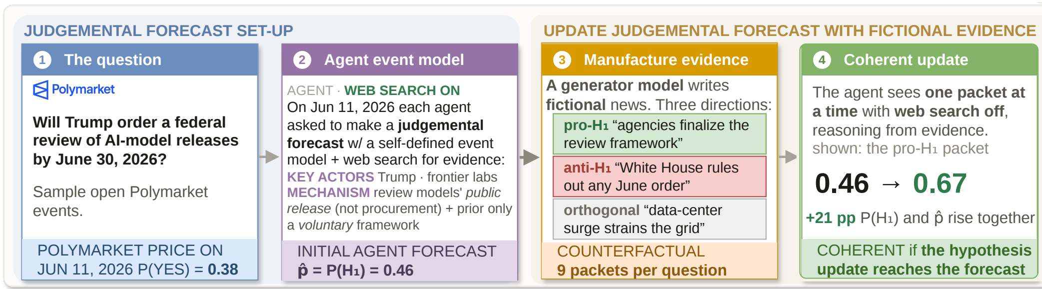


Figure 2: **Prospective forecasting study.** An agent forecasts an unresolved question, then receives one generated evidence packet with search off. The packet's intended direction is hidden from the agent and used only for scoring. Directional correctness asks whether the update follows a pro- or anti-H_1 packet; selectivity compares movement under directional and orthogonal packets.

3 Do forecasts change for the right reason?

We begin on unresolved markets, where the questions are real but lexical cues remain. We then use Coin City, a controlled three-city forecasting task, to display competing response regimes and change which regime the context supports while holding the numerical record fixed. Arbitrary labels remove any stable meaning from the cue. Across studies, contrasts are paired by market or episode, and uncertainty resamples that unit. Repeated generations are not treated as independent observations.

3.1 Separate evidence from topical chatter

Design. We sample 100 unresolved binary Polymarket questions spanning politics, government, elections, economics, technology, and geopolitics. Each model first constructs an event model and forecast with web search available. Search is then disabled, and the model receives one fictional news packet at a time: three support the YES hypothesis, three support NO, and three concern the same topic without bearing on the resolution. The model sees only its prior and the snippet. Packet type, intended direction, and resolution are withheld and used only for scoring. Nine deployments produce 30,094 valid updates (Appendix B).

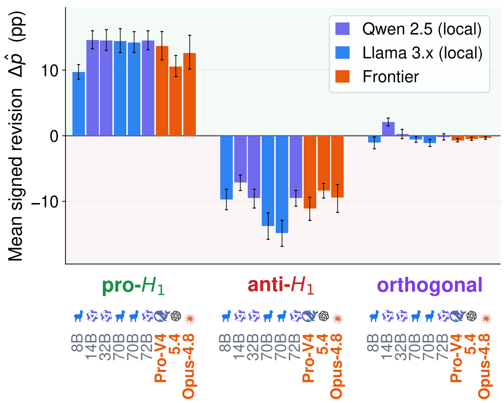
The design separates two questions. *Selectivity* asks whether the model changes its forecast more for outcome-bearing evidence than for a topic-matched control. *Directional consistency* asks whether a sufficiently large change follows the packet's withheld implication for the market outcome. A market-held-out bag-of-words classifier measures how much direction is available from lexical content alone. Figure 2 summarizes the forecast, evidence, and scoring sequence.

Finding: models first decide whether an update is warranted. Figure 3 shows both parts of the result. Every deployment leaves its prior unchanged more often for a topic-matched control than for directional evidence. When they do update, models move 1.9–7.5× farther for directional packets, with every clustered interval above one. They also move in the implied direction: directional consistency spans .887–.990, compared with .688 for the held-out lexical classifier. Eight quality-eligible reviewers match the frozen packet direction on 75.0% of judgments, and the panel majority matches 17 of 18 packets.

The first experiment therefore establishes relevance-sensitive updating. Coin City then tests relationship selection directly by displaying and manipulating generated response regimes, separating the operative relationship from the natural-language packets used above.

3.2 Let context select a relationship, then challenge it

Coin City makes the latent choice observable. We call the task Coin City because it carries Figure 1's unfamiliar-coin problem into city polling. It turns that ambiguity into a controlled test: two candidate relationships are displayed, context favors one, and local observations can overturn the initial choice. Each episode describes three cities whose polls respond to campaign news. Cities A and B provide noisy examples of a strong and a weak response regime. City C is the target. A



(a)

Model	Size	Abst.	Sens.↑	EHC$_{\geq 3}$↑
Hosted API (parameters undisclosed)				
Opus 4.8	—	71.4%	7.0×	.990
GPT-5.4	—	8.4%	7.5×	.987
V4-Pro	—	44.4%	4.9×	.977
Open-weight, descending parameters ↓				
2.5-72B	72B	24.2%	3.2×	.963
3.3-70B	70B	10.7%	5.7×	.965
3.1-70B	70B	2.5%	4.5×	.961
2.5-32B	32B	11.1%	2.8×	.964
2.5-14B	14B	26.1%	2.2×	.943
3.1-8B	8B	1.1%	1.9×	.887

(b)

Figure 3: **Direction-selective updating across nine models and 30,094 valid updates. (a)** Mean signed revision with 95% market-clustered bootstrap intervals. **(b)** Abstention is the unchanged-prior rate on topical controls, sensitivity is the conditional movement ratio, and EHC is directional consistency at a three-point threshold. Appendix B.3.4 reports the threshold sweep.

one-sentence description links City C to one reference environment, and from zero to four City C cases provide increasingly direct local evidence. For city j in episode i,

$$\Delta p_{ij} = g_j x_{ij} + \epsilon_{ij}, \qquad \epsilon_{ij} \sim \mathcal{N}(0, 5^2),$$

where x_{ij} is signed net news and g_j is the city's stable response. No prompt states the coefficient or names a graph. Figure 4 shows the shared query, reference cities, and target-city context in one illustrated arm.

Five prompt arms expose the choice behind the forecast: target cases alone; references without City C context; references with the correct context; the same references with the context inverted; and an arbitrary-label version that replaces familiar media descriptions with KIV and ZOR. In the last arm, which label names the strong regime changes by episode and must be inferred from the reference cases. Semantic arms evaluate six hosted deployments over 250 paired episodes and five evidence depths; the arbitrary-label roster contains fifteen hosted and open systems. Forecast error is measured

Shared prompt in every arm

Each row is a separate polling case rather than a time series. Positive net news favors the candidate and negative net news harms the candidate. Poll change is the ending poll minus the starting poll. Within a city, typical responsiveness to net news is stable across cases, although individual end polls remain noisy. Rows with the same Pair label have equal-sized news in opposite directions.

Query: A new City C case begins with a poll of 53.9 and has net news +5. What poll do you predict at the end of this new case? Think carefully about the expected poll, but do not provide step-by-step working. Respond with one JSON object only: {"rationale": "one short sentence", "predicted_poll": <number from 0 to 100>}

Illustrated arm: References + target context

CITY A

Background: Residents generally encounter campaign developments through **national news coverage**, where campaign stories receive sustained attention and tend to produce larger immediate poll movements.

Pair	Start	News	Delta	End
1	45.4	-8	-6.5	38.9
1	45.4	+8	+12.2	57.6
2	55.1	-6	+0.9	56.0
2	55.1	+6	+11.4	66.5

CITY B

Background: Residents generally encounter campaign developments through **local news coverage**, where campaign stories compete with local issues and tend to produce smaller immediate poll movements.

Pair	Start	News	Delta	End
1	35.0	-10	-5.6	29.4
1	35.0	+10	+9.3	44.3
2	51.8	-10	-5.5	46.3
2	51.8	+10	+7.2	59.0

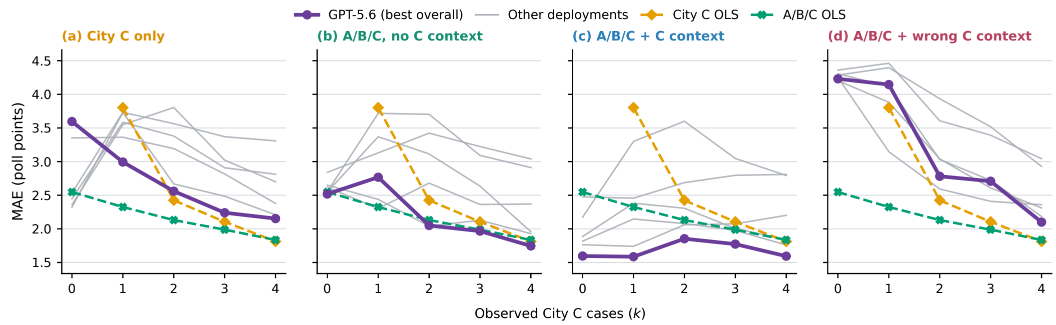
CITY C · TARGET CONTEXT

Background: City C residents generally encounter campaign developments through **national news coverage**.

Pair	Start	News	Delta	End
1	57.8	+8	+9.4	67.2
1	57.8	-8	-1.8	56.0
2	47.4	+10	+8.1	55.5
2	47.4	-10	-10.6	36.8

Figure 4: **Coin City turns contextual reasoning into a paired comparison.** Every arm shares the query and numerical cases. The context arm adds a description connecting City C to one displayed media environment; the opposite-context arm changes only that sentence. Target cases then accumulate, allowing local evidence to confirm or correct the contextual choice.

against the noise-free episode expectation, while the implied response coefficient reveals which reference relationship the forecast follows.



Model	Paired n	Forecast MAE [95% CI]				$\rho(\hat{g}, g_C)$	
		Target only	No C context	+ C context	+ wrong C context	No context	+ context
Opus 4.8	250	2.32 [2.20, 2.46]	2.65 [2.41, 2.89]	**1.82** [1.64, 2.02]	4.20 [3.87, 4.54]	−.06	**.70**
Opus 5	250	2.36 [2.12, 2.59]	2.62 [2.38, 2.84]	**1.76** [1.57, 1.99]	4.25 [3.92, 4.57]	−.01	**.62**
GPT-5.6	250	3.60 [3.28, 3.94]	2.52 [2.33, 2.75]	**1.60** [1.42, 1.77]	4.23 [3.90, 4.55]	−.04	**.68**
V4-Pro	250	**2.45** [2.27, 2.61]	2.84 [2.55, 3.15]	2.47 [2.19, 2.77]	4.36 [4.01, 4.74]	−.03	**.52**
K3	250	3.35 [3.04, 3.67]	2.55 [2.32, 2.78]	**1.89** [1.66, 2.13]	4.31 [3.98, 4.62]	−.02	**.59**
3.6 Flash	250	2.53 [2.35, 2.73]	2.55 [2.31, 2.80]	**2.17** [1.89, 2.48]	4.29 [3.94, 4.66]	.08	**.56**

Figure 5: **Context improves zero-shot forecasts and regime assignment across six deployments.** Panels add references, the correct cue, then an inverted cue as target observations accumulate. The table reports zero-shot MAE and slope correlation with 95% intervals (Appendix C.5).

Finding: the forecast follows the selected analogue. Figure 5 shows how the contextual cue and target observations jointly determine the forecast. At zero target cases, correct context lowers error for every deployment; changing only the context sentence reverses the selected regime. The implied response loads on the reference named by the cue, not on the cue's words alone: after partialling out the true regime, the named-reference correlation is .51–.86, while the other reference is near zero, and inversion transfers that loading to the incorrectly named reference. As City C observations accumulate, the cue's advantage and the harm from inversion shrink. Forecasts weight context when direct evidence is sparse and shift toward local data as it becomes more diagnostic.

3.3 Infer a convention with no stable semantics

The semantic cue still carries a familiar association: national coverage is described as producing larger movements. KIV and ZOR remove that shortcut. Their meaning changes across episodes, so a fixed preference for either token is useless. On the original Coin City population, twelve of fifteen systems recover the local mapping at zero target cases, including all hosted systems tested. A harder population compresses every score toward chance without reordering the checkpoints, so the threshold at which recovery is detectable moves while the ordering holds (Appendix C.8).

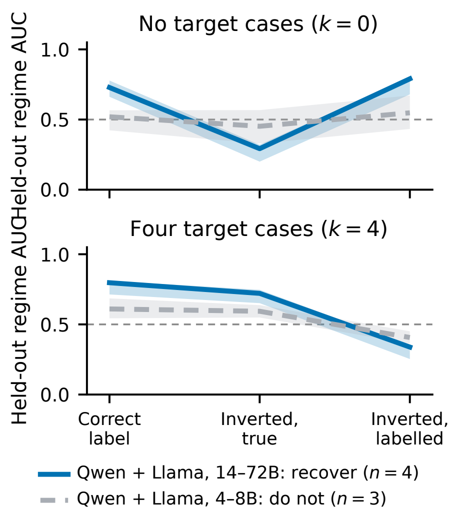


Figure 6: **Across model families, larger checkpoints decode the arbitrary cue, then follow direct evidence.** Lines show group medians and bands span checkpoints. The split appears within both Qwen3 and Llama 3.1: four larger Qwen and Llama checkpoints (14–72B) recover the mapping, while three smaller ones (4–8B) do not. Held-out AUC on 88 held-out episodes.

Figure 6 reports the representation analysis. Across seven open-weight checkpoints from the Qwen and Llama families, the four larger checkpoints (14–72B) recover the arbitrary mapping in their forecasts and encode the selected regime; the three smaller checkpoints (4–8B) do neither. The same size-linked split appears within both Qwen3 and Llama 3.1. Held-out AUC at $k = 0$ is .778 for Qwen2.5-32B, .728 for Llama 3.1-70B, .678 for Qwen3-14B and .662 for Qwen2.5-72B, all rejecting their held-out permutation null at $p < .01$, against .566, .520 and .423 for Qwen3-8B, Qwen3-4B and Llama 3.1-8B, none of which reject. Llama 3.1 gives the cleanest contrast: it holds tokenizer and architecture fixed while varying only scale, and it is the one within-family split whose behavioral classification also survives the harder population. Decodability and behavior remain correlational. Figure 7 shows an activation-patching test in Qwen3-14B. Replacing the label-token representation with the representation induced by the opposite episode-local mapping shifts the zero-shot forecast toward the donor regime (+.333). After four target cases, the estimate is −.035 and lies inside the permutation-null range. At that checkpoint, the episode-local convention causally contributes to the forecast when behavior depends on the cue.

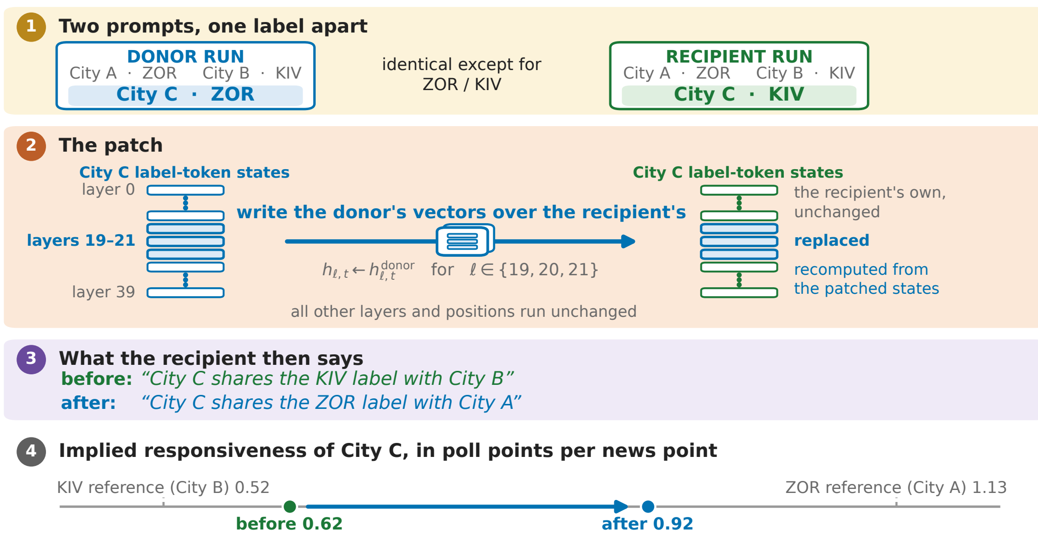


Figure 7: **Patching the induced label representation changes the zero-shot forecast, but not the forecast after direct target evidence arrives.** One sealed $k{=}0$ episode, selected as the episode whose shift is closest to the 88-episode mean. The donor and recipient prompts are identical except for the two City C label tokens. Writing the donor's hidden states at those tokens over the recipient's at layers 19–21, then finishing the forward pass, moves the recipient's forecast from 51.3 to 54.0 — the donor's own forecast — and its stated rationale switches to the donor's reference city. Writing the recipient's own states back in leaves the forecast unchanged, and the reverse direction moves the donor from 54.0 to 51.4. Responsiveness is the implied poll change per news point. Across-episode means and permutation nulls are in the text and Table 34.

Real markets show that models distinguish probative evidence from topical similarity. Coin City shows context selecting between relationships visible in the same prompt, then direct observations revising that choice. Arbitrary labels rule out a stable semantic prior, and the patch connects one induced representation to behavior. Together, these results establish relationship selection and revision within the tested tasks. The next question is whether training carries that behavior into a new world.

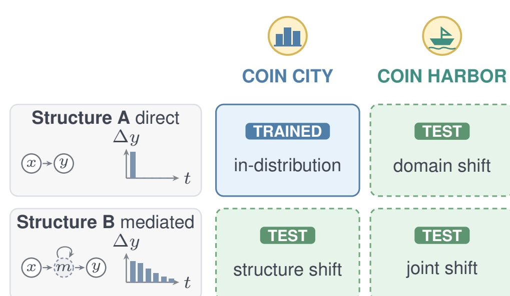


Figure 8: **Transfer grid.** Training uses the shaded cell; all four are evaluated. Columns vary domain; rows vary response mechanism.

4 Can relationship use survive a change of world?

The transfer study asks whether relationship selection remains portable beyond one prompt. Improvements in response format, answer marginals, or generic campaign–poll knowledge do not require matching a particular episode's evidence to its outcome. The transfer study isolates that match.

4.1 Transfer under domain and mechanism shift

Training occurs only in Coin City under the direct, one-period relationship $x_t \rightarrow y_{t+1}$. Every prompt contains two complete reference cases, a partially observed target, a pathway description, and a multi-outcome query. The *episode-matched* arm pairs prompt i with its own answer key. The *population-prior* control uses byte-identical prompts and the same multiset of answer keys, but applies a fixed within-evidence-depth derangement. Thus both arms see the same inputs and answer distribution; only the prompt–outcome correspondence is removed from the control.

At evaluation, we change both the domain and the response mechanism. Coin City uses weekly campaign news and candidate support; held-out Coin Harbor uses shipping bulletins and a cargo-flow index over tide cycles. Coin Harbor is not a relabeling: it also changes the baseline and support, driver and intervention distributions, observation noise, terminology, units, and time scale while preserving the task template and latent response mechanisms. Separately, a mediated, persistent mechanism replaces the direct response. Figure 8 defines the resulting four evaluation cells. Their factorial combination yields in-distribution, domain-shift, mechanism-shift, and joint-shift cells, none of which names a graph to the model. The primary comparison is population-prior minus episode-matched response MAE in the joint cell. Training uses three seeds per arm, and evaluation averages five stochastic draws on 1,440 shared tasks before a paired seed-first bootstrap (Appendix F).

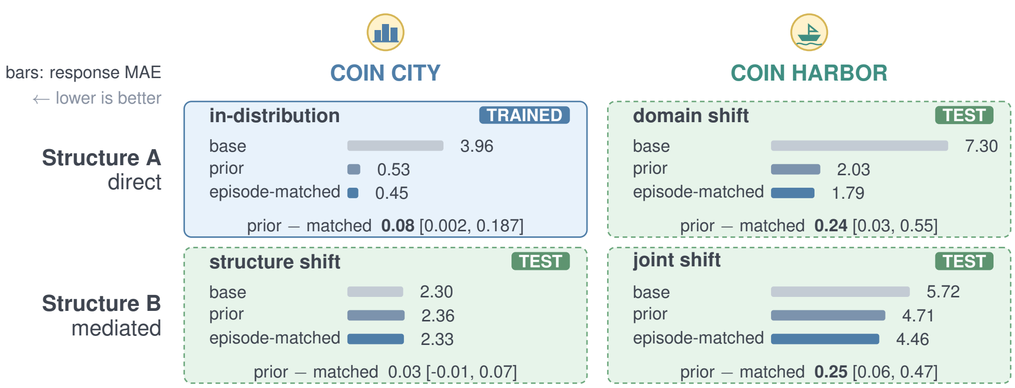


Figure 9: **Qwen3-8B Coin City transfer at round 300 under five-draw stochastic decoding.** Bars show task-averaged response MAE over four evidence depths; trained arms average three seeds. Each cell prints the paired seed-first prior-minus-matched contrast, bold when its 95% interval excludes zero.

Finding: episode matching adds a transfer gain. Figure 9 reports the four-cell comparison. For Qwen3-8B under joint shift, response MAE is 5.72 for the untrained base, 4.71 after population-prior training, and 4.46 after episode-matched training. The matched advantage over the distribution control is $.251$ $[.058, .470]$. It has the lower point estimate in all four cells. Qwen3-4B replicates the joint-cell contrast. Removing or reversing the pathway description also degrades the matched model. Transferred forecasts therefore remain responsive to relational language.

The control also isolates the gain from episode matching. Episode matching accounts for about one fifth of Qwen3-8B's total improvement over base in the joint cell, beyond shared format and population learning. The replicated joint-shift contrast shows that matching each episode's evidence to its outcome adds transfer value across the two tested model scales.

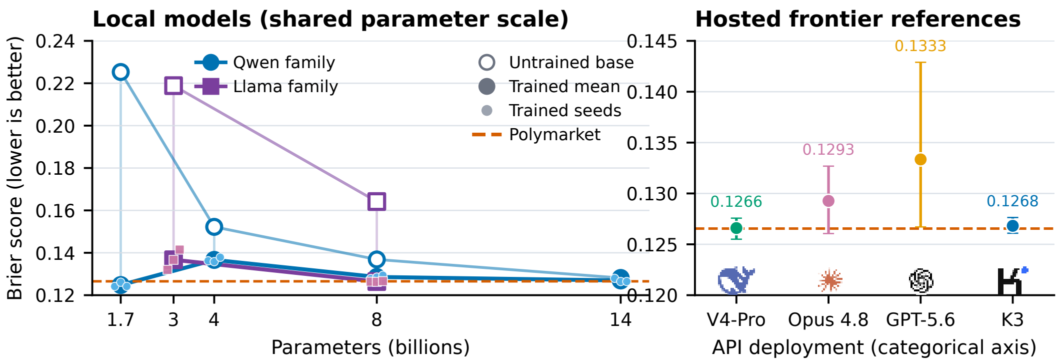


Figure 10: **Outcome-based training improves the tested bases, and several trained checkpoints reach the crowd-level floor.** Performance is nonmonotonic across the tested Qwen sizes. The left panel connects checkpoints only within the Qwen and Llama families; small points are training seeds. Hosted deployments in the right panel are categorical references. Panels use different Brier scales.

4.2 External grounding: held-out historical markets

A separate historical-market study tests outcome-based training on real, held-out events. Resolved political, governmental, macroeconomic, and geopolitical markets are censored before resolution. Prompts contain only the question, rules, cutoff, and genuine pre-cutoff prices; outcomes and post-cutoff fields are withheld. Calendar splits prevent event and normalized question families from crossing training, development, and test. Qwen3 and Llama checkpoints receive rank-32 LoRA training with reward $1 - (\hat{p} - y)^2$, the complement of Brier loss. A 318-market holdout is disjoint from training, development, and the earlier 1,024-market test, and every endpoint is evaluated with five stochastic draws (Appendix E).

Figure 10 separates the local scaling comparison from hosted reference systems. Proper-score training lowers mean Brier relative to every tested base checkpoint. The trained Qwen series is nonmonotonic from 1.7B to 14B. Across both evaluated families, Qwen3-1.7B, Qwen3-14B, and Llama-3.1-8B reach a practical floor set by contemporaneous Polymarket forecasts: each trained-minus-market interval crosses zero. Hosted systems are categorical reference points, not members of the local scaling curve.

The controlled factorial separates episode-level relationship learning from shared format and answer-distribution gains. The historical study measures training effects on held-out real events. Relationship use and external accuracy are separate claims.

5 Implications for forecasting and model evaluation

Consider a forecaster asked to predict public reaction to a policy announcement in an unfamiliar jurisdiction. The forecaster must distinguish outcome-relevant news from topic-matched chatter, choose the comparison population or institution that supplies the response pattern, and revise that choice when direct local evidence arrives. A new convention must be learned from the current case, not imported from a familiar word. Under domain or mechanism shift, training must preserve the relationship, not the answer distribution alone.

The prospective markets test selective updating in the wild. Coin City makes the chosen analogue observable and puts context in conflict with local evidence. Arbitrary labels remove stable semantics; activation patching tests one internal link; and the transfer control breaks the episode-level correspondence without changing the answer distribution. The historical-market study tests accuracy on real outcomes. These studies connect selective updating, relationship choice, local revision, and transfer.

Forecasting makes conditional relationships testable because every probability commits to how evidence, context, and outcomes connect. The relevant test is whether that probability changes when the relationship changes.

These contrasts extend beyond forecasting. For synthetic populations, paired relationship changes test whether conditional structure survives when marginal averages resemble human data [Park et al., 2023, Aher et al., 2023, Argyle et al., 2023, Bisbee et al., 2024]. A persuasion study holds a message fixed while changing who delivers it; a coordination study preserves payoffs while changing what participants know about one another; and a policy analysis preserves surface language while changing the institutional pathway. Local observations test correction of a misleading contextual cue, arbitrary labels separate local induction from word association, and distribution-preserving derangements separate relationship learning from output style or marginal frequencies. These local behavioral tests evaluate conditional structure directly without treating the model as a human proxy.

Parameter count and headline accuracy are insufficient model-selection criteria. Arbitrary-cue recovery changes across populations and scales, the controlled transfer effect replicates at two model sizes, and held-out market performance is nonmonotonic. Validation should therefore be local to the population, mechanism, and contrast that matter for the intended use.

6 Limitations

The experiments test specific forms of relationship use. The real-market study measures selective updating under natural-language packets. Coin City isolates relationship selection between two response regimes and repeats the test in a harder population. Future work can add exact coefficient recovery, unreliable cues, new variables, longer horizons, and interactive multi-agent settings.

Representation analyses cover seven open checkpoints; activation patching tests one checkpoint and one label representation. Training comparisons use three seeds with seed-first inference. The main results use held-out tasks, paired contrasts, and fixed estimators. Historical-market training improves every tested base, and several checkpoints reach the crowd-level floor. A larger model roster and later market cohorts would test the scaling pattern.

7 Ethics, artifacts, and conclusion

The studies use public market metadata and prices, model outputs, synthetic worlds, and an unpaid adult review of synthetic materials. The review collected no identifying or sensitive information and did not constitute human-subjects research under institutional guidelines. Counterfactual news packets were used offline and were never posted to markets or used for trading. The paper reports aggregate model behavior, not a deployment or trading recommendation.

An anonymized artifact contains prompts, fixed manifests, saved generations, analysis code, and figure builders.[1] The supplementary material records censoring, estimators, missingness, hyperparameters, and validation checks.

Across real markets and controlled tasks, LM forecasts select context-dependent predictive relationships and revise them as direct evidence arrives. Episode-matched training carries this relationship use across domain and mechanism shifts at two model scales. On real markets, outcome-based training improves every tested base, and several checkpoints reach the crowd-level floor. These behaviors can be tested directly without assuming a general world model.

A practical evaluation should pair outcome accuracy with targeted contrasts that expose which relationship drives a prediction. Evaluators can hold the topic and answer distribution fixed while changing the relevant analogue, provide local evidence that conflicts with context, and test whether an induced relationship survives changes in domain or mechanism. These tests expose failures hidden by an aggregate score: reacting to lexical resemblance, preserving a population prior when the episode differs, or carrying a valid relationship into a setting where it no longer applies.

A useful forecasting system must meet both standards: accuracy on real outcomes and controlled responsiveness to changes in evidence and context. The experiments here provide a concrete way to test that requirement across the populations, mechanisms, and decisions that define the intended use.

[1] Study data and artifacts; code repository. Links anonymized for review.

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Appendices

The supplement is organized by experiment. A descriptive coin-flip elicitation precedes the Experiment 1 evidence-response details. Experiment 2 documents contextual relationship selection and revision, including a linear-probe representation analysis; Experiments 3–4 document controlled and historical transfer. Shared training details and the artifact map close the appendix.

A Descriptive Coin-Flip Forecast Elicitation

This appendix reports the forecasts used to position the model markers in Figure 1; it is not one of the four experiments. We queried GPT-5.4, Claude Opus 4.8, and DeepSeek V4-Pro eight times each (24 calls total) at temperature 0.7 and a 600-token output ceiling. All responses parsed. The figure places each model's logo at its eight-sample mean; the green $\approx .68$ annotation rounds the pooled mean of $.679$.

The exact user prompt was:

```
You are at a carnival game. A coin you have never seen before is flipped
twice in front of you, and it comes up heads both times. These two flips
are everything you know about this coin.

What is the probability that the next flip comes up heads?

Think about it intuitively, the way you actually would in the
moment -- do NOT use any formula, rule, or calculation, and do not
work out any numbers in your reasoning. Just reason in plain words
about what you'd expect and why. Then END your reply with your gut
answer as a JSON object on its own line:
{"rationale": "one short sentence, no math",
 "p_heads": <number between 0 and 1>}
```

Table 1: Forecasts underlying the three model markers in Figure 1. Each row summarizes eight independent temperature-0.7 calls.

Model	Mean	Median	Range
GPT-5.4	.644	.670	[.60, .67]
Claude Opus 4.8	.681	.675	[.65, .75]
DeepSeek V4-Pro	.712	.750	[.60, .75]
Pooled ($n = 24$)	.679	.670	[.60, .75]

What this probe does and does not separate. It has one condition. The 0.5 and 1.0 endpoints in Figure 1 are reference policies rather than measurements: no fair-coin arm and no no-carnival arm were collected, so the probe cannot attribute the departure from 0.5 to the carnival framing. It also does not separate a prior about carnival coins from ordinary sequential updating on an unknown bias. Laplace's rule of succession gives $3/4$ after two heads from a coin with an unknown bias, using no world knowledge at all, and the observed $.679$ sits below that rather than above it. Finally, the instruction not to use any formula or calculation discourages explicitly deriving either the 0.5 or 0.75 reference policy, changing the elicitation mode and complicating comparison to both. We report the probe as a descriptive illustration of the distinction the paper draws, not as evidence for it; the four experiments carry the argument. `eval/run_coin_probe.py` implements three further arms—a stipulated fair coin, an unfamiliar coin with no carnival, and the carnival prompt without the no-calculation instruction—which would supply the missing contrast.

The prompt, temperature, output limit, parsed probabilities, rationales, and raw completions are stored in `exp2_simulated_worlds/biased_news/data/coin_probe/coin_probe_2026-06-30.jsonl`; the collection and parsing code is `exp2_simulated_worlds/biased_news/eval/run_coin_probe.py`. The prompt names neither a fair nor a rigged coin. It requests qualitative reasoning while retaining a numeric forecast for the plotted quantity.

B Experiment 1: Prospective Forecast Updating

Experiment 1 tests whether forecast revisions follow evidence direction and whether directional packets produce larger changes than orthogonal packets. All reported counts refer to parseable records in the frozen consistency report unless stated otherwise.

The main comparison reports the nine deployments with scale-valid archived revisions. This appendix retains Qwen 2.5-7B for directional-compliance diagnostics and documents why its magnitude-based statistics are excluded.

B.1 Study Design, Sample, and Agent Roster

Protocol summary. Experiment 1 crosses ten forecasting agents with a frozen set of unresolved binary Polymarket questions. The study has four stages: (1) freeze the market sample; (2) obtain one or more initial forecasts from each agent; (3) generate nine synthetic evidence packets for each of the 100 core markets (three pro-H_1, three anti-H_1, and three outcome-orthogonal); and (4) ask the same agent to revise each parseable initial forecast after receiving one packet at a time. The unit of analysis is an *agent* $\times$ *market* $\times$ *initial run* $\times$ *counterfactual packet* update. Requested repeat counts were five initial runs for hosted agents and three for local agents; realized valid counts are uneven because of provider failures, parsing failures, and targeted top-ups. Section B.3 specifies how those records are normalized and aggregated.

B.1.1 Sampling Frame and Market Selection

The primary sampling frame is a Polymarket API snapshot taken on June 9, 2026. Selection was deterministic given that snapshot. We retained markets that were binary, active and unresolved, scheduled to resolve 14–180 days after sampling, priced strictly between 0.10 and 0.90 on the YES contract, and supported by at least $1,000 in USD volume. A keyword-based relevance filter retained politics, government, elections, economics, technology, and geopolitics while excluding sports, cryptocurrency, entertainment, and other out-of-scope topics. When several contracts belonged to the same event, we retained the highest-volume contract.

The recorded filter funnel makes the sampling loss explicit. The snapshot contained 60,989 binary records; 37,386 were active, 8,639 passed the topic filter, 3,570 passed the resolution-window filter, 1,403 passed the price filter, 839 passed the volume filter, and event-level deduplication left 436 candidates. A diversity pass selected the 100-market core using a cosine-similarity threshold of 0.25 and a maximum of 16 markets per topical bucket. The frozen core contains 16 global-election, 16 other, 15 Iran-nuclear, 12 U.S.-federal-policy, 10 AI/technology, 9 U.S. state-election, 7 Russia–Ukraine, 5 Brazil, 5 U.S.-midterm, 3 macroeconomic, and 2 China–Taiwan markets.

Counterfactual generation and Stage 3 updating target this 100-market core for every model, but per-model coverage of it is not uniform. Llama 3.1-70B has 99 analyzable markets because one market has no parseable initial forecast, and Claude Opus 4.8 has usable updates on all three packet arms in 73 markets; Section B.5.3 reports the coverage table and the resulting robustness checks.

Table 2: Representative markets from the frozen 100-market core. Snapshot mid is the stored YES midpoint on June 9, 2026 rather than an eventual outcome.

Question	Topic	Snapshot mid
Will Renan Santos win the 2026 Brazilian presidential election?	Brazil	0.17
US–Iran nuclear deal by June 30?	Middle East	0.20
US $\times$ Iran permanent peace deal by July 31, 2026?	Iran	0.32
Will a new Gemini flagship be released by June 30, 2026?	AI/technology	0.89

B.1.2 Agent Roster and Execution Modes

The roster comprises three hosted API systems and seven open-weight local systems (Table 3). The hosted models ran through Azure AI Foundry and used its native search-supported Responses workflow [Microsoft, 2026]. Local models were served by Ollama [Ollama, 2026] on cluster GPUs; their research turn was supplied by Tavily search. All agents received the same substantive three-step contract: build an event model, gather current evidence, and emit a structured forecast. Provider-specific search orchestration and model runtimes were not identical.

Table 3: Experiment 1 agent roster and requested initial-forecast repeats. Requested k is the collection target rather than an analysis denominator; exact realized coverage appears in Table 6.

Model	Runtime	Research search	Nominal params	Requested k
GPT-5.4	Azure Foundry	Foundry native	—	5
Claude Opus 4.8	Azure Foundry	Foundry native	—	5
DeepSeek V4-Pro	Azure Foundry	Foundry native	—	5
Qwen 2.5-7B	Ollama	Tavily	7B	3
Qwen 2.5-14B	Ollama	Tavily	14B	3
Qwen 2.5-32B	Ollama	Tavily	32B	3
Qwen 2.5-72B	Ollama	Tavily	72B	3
Llama 3.1-8B	Ollama	Tavily	8B	3
Llama 3.1-70B	Ollama	Tavily	70B	3
Llama 3.3-70B	Ollama	Tavily	70B	3

For local models, reasoning calls used the frozen configuration temperature of 0.1 and the final JSON-formatting call used temperature 0.0. Search queries, tool responses, token counts, parse errors, and provider errors were retained with each run. Some smaller models returned probabilities on a 0–100 rather than 0–1 scale. The initial-forecast parser converted those values, but the frozen Stage 3 data retain a mixed-scale Qwen 2.5-7B subset documented in Section B.3.1.

B.2 Forecast, Intervention, and Validation Protocol

B.2.1 Initial Forecast Elicitation

Design rationale. The elicitation asks each agent to restate the resolution criterion, construct explicit YES and NO hypotheses, identify a status-quo trajectory, research current evidence, and map its final YES probability to the posterior assigned to H_1. Question restatement and status-quo anchoring follow prior forecasting-prompt evidence [Halawi et al., 2024, Wilson, 2025].

Executed three-turn sequence. Every frozen initial forecast followed the same substantive sequence, with provider-specific tool plumbing:

1. **Event model.** The agent received the question, full resolution description, days to resolution, and category. It restated what would count as YES versus NO; identified actors, mechanisms, and latent variables; formed H_1 (YES) and H_2 (NO); and described the status-quo outcome.
2. **Current-evidence research.** The agent gathered evidence for and against each hypothesis, checked whether the required chain of events was plausible within the remaining time, and revised its assessment. Hosted systems used Azure Foundry's native search workflow; local systems generated queries that were executed through Tavily and received the returned evidence in context.
3. **Structured forecast.** A final call converted the accumulated analysis to the common JSON contract below. A run entered downstream processing only when this object was parseable.

The research turn included this calibration instruction:

```
Given the days remaining, what chain of events is required for H1, and
is that pace of change plausible? Give extra weight to the status quo;
depart substantially only when concrete directional evidence warrants it.
```

The common output contract was:

```
{
  "event_model": {
    "key_actors": [str],
    "key_mechanisms": [str],
    "latent_variables": [str]
  },
  "hypotheses": [
    {"id": "H1", "description": str,
     "prior_probability": float, "posterior_probability": float,
```

```
      "supporting_evidence": [str], "contradicting_evidence": [str]},
    {"id": "H2", ...}
  ],
  "hypothesis_to_forecast_mapping": str,
  "yes_prob": float,
  "confidence": "low"|"medium"|"high",
  "rationale": str,
  "evidence_sources": [str]
}
```

The contract makes the intended invariant explicit: $\hat{p} = P(H_1)$. The ICS metric tests whether the returned fields obey that invariant after an intervention. The substantive prompt contract was common, but prompts were not byte-identical across runtimes because the hosted and local search interfaces require different orchestration.

B.2.2 Counterfactual Packet Construction and Validation

For each of the 100 core markets, GPT-5.4 generated nine synthetic evidence packets: three pro-H_1, three anti-H_1, and three orthogonal. Each generation call received the market question, resolution description, and a canonical parseable initial event model and forecast. It produced:

- **evidence_headline**: one sentence that states the new development;
- **evidence_text**: a 2–4 sentence synthetic wire-style item;
- **mechanism_targeted**: the causal pathway by which the item should affect, or remain irrelevant to, resolution;
- **direction**: `pro_H1`, `anti_H1`, or `orthogonal`;
- **slot_type**: one of the six roles in Table 4; and
- **plausibility_rating**: the generator's 1–5 self-rating.

Within each direction, sequential exclusion required a new packet to target a different actor and mechanism from earlier packets. Search was disabled so the packets remained controlled synthetic premises rather than paraphrases of live reporting.

Table 4: Counterfactual slot taxonomy. Each market receives one packet of every direction–slot combination shown, for nine packets total.

Slot type	Direction	Role in the intervention
DATA	pro, anti	A measurement, price signal, statistic, or poll that shifts the likelihood of H_1.
DECISION	pro, anti	An actor decision, official statement, policy announcement, commitment, or refusal bearing on H_1.
STRUCTURAL	pro, anti	A technical, logistical, legal, or contextual change that makes H_1 more or less likely.
PROCEDURAL	orthogonal	A same-domain administrative or process update without a direct causal path to H_1 versus H_2.
PERSONNEL	orthogonal	A staffing or leadership change involving a relevant actor but an issue unrelated to H_1 versus H_2.
BACKDROP	orthogonal	Contextual economic, logistical, or social information that sets the scene without shifting the outcome odds.

Packet-level integrity checks. The frozen packet file contains 900/900 parseable records: 100 markets multiplied by nine packets, with exactly 300 pro-H_1, 300 anti-H_1, and 300 orthogonal records. Every packet has a nonempty headline, and the logs contain zero search calls. All 900 generator-assigned plausibility ratings equal 4/5. That field documents the generator's compliance claim but provides neither variation nor independent validation; the separate human materials review is reported below. These checks verify structural completeness; they do not establish that directional strength or difficulty is perfectly matched across packets.

As an illustration, for the Hormuz market the pro-H_1 headline *Oman says Iran agrees daily protected windows for commercial Hormuz transits* elicited the largest single update reported in the viewer:

Claude Opus 4.8 revised from 10% to 45% ($\Delta\hat{p} = +35$ pp). The orthogonal IMO incident-reporting item produced negligible movement.

B.2.3 Stage 3 Update and Pairing

Each packet was presented separately with the agent's own parseable initial structured forecast. The agent was told to reason only from that forecast and the packet and to return the same structured forecast with revised H_1, H_2, and YES probabilities. Search was disabled.

One deployment received an extra instruction. The Stage 3 template was not common across the roster. GPT-5.4 alone was run with the runner's `novelty` option, which appends a rule asking for a `novelty_assessment`: *a single sentence stating whether this evidence was likely already reflected in your initial forecast as background knowledge (suggesting a small update) or is genuinely new and surprising (warranting a larger revision).* The frozen records show the asymmetry exactly: 3,838 of 3,921 GPT-5.4 updates carry the field and no other deployment carries it in any record. The instruction tells a model to modulate revision size by perceived novelty, which is close to the construct Section B.5.3 estimates, and GPT-5.4 is the deployment that behaves differently on exactly that dimension: it returns the prior unchanged on 8.4% of orthogonal packets against 71.4% and 44.4% for the other two hosted systems, writing a small revision where they abstain. We cannot separate that difference from the prompt difference, so GPT-5.4's abstention rate is not comparable to the rest of the roster and we do not read the hosted ordering on abstention as a model difference. Its conditional magnitude, directional correctness, and AUC use the same scoring as every other deployment.

For an agent–market cell with k parseable initial runs, the intended crossed design contains $9k$ update records. Each update is conditionally independent in prompt context: it contains one initial run and one packet, never another packet or another agent's answer. The novelty field is not a reported endpoint and should not be viewed as a validated correction.

Provider interruptions, parse failures, and subsequent top-ups make realized cells uneven. We therefore report exact valid record and market denominators by model in Table 6 rather than treating requested k or the theoretical $9k$ as observed sample sizes.

B.3 Estimands, Record Processing, and Aggregation

B.3.1 Record Construction, Deduplication, and Missingness

The frozen analysis artifact is `exp1_prospective/data/results/consistency_report_2026-06-17.json`; tables in this paper use its stored denominators rather than a fresh scan of the evolving collection directories. During report construction, date suffixes are stripped from task identifiers so daily collection files join to the same market. Counterfactuals are deduplicated by `cf_id`; updates are deduplicated by `update_id`, with a parseable record preferred over an error record when both exist. Files are read newest first for the initial-forecast and market-price lookup. Malformed JSON lines are skipped. Missing values are handled per estimand: a record may therefore contribute to ICS but not EHC, or vice versa, depending on which fields parse.

Stage 3 update fields were evaluated on their stored scale in the frozen report. For Qwen 2.5-7B, 207 of 8,636 deduplicated records store `updated_yes_prob` above 1.5; no other model does. Its directional sign metrics remain usable, but its absolute revision magnitudes and sensitivity ratio are scale-contaminated and are omitted from the paper tables.

B.3.2 Consistency Estimands

For update record i, let $d_i \in \{+1, -1\}$ denote the intended direction ($+1$ for pro-H_1, -1 for anti-H_1). Orthogonal packets do not have an intended sign and are excluded from EHC and HFC. The 3 pp movement threshold excludes cases in which a model effectively leaves the forecast unchanged.

Evidence–Hypothesis Consistency (EHC). Let $\Delta p(H_1)_i = p(H_1)_i^{\text{post}} - \hat{p}_i^{\text{initial}}$. For directional records with $|\Delta p(H_1)_i| \geq 0.03$,

$$\text{EHC}_i = \mathbf{1}[\Delta p(H_1)_i d_i > 0]\,. \tag{1}$$

EHC asks whether the updated hypothesis field moves in the direction implied by the packet; it does not score the size of that movement.

Hypothesis–Forecast Consistency (HFC). Let $\Delta\hat{p}_i = \hat{p}_i^{\text{post}} - \hat{p}_i^{\text{initial}}$. For directional records with $|\Delta\hat{p}_i| \geq 0.03$,

$$\text{HFC}_i = \mathbf{1}[\Delta\hat{p}_i d_i > 0]\,. \tag{2}$$

HFC asks the same directional question of the headline YES probability. Because the output contract requires this probability to equal $P(H_1)$, we retain HFC in the appendix as a compliance diagnostic rather than as a separate main-text result.

Internal Coherence Score (ICS). ICS is defined whenever both updated fields parse and is unconditional on movement:

$$\text{ICS}_i = \mathbf{1}\left[\left|\hat{p}_i^{\text{post}} - p(H_1)_i^{\text{post}}\right| < 0.02\right]. \tag{3}$$

The inequality is strict: a discrepancy of exactly 2 pp does not pass.

Directional sensitivity. Packet selectivity is summarized by

$$\text{Sensitivity} = \frac{\overline{|\Delta\hat{p}|}_{\text{pro+anti}}}{\overline{|\Delta\hat{p}|}_{\text{orthogonal}}}. \tag{4}$$

A value near one indicates comparable movement for directional and orthogonal packets; larger values indicate preferential response to directional packets. This ratio is descriptive. It cancels a uniform multiplicative scale within a model, but does not repair mixed scales across records, hence the Qwen 2.5-7B caveat above.

B.3.3 Aggregation and Uncertainty

For EHC, HFC, and ICS, the evaluator first pools eligible records within each model–market cell to obtain a market-level rate. The reported model score is the unweighted mean of those market rates, so a market with more successful repeat runs does not receive greater weight. Reported standard errors are the sample standard deviation of market-level rates divided by the square root of the number of markets.

Revision magnitudes and sensitivity retain the record-pooled point estimands. Uncertainty uses 20,000 percentile cluster-bootstrap draws over represented markets. All nine packets and every repeated initial-run update from a selected market remain together, and sensitivity is recomputed from pooled directional and orthogonal movement within each draw. This accounts for dependence among records from the same market while preserving the published estimand.

B.3.4 Output Compliance and Threshold Robustness

Output-field coupling. The final-output schema explicitly instructs the agent that $\hat{p} = P(H_1)$, and the Stage 3 prompt requests the same structure after updating. EHC and HFC are therefore not independent tests of an inferred hypothesis-to-forecast link. ICS is a schema-adherence diagnostic, while close EHC–HFC agreement reflects a mixture of directional updating and compliance with the prompted equality. Among the 26,010 relevant records with both updated fields present, 25,984 (99.90%) satisfy the equality exactly after scale normalisation and 25,987 (99.91%) satisfy it within the 2-pp ICS tolerance. We therefore do not treat EHC–HFC parity as independent evidence for a latent representation or for spontaneous propagation between separately elicited quantities.

Unconditional estimand. For threshold t, Table 5 retains a valid directional update when $|\Delta| \geq t$ and reports the fraction for which $d_i\Delta_i > 0$. At $t = 0$, every valid pro- and anti-H_1 update enters the denominator and a zero change is scored incorrect. We first compute correctness within market and then average markets equally, matching the primary analysis. The *Below 3 pp* column is the record-weighted fraction and count excluded by the published 3-pp conditional metric. The table's n is the complete pre-threshold denominator: records require a parseable initial probability and the applicable updated field, but are not selected on the direction or magnitude of movement.

Table 5: Unconditional directional correctness and robustness to 0-, 1-, 3-, and 5-percentage-point movement thresholds. The 0-pp column is unconditional over all valid directional updates and scores zero movement as incorrect; positive-threshold columns condition on the stated magnitude. *Below 3 pp* gives the percentage and count omitted from the corresponding published 3-pp EHC or HFC denominator. Rates are macro-averaged across markets.

			Directional correctness				
Model	**Metric**	n **valid**	**0 pp**	**1 pp**	**3 pp**	**5 pp**	**Below 3 pp**
Claude Opus 4.8	EHC	2,119	98.2%	98.6%	99.0%	99.5%	18.1% (383)
	HFC	2,119	98.3%	98.6%	99.0%	99.5%	18.1% (383)
GPT-5.4	EHC	2,678	98.4%	98.4%	98.7%	98.8%	6.9% (185)
	HFC	2,684	98.4%	98.4%	98.7%	98.8%	6.9% (185)
DeepSeek V4-Pro	EHC	2,929	97.3%	97.6%	97.7%	98.0%	12.7% (371)
	HFC	2,929	97.3%	97.6%	97.7%	98.0%	12.6% (370)
Qwen 2.5-72B	EHC	1,795	96.3%	96.3%	96.3%	96.4%	0.8% (15)
	HFC	1,795	96.3%	96.3%	96.3%	96.4%	0.8% (15)
Llama 3.3-70B	EHC	1,788	96.4%	96.4%	96.5%	97.4%	0.7% (12)
	HFC	1,788	96.4%	96.4%	96.5%	97.4%	0.7% (12)
Llama 3.1-70B	EHC	1,659	96.1%	96.1%	96.1%	96.7%	0.6% (10)
	HFC	1,659	96.1%	96.1%	96.1%	96.7%	0.6% (10)
Qwen 2.5-32B	EHC	1,795	96.4%	96.4%	96.4%	96.5%	1.9% (34)
	HFC	1,795	96.4%	96.4%	96.4%	96.5%	1.9% (34)
Qwen 2.5-14B	EHC	1,800	93.1%	94.4%	94.3%	94.3%	2.9% (52)
	HFC	1,800	93.1%	94.4%	94.3%	94.3%	2.9% (52)
Llama 3.1-8B	EHC	3,601	88.1%	88.7%	88.7%	88.9%	1.7% (62)
	HFC	3,604	88.4%	88.7%	88.7%	88.9%	1.4% (52)
Qwen 2.5-7B	EHC	5,759	90.2%	90.2%	90.6%	90.8%	2.2% (129)
	HFC	5,759	89.9%	90.2%	90.5%	90.8%	2.4% (140)

Threshold-sweep results. The threshold sweep preserves the qualitative conclusion but narrows its wording. Hosted-model unconditional correctness is 97.3–98.4%; at the published 3-pp threshold it is 97.7–99.0%. The 3-pp rule excludes 6.9% of GPT-5.4, 12.6–12.7% of DeepSeek V4-Pro, and 18.1% of Claude Opus 4.8 directional updates. Thus, the near-1.0 EHC/HFC values do not result from discarding a majority of cases, and the conclusion is stable across 0, 1, 3, and 5 pp. The precise statement is *99% directionally correct among updates of at least 3 pp*. It is not *99% coherent updating*. Small local models have fewer sub-3-pp updates (0.6–2.9%) but lower unconditional correctness (88.1–96.4%), so their errors are not hidden primarily by the movement filter.

Normalisation and reproducibility. Before recomputing deltas, the analysis independently normalises the initial headline probability, updated headline probability, and updated H_1 posterior from the documented $[0, 1]$ or $[0, 100]$ formats. This matters for 166 relevant Llama 3.1-8B records in which H_1 was emitted as a percentage while `yes_prob` was emitted as a proportion. The complete eligible record and market counts at every threshold, invalid-field counts, market-level standard errors, input-file SHA-256 hashes, and full-precision estimates are in `exp1_prospective/data/results/threshold_robustness.json`. The table is regenerated by `exp1_prospective/agent/evaluate_threshold_robustness.py`; neither script nor table filters items using realized outcomes.

B.4 Extended Results

B.4.1 Analysis Coverage

Table 6 gives the exact Stage 3 record and market counts. The market column counts markets with at least one valid update. Coverage of all three packet arms is not uniform across models: Llama 3.1-70B has 99 analyzable markets because one market has no parseable initial forecast, and Claude Opus 4.8

has all three arms in 73 of 100. Section B.5.3 gives the per-arm coverage and assesses whether missingness is arm-selective.

Table 6: Valid Stage 3 update records and represented core markets by agent. Hosted systems first; open-weight rows in descending parameter count.

Model	Type	Records	Markets
Claude Opus 4.8	Hosted	3,125	100
GPT-5.4	Hosted	3,913	100
DeepSeek V4-Pro	Hosted	4,395	100
Qwen 2.5-72B	Local 72B	2,692	100
Llama 3.3-70B	Local 70B	2,682	100
Llama 3.1-70B	Local 70B	2,490	99
Qwen 2.5-32B	Local 32B	2,693	100
Qwen 2.5-14B	Local 14B	2,700	100
Llama 3.1-8B	Local 8B	5,404	100
Qwen 2.5-7B	Local 7B	8,636	100
Total		**38,730**	

B.4.2 Prospective Settlement Audit

The frozen-cohort evaluation includes a dated settlement audit that leaves the directional-updating estimand unchanged. On August 24, 2026, we queried the public Polymarket market endpoint for the exact 100 frozen IDs. We admitted a label only when the market was closed and the YES/NO token prices were exactly $[1, 0]$ or $[0, 1]$. This strict rule yielded 60 scoreable markets (21 YES and 39 NO). One additional closed market had an invalid $0.5/0.5$ settlement and 39 remained unresolved at the audit cutoff; all 40 were excluded.

For each agent–market cell, the forecast is the median of its parseable repeated initial YES probabilities, matching the maintained Experiment 1 viewer. The crowd benchmark is the YES price stored in the frozen June 9 selection record. We report threshold accuracy (YES iff $p > 0.5$), Brier score, and natural-log loss. Proper-score contrasts are paired agent-minus-market differences; negative values favor the agent. Percentile intervals use 20,000 market-level bootstrap resamples.

Table 7: Descriptive accuracy on the first 60 strictly settled markets. A/M denotes agent/June 9 market on the same rows; Correct is the number of thresholded calls matching the outcome. Lower Brier and log loss are better, and each Δ is agent minus market. Llama 3.1-70B lacks a parseable initial forecast for one settled market, so its paired denominator is 59.

Model	n	Correct A/M	Brier A/M	ΔBrier [95% CI]	Log A/M	Δlog [95% CI]
Claude Opus 4.8	60	41/39	$.242/.246$	$-.004[-.063, .054]$	$.738/.700$	$.038[-.129, .210]$
GPT-5.4	60	40/39	$.225/.246$	$-.021[-.074, .030]$	$.663/.700$	$-.037[-.178, .103]$
DeepSeek V4-Pro	60	40/39	$.257/.246$	$.011[-.050, .069]$	$.793/.700$	$.093[-.093, .282]$
Qwen 2.5-72B	60	32/39	$.265/.246$	$.019[-.031, .066]$	$.728/.700$	$.028[-.097, .144]$
Llama 3.3-70B	60	30/39	$.266/.246$	$.020[-.038, .078]$	$.732/.700$	$.032[-.113, .177]$
Llama 3.1-70B	59	33/38	$.258/.249$	$.008[-.058, .075]$	$.712/.707$	$.004[-.155, .164]$
Qwen 2.5-32B	60	32/39	$.264/.246$	$.018[-.044, .078]$	$.728/.700$	$.028[-.126, .178]$
Qwen 2.5-14B	60	37/39	$.277/.246$	$.031[-.031, .092]$	$.827/.700$	$.127[-.049, .308]$
Llama 3.1-8B	60	31/39	$.266/.246$	$.020[-.047, .085]$	$.741/.700$	$.041[-.127, .202]$
Qwen 2.5-7B	60	38/39	$.230/.246$	$-.016[-.075, .041]$	$.651/.700$	$-.049[-.195, .090]$

The raw correct-call counts are close for the hosted agents and crowd: Claude Opus 4.8 gets 41/60 correct, GPT-5.4 and DeepSeek V4-Pro each get 40/60, and the June 9 market side gets 39/60. GPT-5.4 has the lowest Brier score (.225 versus .246 for the market), a paired difference of $-.021$ $[-.074, .030]$; its log-loss difference is $-.037$ $[-.178, .103]$. Qwen 2.5-7B has the lowest log loss (.651 versus .700), but gets 38/60 thresholded calls correct. No agent's paired Brier or log-loss interval excludes zero.

This audit describes the outcome-ready slice of the frozen cohort. Settlement by the audit date censors the cohort toward shorter-horizon and early-resolving questions, so the 60 are not a random subset of the frozen 100. Moreover, the benchmark is the June 9 snapshot whereas agent forecasts were collected from June 10 through June 17; later-running agents may have had additional public information. The result therefore does not establish superiority to the crowd. It documents the prospective check; the directional-update estimates do not, by themselves, imply a detectable accuracy advantage.

B.4.3 Revision Magnitudes and Packet Selectivity

Table 8 reports mean absolute forecast revision $\overline{|\Delta \hat{p}|}$ by packet direction. For the nine models whose updates are consistently stored on 0–1, hosted-system sensitivity is 8.2–24.0×, compared with 1.9–6.4× for local systems. These are the unconditional ratios; Table 9 separates the abstention and magnitude factors behind them, and the main text reports those two rather than the product.

Qwen 2.5-7B has mixed 0–1 and 0–100 stored updates (Section B.3.1); we therefore omit its absolute magnitudes and sensitivity ratio rather than present them as comparable estimates.

Table 8: Mean absolute forecast revision by counterfactual direction. Hosted systems first; open-weight rows in descending parameter count. Sensitivity is mean directional divided by mean orthogonal movement; n is the valid Stage 3 record count. Values are percentage points for models with consistently stored 0–1 updates.

Model	**Type**	pro-H_1	anti-H_1	orthogonal	**Sensitivity**	n
Claude Opus 4.8	Hosted	12.7 pp	9.6 pp	0.5 pp	24.0×	3,125
DeepSeek V4-Pro	Hosted	13.9 pp	11.4 pp	1.4 pp	8.9×	4,395
GPT-5.4	Hosted	10.6 pp	8.5 pp	1.2 pp	8.2×	3,913
Qwen 2.5-72B	Local 72B	15.0 pp	10.2 pp	3.0 pp	4.2×	2,692
Llama 3.3-70B	Local 70B	14.8 pp	14.3 pp	2.3 pp	6.4×	2,682
Llama 3.1-70B	Local 70B	14.8 pp	15.5 pp	3.3 pp	4.6×	2,490
Qwen 2.5-32B	Local 32B	15.0 pp	10.3 pp	4.0 pp	3.2×	2,693
Qwen 2.5-14B	Local 14B	14.8 pp	8.6 pp	4.1 pp	2.9×	2,700
Llama 3.1-8B	Local 8B	11.1 pp	12.0 pp	5.9 pp	1.9×	5,404
Qwen 2.5-7B	Local 7B	—	—	—	—	8,636

Qwen 2.5-7B magnitude cells are omitted because 207 deduplicated updates are stored above 1.5, mixing probability scales within the row. Table 8 documents the magnitude contrast; Figure 11 then places directional consistency and packet selectivity on a common visual scale.

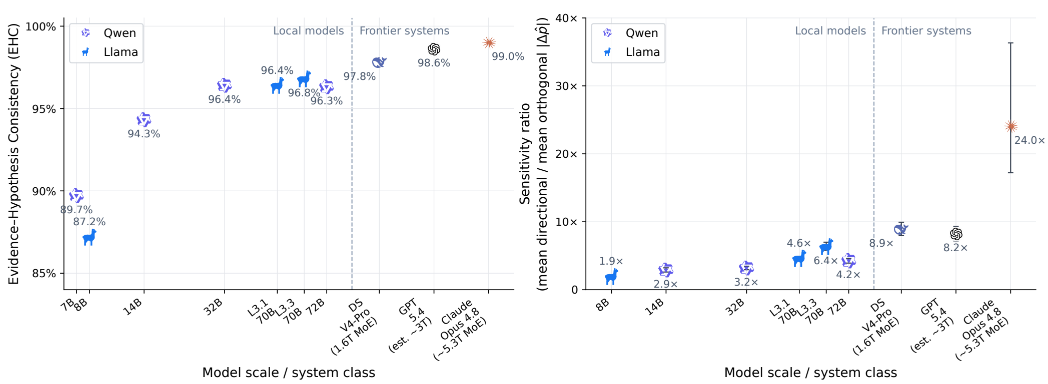


Figure 11: **Directional consistency and packet selectivity by deployment.** *Left:* EHC is 87–90% for 7–8B local models and 98–99% for hosted systems. *Right:* sensitivity is 1.9–6.4× for local models and 8.2–24.0× for hosted systems; Qwen 2.5-7B is omitted from this panel because its archived absolute revisions mix 0–1 and 0–100 scales and therefore do not yield a comparable ratio. The panel plots the unconditional ratio, which Table 9 decomposes.

B.5 Study Limitations

B.5.1 Sampling and Outcome Scope

The sample is a filtered slice of liquid, medium-horizon Polymarket questions rather than a probability sample of forecasting tasks. Topic exclusions, the 0.10–0.90 price window, the 14–180 day horizon, event deduplication, and the diversity cap shape its difficulty distribution. The primary Experiment 1 estimand evaluates conditional updating on questions that were unresolved at collection. The settlement audit in Section B.4.2 is naturally censored to the first 60 strict settlements and is descriptive. Experiment 4 separately evaluates accuracy on a locked sample of resolved historical markets with synchronized forecast cutoffs and market-price baselines.

B.5.2 What the Update Prompt Withholds

The Stage 3 template fills in one packet field, `evidence_text`. The packet record also holds the intended `direction`, `target_hypothesis`, `expected_hypothesis_shift`, `slot_type`, `evidence_headline`, `mechanism_targeted`, and a free-text `rationale`. None of these reaches the agent; all are read only when the update is scored. The agent sees its own prior forecast, the snippet, and the fixed instruction.

We checked this two ways. Over the 12,728 frozen records that kept their prompt string, spanning all three hosted systems, the evidence block is identical to the packet's `evidence_text` and contains no private packet field. Prompt persistence was a per-runner logging choice, so this covers a subset of records by logging coverage rather than by outcome; the runners that stored nothing use the same one-field template. Turning to the snippets, none of the 900 names H_1, H_2, the market, or its resolution. Twenty-two (2.4%, 10 pro-H_1 and 12 anti-H_1) use any likelihood wording—*odds*, *chance of*, *more/less likely*, *probability*, *likelihood*—and each time it describes something in the world, a simulation percentage or a betting line, not the market's resolution. The sign is not stated; it must be inferred from the relation between the reported event and the market outcome. Reproduced by `exp1_prospective/agent/audit_packet_polarity.py`.

B.5.3 Alternative Explanations for the Selectivity Gap

Here we state the competing accounts of the movement gap and what we find for each. All checks come from `exp1_prospective/agent/audit_selectivity_robustness.py`, `audit_orthogonal_abstention.py`, `audit_lexical_sign_baseline.py`, and `audit_group_contrast.py`.

Where the numbers come from. The checks read the frozen June 17 report, which enumerates every scored run, so they use exactly the data behind the published numbers. The enumeration matches the report's own per-arm counts for seven of the nine models; Claude Opus 4.8 and GPT-5.4 carry 14 and 4 extra runs in markets absent from the per-market join, 18 of 38,730. Reproduced sensitivity ratios agree with the published ones to within 0.2. Three checks need packet text and so join runs to the June 10 packet set; 346 runs (1.1%) do not join because they come from a June 7–8 pilot against a superseded 63-packet generation, and they are dropped from those three checks only.

Attrition is much higher for the hosted systems. Claude Opus 4.8 made 9,313 calls and yielded 3,232 usable updates, losing 3,656 to broken pipes, 914 to HTTP 401s after a key expired mid-run, and 1,481 to unparseable responses. Of those, 1,461 come from one malfunctioning June 15 retry pass that failed at 91%; the main June 10 collection parsed at 99.3%, so the published compliance figures are not survivorship. DeepSeek V4-Pro made 6,338 calls for 4,598 usable updates, losing 1,629 to HTTP 401s; GPT-5.4 made 4,269 for 4,057. The six local models lose almost nothing—four record no failed calls, the other two lose 8 and 15 responses to parsing. Content-based exclusion is confined to 28 GPT-5.4 calls across 2 markets. Recorded failures are predominantly transport or parsing failures; the coverage check below assesses arm and market imbalance, not content-specific selection. Separately, two update streams were corrupted by concurrent writes and repaired in August 2026, after the freeze; the original bytes and unrecoverable fragments sit in `exp1_prospective/data/_recovery_audit/`. Nothing in this section reads those files.

The generator saw one agent's world model. All packets were built from a single canonical initial forecast, and it is DeepSeek V4-Pro's: its June 10 YES probability matches the stored generation prior on 100 of 100 markets, where no other agent matches on more than 28. For the other eight models the evidence came from a different system's event model. The seed model gains nothing by it—among hosted systems it is second on sensitivity ($8.9\times$ against Claude's $24.0\times$), third on AUC (.967), and last on EHC (.977)—but its estimates are the least independent in the roster.

Coverage is uneven across arms. Table 10 gives per-arm market coverage. Seven models cover 99 or 100 markets on every arm. Claude Opus 4.8 covers 82 on the orthogonal arm and 77 on the directional arms, 73 in common, so its pooled ratio divides means taken over slightly different market sets. Restricting to its 73 common markets gives $23.7\times$ against $24.2\times$ pooled. As a proxy check, every other model has similar sensitivity on the markets Claude covers and those it misses (8.4 vs. 7.6 for GPT-5.4, 9.7 vs. 7.2 for DeepSeek V4-Pro, 4.3 vs. 4.2 for Qwen 2.5-72B, 1.9 vs. 2.1 for Llama 3.1-8B).

The ratio has an unstable denominator. Claude's orthogonal mean is 0.5 pp, so its $24\times$ comes from a small divisor rather than unusually large directional movement, and its market-clustered interval is correspondingly wide, $[17.3, 36.9]$. Table 11 reports three statistics that avoid that division. The difference of means is 5.6–12.3 pp and does not order the models the way the ratio does. The within-market probability that a directional packet moves the forecast more than an orthogonal one runs from .791 for Llama 3.1-8B to $.994$ for GPT-5.4, putting GPT-5.4 above Claude Opus 4.8 and reversing the ratio ranking. Every model moves further on directional packets in every market it covers.

The denominator is mostly zeros. That instability has a specific cause, and naming it replaces the ratio with two statistics that are easier to read. A revision of exactly zero—the model reproduces the probability shown to it—enters the mean like any other record, so the ratio silently mixes how often a deployment declines to move with how far it moves when it does. Writing A for the share of runs with $\Delta\hat{p} = 0$ and M for the mean absolute revision among the rest, $\overline{|\Delta\hat{p}|} = (1 - A)M$, and the ratio factors exactly:

$$\text{Sensitivity} = \underbrace{\frac{M_{\text{dir}}}{M_{\text{orth}}}}_{\text{magnitude}} \times \underbrace{\frac{1 - A_{\text{dir}}}{1 - A_{\text{orth}}}}_{\text{abstention}} . \tag{5}$$

Table 9 gives both factors. Directional abstention is 0.0–1.4% for every deployment, so the second factor is governed entirely by the orthogonal arm, where abstention runs from 1.1% for Llama 3.1-8B to 71.4% $[63.8, 78.5]$ for Claude Opus 4.8. Claude's $24\times$ is therefore $7.0\times$ $[5.5, 9.1]$ in magnitude multiplied by 3.48 in abstention; DeepSeek V4-Pro's $8.9\times$ is $4.9\times$ times 1.79; GPT-5.4's $8.2\times$ is $7.5\times$ times 1.09, because GPT-5.4 writes a small revision where the other two hosted systems return the prior unchanged. Across the roster the published 1.9–$24.2\times$ becomes 1.9–$7.5\times$ once the zeros are separated out, and the ordering changes: GPT-5.4 rather than Claude Opus 4.8 is highest, and Llama 3.3-70B at $5.7\times$ sits above DeepSeek V4-Pro at $4.9\times$.

We report both factors rather than the product. Declining to revise at all is an appropriate response to an outcome-irrelevant premise, so the abstention rate is a selectivity measure in its own right and a better-behaved one: it is a bounded proportion, it is paired within model across arms, and it needs no division by a small number. It is also the more fragile of the two. An exact copy of a displayed prior is a discrete act adjacent to formatting, and the update prompt does instruct the agent to leave unaffected fields unchanged, so we do not read a high rate as evidence about internal computation. The no-information condition described in Section B.5 is what would separate declining to revise from copying the anchor, and it was not run. Reproduced by `exp1_prospective/agent/audit_orthogonal_abstention.py`.

Table 9: Decomposition of the sensitivity ratio into magnitude and abstention factors (Equation 5). *Orthogonal abstention* is the share of orthogonal runs with $\Delta\hat{p}$ exactly zero; the directional-arm rate is 0.0–1.4% throughout. The two *given a move* columns condition on a nonzero revision, and their ratio is the magnitude factor. Brackets are 95% market-clustered percentile intervals from 20,000 draws, the scheme used for the published movement intervals. Qwen 2.5-7B is omitted for mixed probability scales.

	Orthogonal	**Mean $\lvert\Delta\hat{p}\rvert$ given a move**		**Sens.**	**Sens.**
Model	**abstention**	orthogonal	directional	**given a move**	**published**
Claude Opus 4.8	71.4% [63.8, 78.5]	1.6 pp	11.2 pp	7.0× [5.5, 9.1]	24.2× [17.3, 36.9]
GPT-5.4	8.4% [5.2, 11.9]	1.3 pp	9.6 pp	7.5× [6.6, 8.5]	8.2× [7.2, 9.3]
DeepSeek V4-Pro	44.4% [38.9, 50.0]	2.6 pp	12.7 pp	4.9× [4.4, 5.5]	8.9× [8.0, 9.9]
Qwen 2.5-72B	24.2% [19.6, 29.0]	3.9 pp	12.6 pp	3.2× [3.0, 3.4]	4.2× [3.9, 4.6]
Llama 3.3-70B	10.7% [7.0, 14.8]	2.6 pp	14.6 pp	5.7× [5.2, 6.2]	6.4× [5.8, 7.0]
Llama 3.1-70B	2.5% [0.8, 5.0]	3.4 pp	15.2 pp	4.5× [4.2, 4.8]	4.6× [4.3, 4.9]
Qwen 2.5-32B	11.1% [8.0, 14.6]	4.5 pp	12.6 pp	2.8× [2.6, 3.1]	3.2× [2.9, 3.5]
Qwen 2.5-14B	26.1% [20.7, 31.9]	5.5 pp	11.9 pp	2.2× [2.0, 2.3]	2.9× [2.6, 3.2]
Llama 3.1-8B	1.1% [0.5, 1.9]	6.0 pp	11.6 pp	1.9× [1.8, 2.0]	1.9× [1.8, 2.1]

The deployment contrast depends on the statistic. Comparing the three hosted systems against the six scale-valid open-weight models by market-clustered bootstrap (20,000 replicates, macro-averaged within group so a model with more runs does not dominate), the statistics disagree. On the ratio the difference is $+9.8$ $[7.2, 14.2]$, group means $13.7\times$ and $3.9\times$, and the per-model ranges do not overlap. On AUC it is $+.070$ $[.062, .079]$, which excludes zero, but the ranges do overlap: Llama 3.3-70B at .989 and Llama 3.1-70B at .978 both beat DeepSeek V4-Pro at .967. On raw movement there is no difference at all, $+0.8$ $[-0.0, 1.7]$ pp around means of 10.1 and 9.3.

The decomposition explains why the ratio is the one statistic that separates the groups completely. Both of its factors favour the hosted systems on average and neither separates them on its own: orthogonal abstention differs by $+.286$ $[.244, .329]$ around group means of .413 and .126, with four open-weight deployments at or above GPT-5.4's 8.4%; the magnitude factor differs by $+3.10$ $[2.31, 4.08]$ around $6.5\times$ and $3.4\times$, with Llama 3.3-70B above DeepSeek V4-Pro. Multiplying two partially separating factors yields a product whose ranges happen not to overlap, so we do not report that non-overlap as a finding. The two classes move about equally far on directional evidence; the hosted systems more often decline to move at all on controls, and move somewhat less far when they do. We state the deployment result that way rather than as a general selectivity ordering. Group contrasts come from `exp1_prospective/agent/audit_group_contrast.py`.

The arms differ in wording. By construction directional packets take the DATA, DECISION, and STRUCTURAL slots and orthogonal packets take PROCEDURAL, PERSONNEL, and BACKDROP, and the text differs accordingly: directional snippets average 96.8 words and 7.1 digits, orthogonal ones 85.7 and 3.7. Movement could track density rather than content. Two checks argue otherwise. Within the directional arm, where length cannot indicate sign, Spearman correlation between word count and $|\Delta\hat{p}|$ is $-.06$ to .00 and between digit count and $|\Delta\hat{p}|$ is .01 to .10. And the arms' word counts almost fully overlap (orthogonal 66–114, directional 66–121); restricting to that band keeps 889 of 900 packets and moves every sensitivity estimate by at most 0.1 (Table 11, last column). The arms are not matched on surface form. These checks show that word and digit counts do not explain the effect; other surface differences remain. Directional correctness is not explained by length, since length cannot say which way to move.

A shallow classifier might be enough. We tested it. Multinomial naive Bayes and L2-regularised logistic regression were fit on the 900 packet texts with five-fold cross-validation grouped by market, so no market appears on both sides of a fold. Telling directional from orthogonal packets is nearly free: naive Bayes reaches .989 against a .667 majority rate. That follows from the slot-type plan rather than revealing anything—the most distinctive orthogonal tokens are *appointed*, *appointment*, *oversee*, and *accreditation*, which name the roles the generator was told to write. So a model could route on that marking without weighing content, and the sensitivity ratio alone cannot rule it out. Matching the control arm on evidence type would need a different packet design; the checks above bound only the length part of the problem.

Recovering the *sign* is another matter. On pro-H_1 versus anti-H_1 the same classifiers reach .688 and .675 against .500 chance, and prepending the market question does not help (.665 and .680): these bag-of-words models do not combine question polarity with event valence. The tokens it keys on carry no consistent sign—*vessel*, *venue*, *signature* for pro-H_1, *table*, *materials*, *findings* for anti-H_1. Against this .69 lexical baseline, the human panel reaches .762 on its smaller categorical task and the hosted agents .973–.984 on probability revision. These are not matched performance comparisons. The agent–classifier gap is evidence against this bag-of-words account, not proof that no surface heuristic can recover sign; unlike the arm contrast, it stays within pro- versus anti-H_1 packets.

Does the stated reasoning track the packet? If movement were routed on surface marking, a model's written rationale would not need to bear any particular relation to what it does with the forecast. We tested two versions of this (`exp1_prospective/agent/audit_mechanism_tracking.py`).

The first asks whether a rationale identifies the specific mechanism its packet was built to target, ranking the true `mechanism_targeted` against the other eight from the same market so that question, actors, and domain vocabulary are held fixed. Rationales rank it first on .278–.430 of records against $1/9$ chance, but the snippet alone scores .516, so most of that is echo of the text the model just read. Removing every token the snippet supplied leaves .128–.195 under matched query length, still above chance—but the generator's own rationale for the packet, scored the same way, lands at .119, essentially chance, because it paraphrases the mechanism rather than restating it. With the reference writer at chance the metric has no usable dynamic range, and we draw no conclusion from it.

Models often declare orthogonal packets irrelevant: explicit non-relevance wording appears in 4.9–61.9% of orthogonal rationales against 0.1–2.8% of directional ones. What matters is whether saying it predicts doing it. Within the orthogonal arm, records whose rationale declares irrelevance move less than those that do not, and the market-clustered gap excludes zero for eight of the nine scale-valid deployments: 0.88 pp $[0.56, 1.22]$ for Claude Opus 4.8, 1.70 $[1.45, 1.96]$ for DeepSeek V4-Pro, 2.73 $[2.28, 3.19]$ for Qwen 2.5-72B, down to 0.36 $[0.21, 0.54]$ for GPT-5.4, whose unflagged orthogonal movement is already 1.3 pp. The exception is Llama 3.1-8B at 0.26 $[-0.64, 1.06]$, where no coupling is detectable; it is also the weakest deployment on sensitivity and directional correctness, though we do not test whether the two are related. Qwen 2.5-7B is omitted for mixed probability scales.

For eight of nine deployments, then, the revision a model makes is not independent of the relevance it states, and the one deployment where that link is undetectable is the weakest on every other measure. Two limits: the flag is a hand-built lexicon, so it captures what a rationale declares rather than what the model understood; and consistent self-description is compatible with surface routing, so this does not show relevance was derived from the mechanism.

One deployment was prompted differently. GPT-5.4, and only GPT-5.4, received the `novelty_assessment` rule at Stage 3 (Section B.2.3). Because that rule speaks directly to revision size, its abstention rate is not comparable to the other deployments'. The within-GPT contrast uses one template, but the extra instruction could interact with packet arm, so its contrast should not be attributed to model differences alone.

Two remaining caveats. A single generator wrote all packets and scored its own work at a uniform 4/5 for plausibility, so automated validation establishes structural completeness but not equal difficulty or independent content validity. And the local models ran through Ollama at the default tag for each name, with the served weight precision not recorded per run; rows ordered by parameter count also vary in quantization and inference stack, so we do not attribute the open-weight ordering to parameter count alone.

Table 10: Per-arm market coverage in the frozen June 17 report. Markets counts entries in the model's per-market block, which for three Qwen rows includes 20 carrying no scored update; the arm columns count markets with at least one.

Model	Markets	pro-H_1/anti-H_1	orthogonal	all three
Claude Opus 4.8	100	77	82	73
GPT-5.4	100	99	99	99
DeepSeek V4-Pro	100	100	100	100
Qwen 2.5-72B	120	100	100	100
Llama 3.3-70B	100	100	100	100
Llama 3.1-70B	100	99	99	99
Qwen 2.5-32B	120	100	100	100
Qwen 2.5-14B	120	100	100	100
Llama 3.1-8B	100	100	100	100

Table 11: Selectivity under statistics that do not divide by the orthogonal mean. *Sens.* is the frozen ratio; *Diff.* is mean directional minus mean orthogonal absolute revision; *AUC* is the within-market probability that a directional packet moves the forecast more than an orthogonal one, averaged over markets; *Markets* counts markets where mean directional movement exceeds mean orthogonal movement; *Length-m.* recomputes the ratio on the 66–114-word band the arms share. Qwen 2.5-7B is omitted for mixed probability scales.

Model	Sens.	Diff.	AUC	Markets	Length-m.
Claude Opus 4.8	24.0×	10.7 pp	0.984	77/77	24.3×
GPT-5.4	8.2×	8.4 pp	0.994	99/99	8.1×
DeepSeek V4-Pro	8.9×	11.2 pp	0.967	100/100	8.8×
Qwen 2.5-72B	4.2×	9.6 pp	0.950	100/100	4.2×
Llama 3.3-70B	6.4×	12.3 pp	0.989	100/100	6.4×
Llama 3.1-70B	4.6×	11.9 pp	0.978	99/99	4.6×
Qwen 2.5-32B	3.2×	8.7 pp	0.906	100/100	3.2×
Qwen 2.5-14B	2.9×	7.7 pp	0.852	100/100	2.9×
Llama 3.1-8B	1.9×	5.6 pp	0.791	100/100	1.9×

B.5.4 Human Materials Review

Nine adult reviewers submitted the 18-item materials protocol. It samples one report from each of 18 distinct markets, balanced across six pro-YES, six anti-YES, and six orthogonal packets. One reviewer met the disclosed response-pattern exclusion below, leaving eight quality-eligible reviewers. We treat the study as descriptive materials evidence. Before the synthetic report, every item shows the market question, resolution criteria, and matching summary from the frozen June 10 initial forecast. The consent screen and every item instruct reviewers to treat June 10, 2026 as the current date and use no external information or assistance. Intended directions, source IDs, generator metadata, and model responses remain hidden.

Reviewers record the report's conditional direction, confidence, clarity, real-world plausibility, and usability as an explicit hypothetical premise. We report exact agreement with the frozen direction and Cohen's κ between the panel-majority labels and the frozen key. Reviewer-level Cohen's κ, the component ratings, and every item-level judgment appear below. Ambiguous judgments remain in every denominator.

Practice responses are stored separately and excluded from the analysis.

The analyzed sample contains eight quality-eligible reviewers (R1 and R3–R9), exceeding the six-reviewer target for this descriptive materials check. Table 12 includes every eligible reviewer.

Exact-direction agreement is 108/144 (75.0%). The panel-majority label matches the frozen direction on 17 of 18 items (94.4%; Cohen's $\kappa = .919$): all six anti-H_1 items, all six orthogonal items, and five of six pro-H_1 items. Across individual judgments, exact agreement is 35/48 for pro-H_1, 37/48 for anti-H_1, and 36/48 for orthogonal packets. Reviewers marked the premise usable on 124/144 judgments (86.1%). The corresponding all-reviewer sensitivity is reported below. These descriptive results indicate that readers usually recovered the intended conditional direction and could use the reports as stipulated premises. They also bound how legible that direction is: 75.0% exact agreement and mean pairwise Cohen's $\kappa = .47$ indicate substantial but imperfect recovery across the four response labels. The panel is not a matched baseline for the agents—reviewers emit a categorical label on 18 items, agents emit a revised probability against their own prior on all 900—so we read it as evidence that sign had to be inferred, not as a human–model comparison.

Table 12: **Stage 3 human materials judgments for eight quality-eligible reviewers.** Direction is exact agreement with the frozen conditional direction and κ is Cohen's κ against that key over the four response options. Confidence, clarity, and plausibility (C/C/P) are reviewer-level medians on 1–5 scales. The pooled row is descriptive because reviewers rated the same items; the majority row takes the modal label per item, resolving ties to *ambiguous*.

Reviewer	Direction agreement	κ vs. key	Premise usable	C/C/P
R1	12/18 (66.7%)	.54	10/18	4/4/4
R3	12/18 (66.7%)	.51	13/18	4/4/3
R4	18/18 (100.0%)	1.00	18/18	5/5/5
R5	10/18 (55.6%)	.38	18/18	4/4/4
R6	13/18 (72.2%)	.61	12/18	4/4/5
R7	15/18 (83.3%)	.76	17/18	5/4/3
R8	12/18 (66.7%)	.55	18/18	4/5/4
R9	16/18 (88.9%)	.84	18/18	5/5/5
Pooled	108/144 (75.0%)	—	124/144	—
Majority label	17/18 (94.4%)	.92	—	—

Table 13: **Item-level Stage 3 judgments for eight quality-eligible reviewers.** Key is the frozen conditional direction; Majority is the modal panel response, with ties assigned to *ambiguous*. Agree and Usable give counts out of eight; C/L/P gives median direction confidence, clarity, and plausibility on 1–5 scales. Item numbers follow the frozen public packet order.

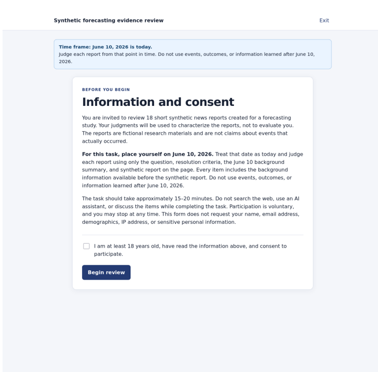
Question	Key	Majority	Agree	C/L/P	Usable
1. Hormuz traffic normal by July 31	More	More	8/8	5/5/4.5	8/8
2. Iran reaches World Cup knockouts	Less	Less	6/8	4.5/4.5/4.5	7/8
3. Israel strikes Yemen by June 30	No effect	No effect	8/8	4.5/4/3.5	6/8
4. Netanyahu exits election by July 31	More	More	6/8	5/5/4.5	8/8
5. Anthropic valuation reaches \$1.1T	Less	Less	7/8	4.5/4.5/5	8/8
6. Oladokun starts Chiefs Week 1	No effect	No effect	6/8	4/4/3.5	7/8
7. S&P 500 reaches 7,700 in June	More	More	4/8	3.5/4/5	7/8
8. Trump unfreezes Iranian assets	Less	Less	5/8	3.5/4/4	7/8
9. May durable-goods orders: 0–2%	No effect	No effect	4/8	3/3/3.5	6/8
10. China Q2 growth: 4.6–4.9%	More	Ambig.	3/8	3/4/3.5	5/8
11. Eisenkot joins Bennett–Lapid alliance	Less	Less	6/8	5/5/5	7/8
12. Russia enters Orikhiv by July 31	No effect	No effect	8/8	5/4/4.5	6/8
13. Romanian PM is independent/technocrat	More	More	6/8	4.5/4.5/4	8/8
14. Russia captures Kostyantynivka	Less	Less	5/8	3.5/5/5	7/8
15. At least four June ChatGPT outages	No effect	No effect	4/8	3/4/4	7/8
16. El-Sayed wins Michigan primary	More	More	8/8	5/5/5	8/8
17. Little is MN-02 Democratic nominee	Less	Less	8/8	5/5/4	7/8
18. OpenAI IPO close: \$1.25–1.5T	No effect	No effect	6/8	4/3.5/3.5	5/8

The response-pattern quality rule excludes any reviewer who selects one direction on at least 17 of 18 balanced items. R2 met this rule by selecting *more likely* on 17 of 18 balanced items, including all six frozen negative items and five of six frozen null items. Raw responses are retained, and no item is removed or relabeled. Retaining R2 gives 115/162 (71.0%) exact agreement across all nine reviewers.

The task requests no identifying or sensitive information. It was conducted as an internal materials review under the authors' institutional guidance; no IRB protocol was opened.

Figure 12 records the reviewer-facing workflow. The capture script at `exp1_prospective/stage3_materials_annotation/capture_review_screenshots.py` loads the frozen item packet against an empty temporary annotation database. Item order was independently randomized by reviewer.

Orthogonal packets are a useful outcome-relevance control, but are not equivalent to receiving no new text. A literal no-information update condition would separate generic prompt-induced revision



(a) Information, temporal frame, and consent.

(b) A complete item with June 10 context, synthetic report, and five judgments.

Figure 12: **Stage 3 materials-review interface.** Reviewers enter a private code before seeing the information-and-consent screen. Each item repeats the June 10 temporal frame, presents the question and pre-report background, labels the fictional evidence as a synthetic report, and collects conditional direction, confidence, clarity, plausibility, and premise usability. Item order was independently randomized by reviewer. Intended direction and private packet metadata are hidden.

from response to topically adjacent content. The very small hosted-system orthogonal revisions (0.5–1.4 pp) do not substitute for that control.

B.5.5 Prompt, Retrieval, and Runtime Confounding

Agents shared a substantive task contract, while execution was not byte-identical. Hosted models used Azure Foundry search, while local models used Tavily results in an Ollama workflow; temperatures and provider behavior also differ. Hosted–local gaps therefore combine model capability with retrieval and runtime differences.

B.5.6 Scale, Missingness, and Dependence

Realized repeat counts are uneven because failures and top-ups differ by provider. Equal-weight market aggregation prevents high-coverage markets from dominating EHC, HFC, and ICS. Movement and sensitivity intervals use 20,000 market-clustered bootstrap draws that retain every packet and repeated initial-run update from each selected market. The mixed-scale Qwen 2.5-7B row described in Section B.3.1 remains an appendix compliance diagnostic and is excluded from the main magnitude comparison.

B.5.7 Premise Acceptance at Stage 3

Search is intentionally unavailable during updating, and agents are told to accept the synthetic headline as true. This design measures conditional premise use rather than source verification or epistemic refusal. The human materials review uses the same instruction and therefore assesses comprehension and premise usability. These results do not establish whether a model should trust or verify the synthetic report.

B.6 Reproducibility and Artifact Map

The paper treats the June 17 model-update report and the August 23 final materials-review summary as the frozen numerical authorities. The relevant artifacts are:

- **Sampling:** `exp1_prospective/fetch_markets/select_markets.py` and the June 9 selection/diversity JSON and manifests in `exp1_prospective/data/selected_markets/`.
- **Initial elicitation:** `exp1_prospective/agent/forecast_agent.py` for hosted runs and `exp1_prospective/agent/run_local_forecast.py` for Ollama runs, with records and manifests in `exp1_prospective/data/initial_forecasts/`.
- **Interventions:** `exp1_prospective/agent/build_counterfactuals.py` and the frozen `exp1_prospective/data/counterfactuals/counterfactuals_2026-06-10.jsonl`.
- **Updates:** hosted and local update runners under `exp1_prospective/agent/`, with update records in `exp1_prospective/data/updated_forecasts/`.
- **Evaluation:** `exp1_prospective/agent/evaluate_consistency.py` and `exp1_prospective/data/results/consistency_report_2026-06-17.json`; `exp1_prospective/agent/evaluate_clustered_uncertainty.py` reconstructs the 20,000-draw market-clustered intervals in `exp1_prospective/data/results/clustered_movement_uncertainty.json`; `exp1_prospective/agent/audit_orthogonal_abstention.py` writes the abstention/magnitude decomposition of Table 9 to `exp1_prospective/data/results/orthogonal_abstention.json` and its table source, under the same resampling scheme.
- **Human materials review:** the frozen protocol, blinded packet, browser application, and tests in `exp1_prospective/stage3_materials_annotation/`; `make_final_summary.py` reconstructs `data/exports/final_summary_current.json`, while `data/exports/agreement_current.json` is the paper-facing agreement export.
- **Figures:** `paper/generate_figures.py` writes the direction-selective panel; `paper/generate_paper_figures.py` writes the combined EHC/sensitivity panel.

The model-update tables are defined by the June 17 frozen JSON. The mutable collection directories contain later records, so evaluating their present contents would define a different dataset. Bit-for-bit reconstruction begins from the frozen JSON; a full pipeline reconstruction uses only the manifests and records included in that freeze.

C Experiment 2: Contextual Relationship Selection and Revision in Coin City

Experiment 2 tests whether context selects which response relationship a model applies and whether direct target observations revise that selection. The six primary deployments and symbol-control comparisons reuse the same 250 seeded episodes, so prompt-arm differences are paired rather than comparisons between independently sampled tests.

C.1 Design and Data-Generating Process

C.1.1 Design summary

The design identifier is `coin_city_stable_relationship_claude_n250_v4`. It contains 250 independent episodes generated from seeds 930000–930249, five target-city prefixes $k \in \{0, 1, 2, 3, 4\}$, and three semantic-selection arms. Cue-inversion and arbitrary-symbol modules reuse those episode identities and numerical inputs; each has a hashed task manifest (Section C.3). Every model received the same episode records, queries, and arm-specific prompt text.

C.1.2 Balanced latent regimes

Each episode crosses two binary factors: whether City A or City B is the strongly responsive reference city, and whether target City C is strongly or weakly responsive. The closest-balanced allocation contains 63 A-strong/C-strong, 62 A-strong/C-weak, 62 B-strong/C-strong, and 63 B-strong/C-weak episodes. Within an episode, each city receives an independently sampled stable slope

$$g_{ec} \sim \begin{cases} \mathcal{N}(0.90, 0.03^2), & \text{strong regime,} \\ \mathcal{N}(0.25, 0.03^2), & \text{weak regime.} \end{cases}$$

Thus the cue identifies a response regime but not the realized coefficient.

C.1.3 Observed cases and forecast query

Every city has four reference cases arranged as two matched pairs. Rows in a pair share a starting poll and news magnitude but have opposite signs; the first sign is random. For reference row r,

$$p_{ecr}^{\text{end}} = \text{round}_{0.1}\left(p_{ecr}^{\text{start}} + g_{ec}x_{ecr} + \epsilon_{ecr}\right), \qquad \epsilon_{ecr} \sim \mathcal{N}(0, 5^2).$$

Starting polls are uniform on $[35, 65]$, and reference-news magnitudes are drawn from $\{6, 7, 8, 9, 10\}$. The query shock is drawn from $\{-10, \ldots, -4, 4, \ldots, 10\}$. Its scoring target is the latent conditional expectation $p_0 + g_C x$, with no additional query-time observation noise.

The prefix k exposes the first k City C rows in their stored order. Consequently, $k = 2$ completes the first matched pair and $k = 4$ completes both; $k = 1$ and $k = 3$ deliberately expose partial pairs. City A and B always contribute all four rows in the ABC arms.

Table 14: Coin City design and evaluation dimensions. Final analytic record counts retain one record per task cell; raw transport-level calls are discussed in Section C.9.

Quantity	Value
Independent episodes	250
Factorial cells	63, 62, 62, 63 across the four A/B-by-C regimes
Target-city prefixes	$k \in \{0, 1, 2, 3, 4\}$
Primary prompt arms	target only; ABC/no C context; ABC/correct C context
Intervention modules	ABC/misleading C context; ABC/episode-randomized arbitrary labels
Primary deployment roster	Claude Opus 4.8; GPT-5.6; DeepSeek-V4-Pro; Kimi K3; Gemini 3.6 Flash; Claude Opus 5
Symbol-comparison roster	Qwen3 4B/8B/14B/32B; Qwen2.5 7B/14B/32B/72B; Llama 3.1 8B/70B; DeepSeek-V4-Pro; Kimi K3; Claude Opus 4.8/5; GPT-5.6 Sol
Primary task cells / final records	22,500 / 22,500 (22,497 parseable)
Misleading task cells / final records	7,500 / 7,500 (all parseable)
Symbol-comparison task cells / response records	56,250 / 56,250 (15 models)
Strong / weak slope means	0.90 / 0.25 poll points per news point
Slope standard deviation	0.03 within each regime
Reference-case noise	5 poll points
Reference cases per city	4 (two matched pairs)

C.1.4 Preflight validation and task integrity

The GPT-5.6, DeepSeek-V4-Pro, Kimi K3, Gemini 3.6 Flash, and Claude Opus 5 replications reuse the exact task files and prompt hashes created for the original Claude Opus 4.8 evaluation; no episode was regenerated. Before model calls, the manifest hashed the engine, task files, answer and scoring keys, validator, runner, renderer, and launcher. Automated validation confirmed the intended dimensions and factorial balance, prompt contracts and hashes, correct cue-to-regime mapping, visible matched pairs, and benchmark reconstruction. No model response or performance statistic was used by this preflight validator.

C.2 Prompt Arms and Information Available

C.2.1 Shared task instructions

All prompts state that rows are separate polling cases rather than a time series, positive news favors the candidate, poll change is end minus start, and within-city responsiveness is stable even though individual polls are noisy. Rows sharing a pair label are described as having the same starting poll and equal-magnitude, opposite-sign news. Every displayed row includes the starting poll, news value, change, and ending poll.

The required answer is a JSON object with exactly a one-sentence `rationale` and a numeric `predicted_poll` between 0 and 100. The prompt asks for no step-by-step derivation.

C.2.2 Five information conditions

Target only. The model sees the first k City C cases and is explicitly told that no additional background information is available. This arm measures recovery from target-city evidence alone.

ABC, no City C context. The model also sees four cases from each reference city, each introduced by a background sentence. The strongly responsive reference city is described verbatim as receiving "national news coverage, where campaign stories receive sustained attention and tend to produce larger immediate poll movements", and the weakly responsive city as receiving "local news coverage, where campaign stories compete with local issues and tend to produce smaller immediate poll movements". The direction of the coverage-to-response mapping is therefore stated in words, though no numerical value is. City C receives no background description. This arm tests numeric transfer without a cue that aligns C to either reference regime.

ABC plus correct City C context. The numeric rows and query are identical to the preceding arm. The sole added information is that City C receives national or local coverage, correctly aligned with its latent strong or weak regime. This is the primary cue-transfer condition.

ABC plus misleading City C context. This intervention arm is identical to the correct-context prompt except that its City C sentence names the opposite coverage regime. Every number, case row, query, and other instruction is unchanged. This arm tests whether the model follows the semantic cue when it conflicts with the truth.

ABC plus an arbitrary City C label. Every national/local description is replaced by one of two arbitrary labels, `KIV` or `ZOR`. The prompt states only that cities sharing a label share a typical response regime and that the labels have no inherent meaning or ordering. Their higher/lower-response mapping reverses on alternating episodes; City A and B's numerical cases are therefore the only evidence from which the episode-local mapping can be induced before applying City C's label.

No prompt contains a fitted coefficient, an estimator, the strong or weak numerical means, the case-noise scale, a rule for pooling cities, or any analyst benchmark. The correct-versus-no-context contrast therefore isolates the one City C background sentence while holding all displayed numeric evidence fixed; the correct-versus-misleading comparison isolates the sentence's regime assignment. The symbol-versus-no-context contrast asks whether the same assignment can be induced without lexical knowledge of national and local coverage.

Because the reference sentences state the direction of the mapping, the semantic arms are not by themselves a test of whether that direction can be induced. A policy that reads no displayed number and assigns slope 1.0 to the nationally covered regime and 0.2 to the locally covered one attains $k = 0$ forecast MAE 0.523 and implied-slope correlation .996, better than any deployment and better than the analogical-regression benchmark of Table 16; 42 of 108 prior pairs on the grid $\{0.4, \ldots, 1.5\} \times \{0.0, \ldots, 0.8\}$ beat every deployment's $k = 0$ MAE. The semantic arms therefore measure how a stated regime assignment is converted into a numerical response. Two separate controls carry the induction claim: the arbitrary-symbol arm, whose labels have no stated direction and whose mapping reverses across episodes, and the reference-selection analysis of Section C.6, which shows that the forecast tracks the named reference city's own sampling error rather than any fixed prior.

C.3 Model Execution and Response Processing

C.3.1 Deployment configurations

Table 15 reports the effective API settings used for production. Temperature refers to the value sent to the provider: GPT-5.6 and Claude Opus 5 omit it even though the local GPT run manifest records a default of zero. No deployment used tools or retrieval.

Table 15: Production deployment configuration and primary-arm parse counts.

Deployment	Interface	Reasoning / sampling	Output ceiling	Parsed
Claude Opus 4.8	Anthropic Messages via Azure	temperature 0	512	3,750
GPT-5.6	OpenAI Responses	low reasoning; temperature omitted	4,096	3,749
DeepSeek-V4-Pro	OpenAI Chat Completions	temperature 0	4,096	3,750
Kimi K3	OpenAI Chat Completions	low reasoning; temperature 0	16,384	3,750
Gemini 3.6 Flash	Gemini OpenAI-compatible Chat	low reasoning; temperature 0	16,384	3,750
Claude Opus 5	Foundry Anthropic Messages	default extended thinking; temperature omitted	4,096	3,748

Claude Opus 5 returns extended thinking before its answer; the runner therefore selects its first text block. Gemini is the sole fast-tier deployment, so the six systems should be read as a replication set rather than a tier-matched leaderboard. Authentication material was supplied at runtime and was not written to source, manifests, logs, or response files.

C.3.2 Deployment attempts, retries, and pilots

Every primary model–prompt pair received one application-level attempt. A failed or answerless task was not re-asked. HTTP-client retries were disabled for GPT, DeepSeek, Kimi, and Opus 5. The original Opus 4.8 client allowed up to two transport retries. Gemini's runner allowed adaptive pre-model transport retries but recorded none in the completed shards. These transport policies do not alter the one-answer-per-task analytic rule.

Before production, 119 Kimi, 15 Gemini, and 17 Opus 5 task-shaped calls were used only to select configurations that reliably returned the required JSON schema. Forecast accuracy was not inspected. The calls are archived separately and are excluded from every denominator and analysis; each production namespace began empty. Production tasks were sharded only for throughput.

C.3.3 Response parsing and record selection

The parser scans the raw reply for JSON objects, retains the last object with an exact `predicted_poll` key whose value converts to a number in $[0, 100]$, truncates the retained rationale to 500 characters, and stores at most 8,000 raw characters. When duplicate task records exist, the analysis keeps the latest parseable response, allowing a valid response to supersede an earlier error. A single offline normalization removed whitespace from JSON keys in one otherwise valid Opus 4.8 reply; this recovered one primary record without another model call. Other record-level corrections are documented in Section C.9.

C.3.4 Misleading-context design

The cue-inversion module begins from the frozen semantic-selection tasks. Before any inversion call, its builder verifies that every prompt equals its correct-context counterpart under exactly one sentence substitution and that all numeric content is unchanged. The cue is inverted for all 1,250 episode-prefix tasks: 625 strong-city tasks are described as weak and 625 weak-city tasks as strong. Each deployment uses the same client and production configuration as its correct-context arm.

C.3.5 Arbitrary-symbol design

The symbol builder generates 1,250 matched prompts from the same frozen episodes and verifies that their complete sequence of numeric tokens is identical to the correct-context arm. The higher-response label is `KIV` in 125 episodes and `ZOR` in 125; the target regime-by-label cells contain 62–63 episodes each. A prompt validator rejects national/local and strong/weak semantic descriptions. At $k = 0$, City C supplies no numerical cases, so its correct regime can be recovered only by decoding the label from City A and B's examples.

For DeepSeek-V4-Pro, we collected all 1,250 symbol tasks and contemporaneously re-collected all 2,500 no-context and semantic-context tasks on the same reachable Azure resource, using one deployment identifier and configuration. The released 15-model comparison combines that triplet with Qwen3-4B, Qwen3-8B, Qwen3-14B, Qwen3-32B, Qwen2.5-7B, Qwen2.5-14B, Qwen2.5-32B, Qwen2.5-72B, Llama 3.1-8B, Llama 3.1-70B, Kimi K3, Claude Opus 4.8, Claude Opus 5, and GPT-5.6 Sol.

The eight Qwen and two Llama checkpoints received all 3,750 prompts through vLLM in bfloat16, with temperature-zero decoding, a 3,072-token model context, and a 512-token output ceiling. Qwen uses the no-thinking chat template; Llama uses its native instruction template. Qwen2.5-14B uses two-way and Qwen3-32B and Qwen2.5-32B four-way tensor parallelism, while Qwen2.5-72B and Llama 3.1-70B use eight-way tensor parallelism.

For Kimi K3, Claude Opus 5, and GPT-5.6 Sol, the comparison reuses each deployment's preserved no-context and semantic-context responses and adds only the 1,250 symbol prompts under the same model-specific production protocol. Claude Opus 4.8's symbol arm instead ran on a second Azure resource serving the same deployment and used a 1,024-token rather than 512-token ceiling; all 1,250 responses returned valid JSON. We therefore treat its arm contrast as a cross-run robustness result and do not use it to rank providers.

Every runner verifies the frozen task hash before generation and writes a timestamped manifest. Each released model has 1,250 unique response records in every arm; this count does not imply that every prediction parsed. No primary response file or frozen task is overwritten. Original outputs are preserved, and the final tables use matched complete cases after the no-call arithmetic reparse described in Section C.9.

C.4 Estimands, Benchmarks, and Statistical Inference

C.4.1 Forecast and latent-response estimands

Forecast error is measured against the latent conditional expectation rather than a newly sampled noisy ending poll:

$$\mathrm{MAE} = \frac{1}{n}\sum_{e=1}^{n} \left|\widehat{p}_e - (p_{0e} + g_{Ce}x_e)\right|.$$

For every nonzero query shock, a forecast also identifies an implied slope $\widehat{g}_e = (\widehat{p}_e - p_{0e})/x_e$. We report its mean absolute error from g_{Ce} and its Pearson correlation with g_{Ce} across episodes. Correlation is needed because a nearly constant prior-mean slope can attain moderate MAE while failing to distinguish strong from weak cities.

Within each deployment and prefix, the three primary-arm metrics use the common intersection of episodes with parseable responses in all three arms. Arm contrasts are paired by episode and prefix. The misleading-arm table uses its final unique parseable record for every task. The arbitrary-symbol analysis uses each model's corresponding common intersection across its three matched arms.

C.4.2 Analyst-only benchmarks

The ABC OLS benchmark fits poll change on news through the origin using every displayed A, B, and available C row. The target-only OLS benchmark uses only displayed City C rows and is undefined at $k = 0$. Both are reconstructed from the visible data by the analyst; neither the estimator nor its output appears in a model prompt.

The label-conditioned analogical regression implements the minimal structured inductive policy defined in Methods. It fits separate through-origin slopes to City A and City B, uses City C's coverage label to select the reference assigned to the same response regime, and predicts with that slope. At $k > 0$, it pools the selected reference rows with the visible City C rows before refitting. In the misleading arm, the inverted label selects the opposite reference. The implementation uses only displayed rows; neither the estimator nor its output appears in a model prompt.

Table 16: Zero-shot behavior of the analogical policy and the six model deployments. At $k = 0$, City C has no numerical cases. Model columns give the range across deployments; forecast MAE is in poll points.

Cue	Policy MAE	Policy ρ	Model MAE range	Model ρ range
Correct	1.842	.656	1.596–2.474	$[.52, .70]$
Inverted	4.561	$-.667$	4.204–4.361	$[-.68, -.50]$

C.4.3 Uncertainty and comparison families

Primary arm contrasts use percentile intervals from 2,000 bootstrap resamples of the paired episode-level absolute-error difference. Deterministic, model-specific seeds make these intervals reproducible. The $k = 0$ pairwise model table uses 4,000 paired bootstrap resamples with seed 20260812 and the common episode intersection for each model pair. Negative differences favor the first-listed condition or model.

The episodes are the resampling units. Intervals are pointwise descriptions of this fixed seeded synthetic evaluation: they are not simultaneous bands across prefixes, and no multiple-comparison correction is applied. Accordingly, the number of pairwise intervals excluding zero is not used as an estimand; each interval supports only its displayed cell.

C.5 Response-Inference Diagnostics

The tables below retain a common arm and prefix ordering. Within a deployment and prefix, all three primary arms use the same complete-case episode set; n is repeated so that every exception is visible. Boldface identifies the lowest forecast MAE or slope MAE, and the highest slope correlation, within that deployment and prefix.

C.5.1 Per-deployment primary-arm recovery

Table 17: Claude Opus 4.8 forecast and latent-response diagnostics for every prompt arm. Forecast MAE is in poll points; slope MAE and ρ compare the implied response with the true City C slope.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.321	.332	.019
	ABC, no C context	250	2.651	.379	-.063
	ABC + C context	250	**1.817**	**.265**	**.700**
1	Target only	250	3.730	.549	.371
	ABC, no C context	250	2.438	.358	.322
	ABC + C context	250	**2.147**	**.316**	**.584**
2	Target only	250	2.668	.403	.383
	ABC, no C context	250	**2.041**	**.296**	**.492**
	ABC + C context	250	2.079	.308	.491
3	Target only	250	2.485	.358	.449
	ABC, no C context	250	2.128	.305	.513
	ABC + C context	250	**1.977**	**.291**	**.627**
4	Target only	250	2.209	.324	.498
	ABC, no C context	250	1.929	.281	.542
	ABC + C context	250	**1.760**	**.257**	**.658**

Table 18: GPT-5.6 forecast and latent-response diagnostics on the same frozen prompts. Metrics use episodes parseable in all three arms at each prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	3.596	.511	.054
	ABC, no C context	250	2.518	.352	-.041
	ABC + C context	250	**1.596**	**.224**	**.678**
1	Target only	249	2.993	.439	.394
	ABC, no C context	249	2.768	.399	.302
	ABC + C context	249	**1.586**	**.225**	**.669**
2	Target only	250	2.560	.387	.459
	ABC, no C context	250	2.050	.299	.534
	ABC + C context	250	**1.854**	**.268**	**.606**
3	Target only	250	2.240	.328	.528
	ABC, no C context	250	1.967	.286	.584
	ABC + C context	250	**1.773**	**.256**	**.600**
4	Target only	250	2.153	.318	.577
	ABC, no C context	250	1.746	.253	.653
	ABC + C context	250	**1.594**	**.228**	**.664**

Table 19: DeepSeek-V4-Pro forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	**2.447**	.353	-.027
	ABC, no C context	250	2.841	.401	-.027
	ABC + C context	250	2.474	**.344**	**.523**
1	Target only	250	3.585	.525	.334
	ABC, no C context	250	3.130	.458	.302
	ABC + C context	250	**2.456**	**.347**	**.578**
2	Target only	250	3.380	.501	.319
	ABC, no C context	250	3.423	.502	.365
	ABC + C context	250	**2.687**	**.391**	**.492**
3	Target only	250	2.908	.424	.454
	ABC, no C context	250	3.224	.463	.403
	ABC + C context	250	**2.797**	**.403**	**.504**
4	Target only	250	2.812	.415	.430
	ABC, no C context	250	3.040	.459	.366
	ABC + C context	250	**2.809**	**.409**	**.470**

Table 20: Kimi K3 forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	3.352	.476	-.016
	ABC, no C context	250	2.548	.357	-.021
	ABC + C context	250	**1.886**	**.266**	**.593**
1	Target only	250	3.362	.494	.365
	ABC, no C context	250	3.370	.490	.345
	ABC + C context	250	**2.382**	**.341**	**.579**
2	Target only	250	3.194	.491	.387
	ABC, no C context	250	3.114	.487	.352
	ABC + C context	250	**2.306**	**.349**	**.455**
3	Target only	250	2.816	.412	.430
	ABC, no C context	250	2.636	.407	.410
	ABC + C context	250	**2.005**	**.292**	**.609**
4	Target only	250	2.378	.357	.444
	ABC, no C context	250	1.957	.284	**.622**
	ABC + C context	250	**1.840**	**.263**	.601

Table 21: Gemini 3.6 Flash forecast and latent-response diagnostics on the same frozen prompts. All 250 episodes parsed in every arm and prefix.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.534	.370	.002
	ABC, no C context	250	2.552	.356	.078
	ABC + C context	250	**2.174**	**.302**	**.559**
1	Target only	250	3.733	.550	**.403**
	ABC, no C context	250	3.717	.548	.361
	ABC + C context	250	**3.300**	**.500**	.292
2	Target only	250	**3.564**	**.558**	**.348**
	ABC, no C context	250	3.703	.579	.335
	ABC + C context	250	3.600	.569	.330
3	Target only	250	3.370	.513	.314
	ABC, no C context	250	3.092	.460	.397
	ABC + C context	250	**3.047**	**.443**	**.471**
4	Target only	250	3.308	.498	.389
	ABC, no C context	250	2.911	.447	.433
	ABC + C context	250	**2.793**	**.419**	**.447**

Table 22: Claude Opus 5 forecast and latent-response diagnostics on the same frozen prompts. Two responses exhausted the output budget on extended thinking and returned no answer, one in each ABC arm.

k	Arm	n	Forecast MAE	Slope MAE	$\rho(\hat{g}, g_C)$
0	Target only	250	2.359	.333	-.018
	ABC, no C context	250	2.619	.371	-.007
	ABC + C context	250	**1.764**	**.247**	**.617**
1	Target only	249	3.549	.520	.391
	ABC, no C context	249	2.303	.338	.370
	ABC + C context	249	**1.741**	**.254**	**.601**
2	Target only	250	3.804	.596	.309
	ABC, no C context	250	2.680	.412	.389
	ABC + C context	250	**2.060**	**.314**	**.524**
3	Target only	250	3.022	.461	.379
	ABC, no C context	250	2.363	.356	.459
	ABC + C context	250	**2.075**	**.309**	**.550**
4	Target only	249	2.700	.418	.443
	ABC, no C context	249	2.369	.366	.459
	ABC + C context	249	**2.201**	**.335**	**.499**

C.5.2 Misleading-cue results

Table 23: Misleading-cue arm: forecast MAE and latent-slope correlation at each City C prefix. The prompt is byte-identical to the correct-context arm except that the City C sentence names the opposite response regime. Negative ρ means the implied response is anti-correlated with the truth, i.e. the deployment has adopted the stated regime. Each of the six deployments contributes all 1,250 prompts.

	k=0		k=1		k=2		k=3		k=4	
Model	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ	MAE	ρ
Claude Opus 4.8	4.20	$-.66$	3.88	$-.51$	3.04	.09	2.61	.22	2.31	.34
GPT-5.6	4.23	$-.68$	4.15	$-.63$	2.78	.17	2.71	.25	2.10	.46
DeepSeek-V4-Pro	4.36	$-.62$	4.46	$-.48$	3.61	.07	3.39	.14	3.04	.32
Kimi K3	4.31	$-.62$	4.12	$-.25$	3.03	.15	2.72	.14	2.18	.44
Gemini 3.6 Flash	4.29	$-.50$	4.40	$-.06$	3.93	.24	3.52	.25	2.93	.36
Claude Opus 5	4.25	$-.62$	3.15	$-.21$	2.59	.27	2.41	.41	2.36	.47

C.5.3 Arbitrary-symbol induction results

Table 24: **Fifteen-system episode-randomized arbitrary-symbol comparison at $k = 0$.** Panel A reports forecast calibration; Panel B reports response-regime discrimination. All arms share the frozen numeric task content; Section C.3 documents model-specific execution and the Opus 4.8 cross-resource deviation. MAE is in poll points; ρ correlates implied and true latent slopes; Acc. classifies the response regime at slope 0.575. Each Δ is symbol minus no context, with a paired 95% episode-bootstrap interval. With no City C examples, its label can be decoded only from City A/B examples.

Panel A: Forecast calibration

Model	n	No context	Semantic	Symbol	ΔMAE [95% CI]
Qwen3-4B	250	2.896	2.365	2.892	$-.005$ $[-.088, .074]$
Qwen2.5-7B	250	2.911	2.669	3.248	$+.338$ $[.027, .682]$
Qwen3-8B	250	2.985	2.859	2.973	$-.011$ $[-.239, .207]$
Qwen2.5-14B	250	2.606	2.548	2.711	$+.105$ $[-.178, .388]$
Qwen3-14B	250	2.632	2.241	2.812	$+.180$ $[-.188, .548]$
Qwen3-32B	250	2.482	2.069	2.914	$+.432$ $[.119, .752]$
Qwen2.5-32B	250	2.679	2.404	3.208	$+.529$ $[.169, .924]$
Qwen2.5-72B	250	2.474	2.383	3.398	$+.924$ $[.582, 1.310]$
Llama 3.1 8B	242	3.113	3.386	3.605	$+.492$ $[.029, .980]$
Llama 3.1 70B	250	3.483	3.761	3.931	$+.448$ $[.030, .886]$
DeepSeek V4-Pro	250	2.622	2.588	2.729	$+.107$ $[-.222, .452]$
Kimi K3	250	2.548	1.886	2.051	$-.497$ $[-.765, -.211]$
Claude Opus 4.8	250	2.651	1.817	2.067	$-.584$ $[-.858, -.314]$
Claude Opus 5	250	2.619	1.764	2.017	$-.603$ $[-.869, -.336]$
GPT-5.6 Sol	250	2.518	1.596	1.744	$-.774$ $[-1.011, -.533]$

Panel B: Response-regime discrimination

		$\rho(\hat{g}, g_C)$			Regime accuracy		
Model	n	No context	Symbol	$\Delta\rho$ [95% CI]	No context	Symbol	ΔAcc. [95% CI]
Qwen3-4B	250	-.011	.085	$+.095$ $[-.060, .230]$	.496	.512	$+.016$ $[-.008, .044]$
Qwen2.5-7B	250	-.034	.230	$+.264$ $[.112, .406]$	.500	.572	$+.072$ $[.020, .128]$
Qwen3-8B	250	-.014	.009	$+.023$ $[-.128, .171]$	.492	.516	$+.024$ $[-.012, .060]$
Qwen2.5-14B	250	-.029	.208	$+.236$ $[.107, .356]$	.492	.580	$+.088$ $[.020, .152]$
Qwen3-14B	250	-.057	.342	$+.398$ $[.253, .541]$	.496	.660	$+.164$ $[.088, .240]$
Qwen3-32B	250	.075	.377	$+.302$ $[.177, .433]$	.524	.656	$+.132$ $[.060, .200]$
Qwen2.5-32B	250	-.015	.358	$+.372$ $[.226, .502]$	.500	.648	$+.148$ $[.088, .208]$
Qwen2.5-72B	250	.027	.408	$+.381$ $[.250, .508]$	.516	.632	$+.116$ $[.056, .176]$
Llama 3.1 8B	242	-.042	.003	$+.045$ $[-.123, .250]$	.467	.512	$+.045$ $[-.029, .120]$
Llama 3.1 70B	250	-.039	.385	$+.424$ $[.314, .543]$	.500	.592	$+.092$ $[.036, .152]$
DeepSeek V4-Pro	250	.018	.454	$+.436$ $[.323, .558]$	.520	.692	$+.172$ $[.104, .240]$
Kimi K3	250	-.021	.612	$+.633$ $[.520, .742]$	.468	.804	$+.336$ $[.272, .400]$
Claude Opus 4.8	250	-.063	.560	$+.623$ $[.492, .744]$	.448	.760	$+.312$ $[.240, .380]$
Claude Opus 5	250	-.007	.583	$+.591$ $[.485, .697]$	.496	.780	$+.284$ $[.220, .348]$
GPT-5.6 Sol	250	-.041	.655	$+.696$ $[.597, .807]$	.464	.836	$+.372$ $[.312, .432]$

Table 25: **Matched open-weight scaling across target-city observations.** Symbol-minus-no-context effects are reported for Qwen2.5 32B/72B and Llama 3.1 8B/70B at each City C prefix. Negative ΔMAE favors the induced symbol mapping; positive $\Delta\rho$ and Δaccuracy indicate better regime discrimination.

Model	k	n	ΔMAE [95% CI]	$\Delta\rho$ [95% CI]	ΔAcc. [95% CI]
Qwen2.5-32B	0	250	$+.529$ [.169, .924]	$+.372$ [.226, .502]	$+.148$ [.088, .208]
	1	250	$+.124$ [$-.188$, .441]	$+.096$ [.010, .190]	$+.068$ [.016, .124]
	2	250	$+.184$ [$-.044$, .421]	$-.010$ [$-.059$, .037]	$-.012$ [$-.048$, .024]
	3	250	$+.462$ [.222, .710]	$-.033$ [$-.096$, .024]	$-.004$ [$-.044$, .036]
	4	250	$+.255$ [$-.001$, .524]	$-.010$ [$-.059$, .033]	$-.016$ [$-.048$, .016]
Qwen2.5-72B	0	250	$+.924$ [.582, 1.310]	$+.381$ [.250, .508]	$+.116$ [.056, .176]
	1	250	$+.189$ [$-.148$, .520]	$+.166$ [.075, .265]	$+.076$ [.024, .128]
	2	250	$-.024$ [$-.286$, .239]	$+.070$ [.017, .127]	$+.016$ [$-.028$, .060]
	3	250	$+.252$ [$-.050$, .549]	$+.019$ [$-.051$, .092]	$+.024$ [$-.024$, .068]
	4	250	$+.019$ [$-.280$, .322]	$+.011$ [$-.067$, .088]	$+.020$ [$-.024$, .064]
Llama 3.1 8B	0	242	$+.492$ [.029, .980]	$+.045$ [$-.123$, .250]	$+.045$ [$-.029$, .120]
	1	240	-2.318 [-3.256, -1.373]	$+.120$ [$-.022$, .263]	$-.050$ [$-.129$, .029]
	2	238	-2.454 [-3.433, -1.529]	$+.106$ [$-.038$, .260]	$-.017$ [$-.097$, .055]
	3	244	$-.867$ [-1.739, .085]	$-.088$ [$-.238$, .061]	$-.078$ [$-.156$, .000]
	4	246	-1.396 [-2.321, $-.489$]	$+.100$ [$-.067$, .245]	$-.049$ [$-.122$, .024]
Llama 3.1 70B	0	250	$+.448$ [.030, .886]	$+.424$ [.314, .543]	$+.092$ [.036, .152]
	1	250	$+.194$ [$-.245$, .658]	$+.077$ [$-.026$, .190]	$-.016$ [$-.076$, .040]
	2	250	$-.200$ [$-.548$, .149]	$+.039$ [$-.026$, .106]	$-.016$ [$-.068$, .036]
	3	250	$-.345$ [$-.621$, $-.075$]	$+.037$ [$-.015$, .087]	$+.020$ [$-.016$, .056]
	4	250	$-.079$ [$-.306$, .158]	$+.034$ [$-.006$, .082]	$+.012$ [$-.020$, .048]

At $k = 0$, Qwen3 provides a within-family contrast. Symbol-minus-no-context $\Delta\rho$ is $.095$ $[-.060, .230]$ at 4B and $.023$ $[-.128, .171]$ at 8B, but $.398$ $[.253, .541]$ at 14B and $.302$ $[.177, .433]$ at 32B; the regime-accuracy contrasts follow the same pattern. The 4B and 8B intervals span zero, whereas the 14B and 32B intervals do not. The 32B point estimate sits below the 14B one, so this is a boundary among the tested Qwen3 checkpoints rather than a monotone or universal parameter threshold. The smaller checkpoints are not wholesale task failures: their semantic-context slopes correlate .370 and .286 with truth, versus $-.011$ and $-.014$ without context.

All four tested Qwen2.5 checkpoints discriminate under arbitrary labels ($\Delta\rho = .264/.236/.372/.381$ at 7B/14B/32B/72B), reinforcing the family qualification. Their symbol-minus-no-context MAE changes are $+.338/+.105/+.529/+.924$: mapping discrimination does not ensure calibrated numerical use. Direct City C examples make the large-Qwen symbol advantage redundant; by $k = 3$–4, the 32B and 72B $\Delta\rho$ and Δaccuracy intervals span zero.

Llama 3.1 provides a second within-family contrast: $\Delta\rho = .045$ $[-.123, .250]$ at 8B and $.424$ $[.314, .543]$ at 70B. The 70B regime-accuracy contrast is $.092$ $[.036, .152]$, although its MAE rises by $.448$ $[.030, .886]$. In Qwen3 the mapping-recovery interval excludes zero only from 14B upward, and in Llama only at 70B. The positive Qwen2.5-7B and Qwen2.5-14B estimates, despite Qwen3-8B's interval spanning zero, indicate that the tested transition depends on model family rather than parameter count. Mapping recovery and numerical calibration remain separate outcomes.

DeepSeek improves discrimination ($\Delta\rho = .436$ $[.323, .558]$) without a detectable zero-shot MAE change. Kimi, both Opus deployments, and GPT combine mapping recovery with lower forecast error: their $\Delta\rho$ values are $.633/.623/.591/.696$, while MAE falls by $.497/.584/.603/.774$ points. Cross-provider magnitudes remain descriptive because runtimes, reasoning settings, and output ceilings differ.

C.5.4 Paired cue effects and interpretation

Table 26: Paired forecast-MAE difference between the correct-context arm and the context-blind ABC OLS benchmark, which the analyst fits through the origin on every displayed A, B and available City C row. Negative values favour the deployment. Boldface marks intervals excluding zero.

Model	k=0	k=1	k=2	k=3	k=4
Claude Opus 4.8	$-.73$	$-.18$	$-.05$	$-.01$	$-.08$
GPT-5.6	$-.95$	$-.74$	$-.28$	$-.21$	$-.24$
DeepSeek-V4-Pro	$-.07$	.13	.56	.81	.97
Kimi K3	$-.66$	.06	.18	.02	.01
Gemini 3.6 Flash	$-.37$	.98	1.47	1.06	.96
Claude Opus 5	$-.78$	$-.57$	$-.07$	.09	.36

At $k = 0$ every deployment matches or beats the context-blind benchmark, and five of the six intervals exclude zero: with no City C cases, the only way to separate City C from the pooled A/B fit is to use its description. The advantage does not persist once City C supplies its own rows. By $k = 2$ DeepSeek-V4-Pro and Gemini 3.6 Flash are significantly worse than the benchmark, Claude Opus 5 becomes so at $k = 4$, and the remaining deployments are indistinguishable from it. The cue effect this experiment measures is therefore a zero-shot phenomenon, and a plain through-origin estimator remains competitive once direct target evidence accumulates.

In the zero-shot matched-reference comparison, the target-city description changes the influence of the A/B cases on forecasts. All six $k = 0$ bootstrap intervals exclude zero. A label-conditioned benchmark that selects a reference, estimates its slope, and transfers that estimate reproduces the aggregate pattern; the comparison does not identify the model's internal algorithm. More target rows do not monotonically reduce error: DeepSeek and Opus 5 drift by 0.34 and 0.44 points from $k = 0$ to $k = 4$, and Gemini's context advantage is significant only at $k \in \{0, 1\}$. Short, noisy prefixes can temporarily mislead either arm.

C.6 Does the Forecast Track the Cue-Named Reference?

Two accounts explain the correct-versus-no-context contrast equally well. Under a lexical-prior account the model maps the City C coverage sentence straight to a remembered response magnitude and never reads City A or City B's numbers. Under a reference-selection account it uses the cue to choose a reference city, estimates that city's slope from its four displayed cases, and transfers the estimate. Both predict a regime-level correlation, because the cue and the latent regime coincide.

The design separates them. A through-origin fit to four noisy reference cases has a within-regime standard deviation of .332 against a generator standard deviation of .03, so the matched reference's fitted slope correlates only .656 with the true City C slope where the regime indicator correlates .996. That sampling error is the discriminating signal: a forecaster that is deterministic given the cue cannot reproduce it, whereas one that transfers a fitted reference slope inherits it. For every deployment, arm and prefix we therefore report

$$r\left(\widehat{g}_e,\ \widehat{\beta}_e^{\mathrm{ref}}\ \middle|\ R_e\right),$$

the partial correlation between the forecast-implied slope and a reference city's through-origin OLS slope, controlling for the response regime R_e that the City C label names. Arms with no City C label are scored against the true regime and report the regime-matched and regime-mismatched reference instead. Intervals are 2,000-draw episode bootstraps with deterministic per-cell seeds, and the analysis makes no model call.

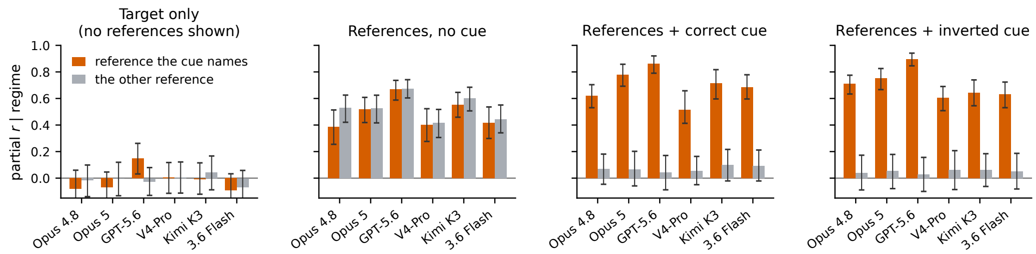


Figure 13: **The cue selects one of the two displayed references.** Partial correlation between each deployment's forecast-implied City C slope and a reference city's own fitted slope at $k = 0$, controlling for the response regime the City C sentence names; bars are 95% episode-bootstrap intervals. Orange is the reference that sentence points to, gray the other. With no reference cases displayed neither loads; with both displayed and no cue they load equally, the signature of pooling; the correct cue concentrates loading on the reference it names; and an inverted cue moves that loading onto the reference it falsely names. A prior attached to the words *national* and *local* predicts no loading in any panel. Per-deployment values and the arbitrary-symbol version are in Appendix C.6.

Table 27: Partial correlation between the forecast-implied City C slope and each reference city's fitted slope at $k = 0$, controlling for the regime the City C label names. "Named" is the reference the label points to, or the regime-matched reference in the two arms without a label. Boldface marks named-reference intervals excluding zero.

	Target only		ABC, no label		ABC + correct		ABC + inverted	
Model	Named	Other	Named	Other	Named	Other	Named	Other
Claude Opus 4.8	−.08	−.02	.39	.53	.62	.07	.71	.04
GPT-5.6	.15	−.03	.67	.68	.86	.04	.90	.03
DeepSeek V4-Pro	.00	.01	.40	.42	.51	.05	.61	.06
Kimi K3	−.01	.04	.55	.60	.71	.10	.64	.06
Gemini 3.6 Flash	−.09	−.07	.42	.44	.69	.09	.63	.05
Claude Opus 5	−.07	−.00	.52	.53	.78	.07	.75	.05

The four arms make four different predictions and all four hold. With no reference cases displayed at all, the target-only arm loads on neither reference: eleven of its twelve intervals span zero. With both references displayed but no City C label, every deployment loads on the two references almost equally, the pooling signature of a model that cannot tell which reference applies. Adding the correct label concentrates the loading on the named reference and removes it from the other, every "other" interval spanning zero. Inverting the label moves the loading wholesale onto the reference that the false cue names, leaving the truth-matched reference near zero.

The forecast therefore tracks the particular reference city the label names, including that city's sampling error, whether or not the label is correct. No prior attached to the words *national* and *local* reproduces this. It also explains why implied-slope correlations plateau near .66 rather than near .996: faithful transfer of a noisy reference estimate is capped at that reference's own .656 correlation with the truth. GPT-5.6 and Claude Opus 4.8 slightly exceed the cap at $k = 0$, which is what blending a reference estimate with a regime prior produces.

The estimator only detects an inefficiency. The signal this analysis reads is noise the forecast did not have to import. Each city draws its slope independently, so a reference city's deviation from its regime mean is uncorrelated with City C's: simulating the generator gives $\operatorname{corr}(\widehat{\beta}^{\text{ref}} - \mu,\ g_C - \mu) = -.0000$ (Section C.7.4). Within a regime the reference's fitted slope is therefore pure sampling error about a mean the regime already supplies, and the response-recovery-optimal policy is to read the cue, ignore the reference numbers, and predict the regime mean—which attains $\rho = .996$ against the .656 a faithful transfer is capped at. A forecaster that did this would show no loading on either reference and would be indistinguishable, under this estimator, from the lexical-prior account the section argues against. The estimator thus identifies reference selection only in forecasters that fail to shrink, and has least power on those closest to the shrinking policy: the two deployments above the transfer cap are the two with the highest slope correlation, one of which also has the lowest $k = 0$ forecast error of the six. We read a positive loading as evidence for reference selection, and a null under this estimator as uninformative rather than as evidence for a prior.

Table 28: The same partial correlation in the arbitrary-symbol arm at $k = 0$, where `KIV` and `ZOR` carry no stated direction and their mapping reverses across episodes. The no-label columns repeat each model's pooling control. Boldface marks symbol named-reference intervals excluding zero.

		ABC, no label		ABC + symbol	
Model	n	Named	Other	Named	Other
Qwen3-4B	250	.05	$-.09$	.00	$-.03$
Qwen2.5-7B	250	.11	.15	.28	.10
Qwen3-8B	250	.12	.06	.09	.11
Qwen2.5-14B	250	.31	.20	.32	.03
Qwen3-14B	250	.26	.20	.46	.06
Qwen3-32B	250	.20	.19	.40	.03
Qwen2.5-32B	250	.22	.15	.38	.02
Qwen2.5-72B	250	.25	.18	.53	.04
Llama 3.1 8B	249	$-.02$	.07	$-.00$	$-.08$
Llama 3.1 70B	250	.28	.19	.54	.00
DeepSeek V4-Pro	250	.40	.42	.51	.07
Kimi K3	250	.55	.60	.86	.01
Claude Opus 4.8	250	.39	.53	.71	$-.05$
Claude Opus 5	250	.52	.53	.83	.02
GPT-5.6 Sol	250	.67	.68	.98	.02

Applying the estimator to the arbitrary-symbol arm removes the lexical account entirely. Twelve of the fifteen released systems have a named-reference interval excluding zero, and the three that do not—Qwen3-4B, Qwen3-8B and Llama 3.1-8B—are exactly the three whose symbol-minus-no-context $\Delta\rho$ intervals span zero in Table 24. The two statistics are computed from different quantities and agree on all fifteen systems.

C.7 Linear-Probe Analysis of Cue-Conditioned Representations

C.7.1 Behavioral Screening Criteria

We screen the same held-out prompts with Qwen3-8B, Qwen3-14B, dense Qwen3-32B, and Qwen2.5-72B-Instruct. Each checkpoint received 16 test episodes crossed with $k \in \{0, 4\}$ and correct, absent, or inverted City C context (96 prompts), with 96/96 parseable forecasts.

Table 29: **Behavioral screening results for the linear-probe analysis.** Entries are paired mean differences with 95% episode-bootstrap intervals ($n = 16$). Evidence is no-context $k = 4$ minus $k = 0$ aligned slope; selection is correct minus absent context at $k = 0$; reversal is inverted minus correct context at $k = 0$; forecast penalty is inverted minus correct poll MAE at $k = 0$. Positive evidence and selection, negative reversal, and a positive forecast penalty are the predicted directions.

Model	Evidence	Selection	Reversal	Forecast penalty
Qwen3-8B	$+.119$ $[-.094, +.374]$	$+.078$ $[-.012, +.219]$	$-.140$ $[-.315, +.007]$	$+.812$ $[-.063, +2.023]$
Qwen3-14B	$+.394$ $[+.152, +.734]$	$+.261$ $[+.134, +.400]$	$-.380$ $[-.597, -.172]$	$+1.662$ $[+.252, +3.155]$
Qwen3-32B	$+.315$ $[+.017, +.646]$	$+.234$ $[+.111, +.407]$	$-.390$ $[-.554, -.221]$	$+1.965$ $[+1.013, +3.083]$
Qwen2.5-72B	$+.246$ $[+.001, +.529]$	$+.096$ $[-.077, +.259]$	$-.433$ $[-.674, -.220]$	$+2.206$ $[+.957, +3.620]$

The 14B and 32B intervals exclude zero in all four predicted directions. For 72B, the evidence, reversal, and forecast-penalty intervals exclude zero, while the correct-versus-absent selection interval spans zero. All four 8B intervals span zero. We use 32B for the dense Qwen3 scaling comparison and 72B for a cross-generation comparison; neither is interpreted as another point on a monotonic size curve. Qwen3-4B and Llama-3.1-8B use the identical task file, held-out split, and model, layer, target, and endpoint selection rules.

C.7.2 Activation and Probe Protocol

We selected 64 episodes without replacement, balanced over which reference city was strongly responsive and whether City C's latent regime was strong or weak. Forty-eight episodes form a development set and 16 form a sealed test set. Each episode is rendered at $k = 0$ and $k = 4$ under correct, absent, and inverted context, giving 384 activation records. Within an episode-prefix pair the three prompts have identical numeric content. The only intervention is the City C relationship description.

For the final input token, we save the embedding output and every transformer-block output. The resulting tensor dimensions are $[384, 37, 2560]$ for Qwen3-4B; $[384, 37, 4096]$ for Qwen3-8B; $[384, 33, 4096]$ for Llama-3.1-8B; $[384, 41, 5120]$ for Qwen3-14B; $[384, 65, 5120]$ for Qwen3-32B; and $[384, 81, 8192]$ for Qwen2.5-72B-Instruct. We fit separate dual-ridge probes at each layer and k, using only correct-context development episodes. Four cell-balanced development folds select the ridge strength from $\{10^{-4}, 3{\times}10^{-3}, 10^{-1}, 3, 100\}$ and then select one reported layer. The untouched test episodes are evaluated in all three context arms ($n = 16$ per arm). A 10,000-repeat held-out label permutation test keeps the development-selected layer, ridge strength, and fitted probe fixed. Thus neither the test labels nor test-arm scores select the reported layer.

We probe three increasingly strict targets: the binary strong/weak regime; the full realized slope g_C; and the within-regime residual $g_C - .90$ for strong episodes or $g_C - .25$ for weak episodes. The first two are recoverable from the displayed evidence and the third is not, by construction: Section C.7.4 derives what a Bayes-optimal reader of the same prompt could score on each, and the residual is included as a specificity control rather than as a target a model could hit. Mean-pooled embedding and final-layer states are additional pooling controls.

Table 30: **Six-model correct-context probe summary.** Each entry gives $k = 0/k = 4$ on the 16 sealed episodes. Layers and ridge strengths were selected on development episodes only. Residual p is the one-sided 10,000-repeat held-out label-permutation value.

Model	Regime AUC	Full-slope R^2	Residual-slope R^2	Residual p
Qwen3-4B	1.000/1.000	.960/.885	-.389/-.227	.927/.614
Qwen3-8B	.844/1.000	.713/.653	-.359/-.374	.502/.735
Llama-3.1-8B	1.000/.984	.913/.731	-.126/-.207	.380/.572
Qwen3-14B	1.000/.984	.898/.627	-.531/-.128	.315/.450
Qwen3-32B	1.000/1.000	.967/.811	-.269/.003	.477/.221
Qwen2.5-72B	1.000/1.000	.975/.869	-.451/-.053	.675/.447

Table 31: **Held-out layer-wise open-weight probe.** AUC is reported for the regime target and R^2 for both slope targets. Correct, none, and wrong score the same development-selected probe against the true City C target under the three context arms; wrong/cue instead scores the inverted arm against the relationship named by its cue. p is the one-sided held-out label-permutation value on the correct-context test episodes.

Model	k	Target	Layer	Dev CV	Correct	None	Wrong/true	Wrong/cue	p
Qwen3-4B	0	Regime (AUC)	1	1.000	1.000	.562	.000	1.000	$<$ .001
		Full slope (R^2)	24	.936	.960	-.517	-2.614	.956	$<$ .001
		Residual slope (R^2)	15	.054	-.389	-.296	-.258	–	.927
	4	Regime (AUC)	19	1.000	1.000	.609	.094	.906	$<$ .001
		Full slope (R^2)	20	.805	.885	-.065	-1.147	.492	$<$ .001
		Residual slope (R^2)	14	.004	-.227	.089	-.183	–	.614
Qwen3-8B	0	Regime (AUC)	19	1.000	.844	.453	.078	.922	.009
		Full slope (R^2)	23	.870	.713	-.697	-2.344	.771	$<$ .001
		Residual slope (R^2)	1	.063	-.359	-.673	-.378	–	.502
	4	Regime (AUC)	23	.925	1.000	.578	.000	1.000	$<$ .001
		Full slope (R^2)	23	.462	.653	-.512	-1.550	.693	$<$ .001
		Residual slope (R^2)	33	.030	-.374	-1.003	-.029	–	.735
Llama-3.1-8B	0	Regime (AUC)	1	1.000	1.000	.359	.000	1.000	$<$ .001
		Full slope (R^2)	1	.907	.913	-18.953	-2.629	.972	$<$ .001
		Residual slope (R^2)	12	.018	-.126	-.458	-.227	–	.380
	4	Regime (AUC)	1	1.000	.984	.438	.000	1.000	$<$ .001
		Full slope (R^2)	1	.751	.731	-15.450	-2.082	.808	$<$ .001
		Residual slope (R^2)	10	.006	-.207	-.148	-.150	–	.572
Qwen3-14B	0	Regime (AUC)	2	1.000	1.000	.422	.000	1.000	$<$ .001
		Full slope (R^2)	26	.874	.898	-.522	-2.423	.943	$<$ .001
		Residual slope (R^2)	34	.159	-.531	-9.255	-.096	–	.315
	4	Regime (AUC)	26	.880	.984	.688	.109	.891	$<$ .001
		Full slope (R^2)	26	.408	.627	-.411	-1.191	.409	$<$ .001
		Residual slope (R^2)	1	-.013	-.128	-.605	-.159	–	.450
Qwen3-32B	0	Regime (AUC)	27	1.000	1.000	.359	.000	1.000	$<$ .001
		Full slope (R^2)	47	.930	.967	-.027	-2.570	.958	$<$ .001
		Residual slope (R^2)	1	.062	-.269	-.232	-.259	–	.477
	4	Regime (AUC)	44	1.000	1.000	.562	.000	1.000	$<$ .001
		Full slope (R^2)	47	.797	.811	-.160	-1.694	.666	$<$ .001
		Residual slope (R^2)	1	.043	.003	.062	-.000	–	.221
Qwen2.5-72B	0	Regime (AUC)	53	1.000	1.000	.469	.000	1.000	$<$ .001
		Full slope (R^2)	58	.955	.975	-1.050	-2.958	.969	$<$ .001
		Residual slope (R^2)	32	.061	-.451	-1.192	-.401	–	.675
	4	Regime (AUC)	58	.995	1.000	.922	.062	.938	$<$ .001
		Full slope (R^2)	58	.678	.869	-.468	-1.072	.493	$<$ .001
		Residual slope (R^2)	1	-.017	-.053	-.063	-.053	–	.447

C.7.3 Regime and Full-Slope Decodability

Across the six checkpoints and two evidence depths, correct-context regime AUC is $.844$–1.000 and full-slope R^2 is $.627$–$.975$ (Table 30). This information is not a context-invariant encoding of the true coefficient: removing the City C description drives selected-layer full-slope R^2 below zero in all 12 model–k cells. Under inversion, true-target regime AUC falls to .000–.109 while cue-target AUC is .891–1.000; the full-slope probe likewise fits the cue-assigned relationship better than the true one. The Llama replication shows that this pattern is not specific to the Qwen checkpoints. Across both

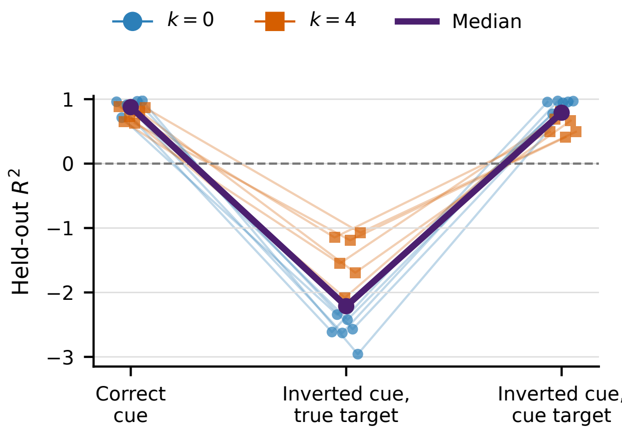


Figure 14: **Semantic-arm slope decoding follows cue assignment.** Each line is a sealed model–k cell across the six open checkpoints of the 16-episode roster; purple is the median. Probes are fit on correct-context development episodes. Under inversion, held-out R^2 collapses against the true slope and recovers against the cue-assigned slope. The input-embedding control below shows this signal is available before any computation, which is why the arbitrary-label replication rather than this figure carries the representational claim.

model families, the linearly available signal therefore tracks the response family that context assigns to City C.

C.7.4 What Each Probe Target Can Be Scored Against

A probe is scored against a latent generator quantity, so a null is only a fact about a model if the displayed evidence determines that quantity. We therefore derive, for each target and evidence depth, the score a Bayes-optimal reader of the same prompt would obtain. This is an analytic property of Equation 3.2 and its row construction, computed in closed form and re-derived by simulating the generator's own row logic; the two agree to within .005 and no model output enters; the derivation is `exp2_v2/biased_news/analysis/compute_bayes_ceilings.py`. Table 32 gives the ceilings. The same module reports forecast-MAE floors: 2.28 poll points at $k = 0$ with no cue, 0.17 for an oracle told the regime means, and 1.78 for a reader who must instead estimate the named regime's slope from its four displayed rows.

The same simulation records one quantity the reference-selection analysis of Section C.6 depends on. Cities draw their slopes independently, so a reference city's deviation from its regime mean carries no information about City C's: $\mathrm{corr}(\widehat{\beta}^{\mathrm{ref}} - \mu,\ g_C - \mu) = -.0000$ over 400,000 simulated episodes, against a reference-fit standard deviation of .318 within regime. That is why a loading on the reference's own sampling error identifies reference selection, and equally why a forecaster that shrank optimally would show none.

Table 32: Maximum score a Bayes-optimal reader of the same prompt can obtain on each probe target, under the frozen data-generating process. The regime and the full slope are recoverable; the within-regime residual is not, because it is generated with standard deviation .03 while a through-origin fit to City C's displayed rows carries a standard deviation of .318. At $k = 0$ City C displays no rows at all and the residual ceiling is exactly zero.

Probe target		$k = 0$	$k = 4$
Regime, correct semantic cue	AUC	1.000	1.000
Regime, arbitrary label	AUC	0.979	—
Full slope g_C	R^2	0.992	0.992
Residual slope $g_C - \mu$	R^2	0.000	0.009

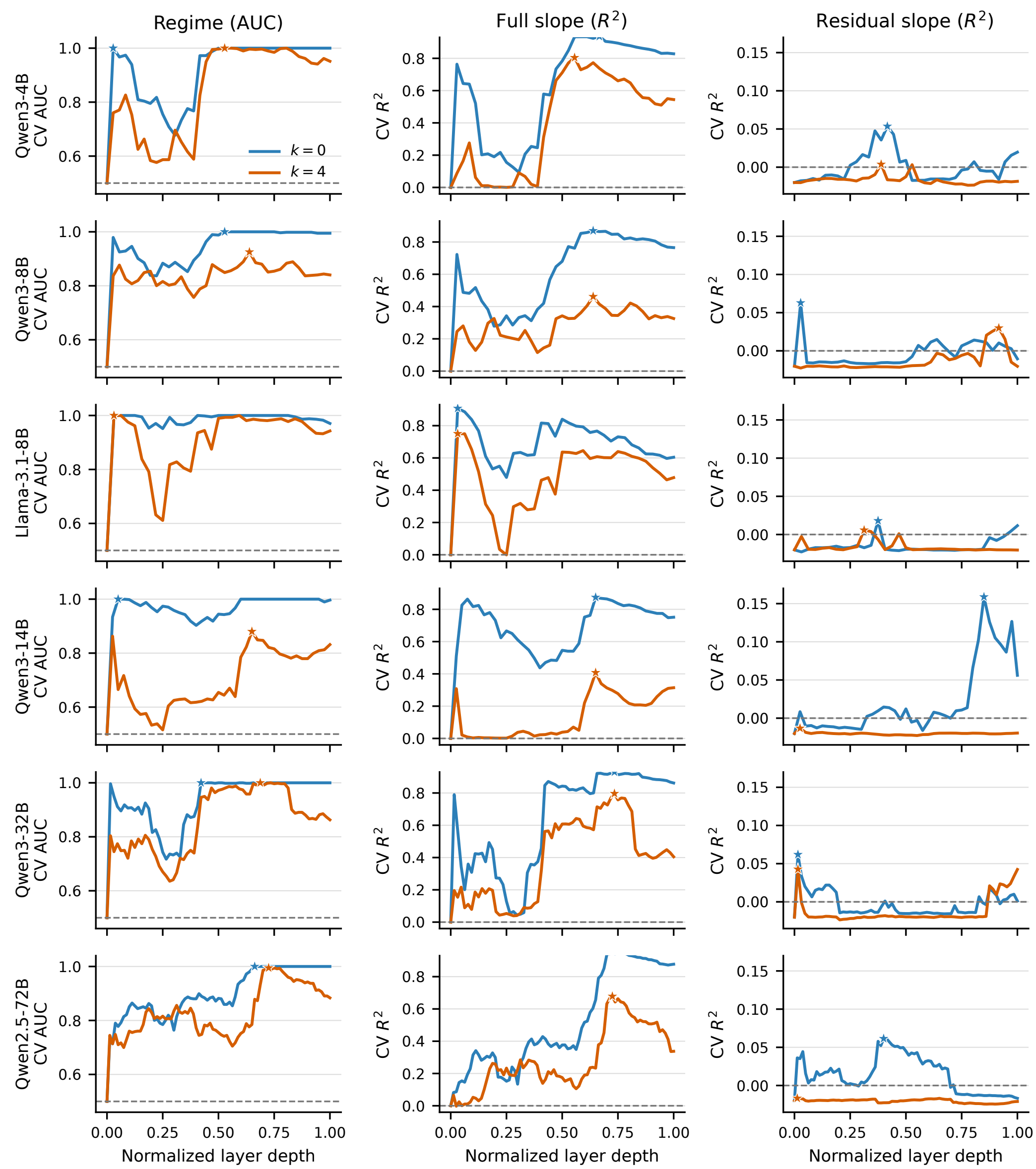


Figure 15: **Development layer trajectories for the six-model probe.** Curves show four-fold development CV rather than test performance; stars mark the development-selected layer. Layer depth is normalized because the checkpoints have 32–80 transformer blocks. The regime and full-slope signals are broad, whereas residual-slope CV is weak and does not survive held-out evaluation.

C.7.5 Specificity: Within-Regime Residuals

The residual target is the specificity control, and it behaves as a target with no available signal should. The 12 correct-context residual-slope R^2 values range from $-.531$ to $.003$, none of their permutation tests rejects the null ($p = .221$–$.927$), and 33 of 36 arm-level residual scores are nonpositive. Against a ceiling of $.000$ at $k = 0$ and $.009$ at $k = 4$, that is the result the design permits: no reader of these prompts, and no representation computed from them, could score materially above zero. We therefore read the residual null as evidence that the probe does not manufacture structure where none is available, and not as a limit on what these checkpoints encode. A probe that returned a positive residual score here would be the finding, and it would be a warning about the probe.

The two recoverable targets do carry information, and the input-embedding control is what bounds their interpretation. Mean-pooled input embeddings for every Qwen checkpoint already give perfect regime AUC and full-slope $R^2 = .988$–$.993$ under correct context—at the $.992$ ceiling—because the explicit description names a regime and the regime means account for essentially all slope variance. Llama input embeddings likewise give perfect regime AUC and full-slope $R^2 = .756/.827$. The layer-wise semantic result is therefore evidence that the cue-selected response family is linearly represented, and reaches nothing further about the continuous g_C: the part of g_C beyond the family is not in the prompt to begin with. The original behavioral inversion screen connects this representation to output sensitivity in the three larger screened checkpoints: inverted context raises $k = 0$ forecast MAE by 1.66, 1.96, and 2.21 points for 14B, 32B, and 72B, respectively.

C.7.6 Power-Matched and Cross-Model Arbitrary-Symbol Replication

The six-checkpoint roster above seals 16 test episodes per arm. Its exact label-permutation null spans ROC AUC $[.203, .797]$, so it can separate a score from chance only above about $.75$. That resolution suffices for the semantic arm, whose scores sit at ceiling, but it cannot evaluate a weaker signal at all. We therefore evaluate Qwen3-14B at the design's balanced maximum—248 episodes, 160 development and 88 sealed test, 1,488 activation records—once with the semantic cue and once with the episode-randomized `KIV`/`ZOR` labels. Anchor, targets, fold structure, ridge grid, layer-selection rule, pooling controls and the 10,000-repeat held-out permutation test are unchanged; only the number of episodes differs. The 88-episode null reaches AUC $.603$. The enlarged development set is disjoint from every episode used by the 16-episode analysis, providing an independent estimate rather than a re-slice.

Table 33: **Power-matched Qwen3-14B probe, 88 sealed episodes per arm.** Correct, none and wrong score the development-selected probe against the true City C target under the three cue arms; wrong/cue scores the inverted arm against the relationship its cue names. “Layer” is the development-selected block of 40 and “Embed.” is the mean-pooled input-embedding control under correct context. p is the one-sided held-out label-permutation value.

Run	Target	Correct	None	Wrong	Wrong/cue	Layer	Embed.	p	Behav. $\Delta\rho$
Qwen3-14B, semantic	k=0 regime AUC	1.000	.429	.000	1.000	1	1.000	<.001	—
	k=4 regime AUC	1.000	.560	.000	1.000	1	1.000	<.001	
	k=0 slope R^2	.942	−.300	−2.806	.961	27	.989	<.001	
	k=4 slope R^2	.910	−24.039	−2.387	.906	1	.989	<.001	
Qwen3-14B, symbol	k=0 regime AUC	.678	.467	.321	.679	26	.555	.0016	.398
	k=4 regime AUC	.713	.689	.747	.253	40	.582	<.001	
	k=0 slope R^2	.016	−.094	−.130	.065	26	.003	.1162	
	k=4 slope R^2	.030	−.237	.126	−.855	40	−.032	<.001	
Qwen3-8B, symbol	k=0 regime AUC	.566	.535	.567	.433	10	.534	.1471	.023
	k=4 regime AUC	.610	.545	.593	.407	19	.621	.0401	
	k=0 slope R^2	−.001	.005	.004	−.007	25	.003	.4883	
	k=4 slope R^2	−.151	−.219	−.188	−.504	20	−.018	.0314	
Qwen3-4B, symbol	k=0 regime AUC	.520	.419	.329	.671	23	.498	.3756	.095
	k=4 regime AUC	.685	.689	.637	.363	22	.585	.0017	
	k=0 slope R^2	−.009	−.065	−.079	.023	23	.004	.2899	
	k=4 slope R^2	−.021	−.265	−.031	−.687	23	.004	.0014	
Qwen2.5-32B, symbol	k=0 regime AUC	.778	.571	.200	.800	50	.550	<.001	.372
	k=4 regime AUC	.797	.831	.662	.338	46	.589	<.001	
	k=0 slope R^2	.217	−.256	−1.033	.254	50	.004	<.001	
	k=4 slope R^2	.263	−.563	−.044	−.774	46	−.029	<.001	
Qwen2.5-72B, symbol	k=0 regime AUC	.662	.492	.210	.790	63	.556	.0026	.381
	k=4 regime AUC	.793	.745	.651	.349	61	.575	<.001	
	k=0 slope R^2	−.005	−.490	−1.034	.187	62	.004	.0072	
	k=4 slope R^2	.222	−.624	−.129	−.870	61	−.047	<.001	
Llama 3.1-8B, symbol	k=0 regime AUC	.423	.447	.452	.548	11	.539	.8925	.045
	k=4 regime AUC	.543	.604	.549	.451	31	.604	.2470	
	k=0 slope R^2	−.097	−.089	−.086	−.001	11	.003	.8607	
	k=4 slope R^2	−.275	−.091	−.156	−.358	31	.009	.4529	
Llama 3.1-70B, symbol	k=0 regime AUC	.728	.468	.292	.708	75	.495	<.001	.424
	k=4 regime AUC	.810	.740	.721	.279	42	.606	<.001	
	k=0 slope R^2	.076	−8.200	−.961	.082	75	.003	<.001	
	k=4 slope R^2	.222	−3.224	.087	−1.007	55	.007	<.001	

The two arms separate sharply, and the input-embedding control is what separates them. Under the semantic cue the development-selected layer is the first transformer block for three of the four conditions, and mean-pooled input embeddings alone already reach regime AUC 1.000 and full-slope $R^2 = .989$: the decodable quantity is the cue token itself. Under arbitrary labels the same control collapses to AUC .555 and $R^2 = .003$, the selected layers move to blocks 26 and 40 of 40, and held-out regime AUC is .678 at $k = 0$ ($p = .0016$) and .713 at $k = 4$ ($p < .001$). The induced label-to-regime mapping is therefore linearly represented, but only after computation, far more weakly than a regime the model was told outright, and below the .979 AUC a Bayes-optimal reader of the same two reference cities could reach. Unlike the residual target, the full slope is recoverable here—its ceiling is .992—so its symbol-arm score of $R^2 = .016$ at $k = 0$, which does not reject its null, is a genuine gap rather than a property of the design.

Representation and behavior agree closely once the cue must be induced. On the same 88 sealed episodes the model’s own forecasts give regime AUC .702, .518 and .331 under correct, absent and inverted labels, against probe AUCs of .678, .467 and .321; inversion raises $k = 0$ poll MAE from 2.49 to 4.28. With four City C cases the inverted-arm representation reverses: true-target regime AUC rises from .321 to .747, and the full-slope probe fits the cue-named relationship at $R^2 = -.855$ while fitting the true one at $+.126$. Direct target evidence overrides the induced label in the representation as it does in the forecast.

Six further checkpoints received the identical symbol protocol at the same 88 sealed episodes: Qwen2.5-32B and Qwen2.5-72B, which change model generation; Qwen3-8B and Qwen3-4B, which hold family, tokenizer and architecture fixed and change only scale; and Llama 3.1-8B and Llama 3.1-

70B, which do the same within a second vendor and architecture. Held-out regime AUC at $k = 0$ is .778 for Qwen2.5-32B ($p < .001$, selected layer 50 of 64), .728 for Llama 3.1-70B ($p < .001$, layer 75 of 81), .662 for Qwen2.5-72B ($p = .003$, layer 63 of 80), .566 for Qwen3-8B ($p = .15$), .520 for Qwen3-4B ($p = .38$) and .423 for Llama 3.1-8B ($p = .89$). No null checkpoint rejects in any $k = 0$ condition—regime, slope or residual slope. Mean-pooled input embeddings stay between .495 and .556 in every symbol run, so no symbol arm carries the mapping before computation.

Decodability tracks behavioral recovery across all seven checkpoints, four positive and three null. The four whose forecasts recover the mapping are the four in which it decodes (Llama 3.1-70B, $\Delta\rho = .424$; Qwen3-14B, .398; Qwen2.5-72B, .381; Qwen2.5-32B, .372), and the three whose forecasts do not are the three in which it does not (Qwen3-4B, .095; Llama 3.1-8B, .045; Qwen3-8B, .023; all three intervals spanning zero). The Llama 3.1 contrast carries the weight, because it holds family, tokenizer and architecture fixed, varies only scale, and is the one within-family contrast whose behavioral classification survives the harder generator population of Section C.8: 8B is null on both populations and 70B positive on both, while the probe runs .423 to .728 across them. Because the null and the positive share an architecture and a tokenizer, the split is not an artifact of either.

The Qwen3 sequence—.520, .566, .678 from 4B to 14B—is consistent with this but does not add independent weight, because the population that would test it is the one on which its behavioral boundary moves. That movement is smaller than the $\Delta\rho$ column alone suggests. Symbol-arm regime accuracy orders the four Qwen3 checkpoints identically in both populations, .512, .516, .660, .656 originally against .528, .554, .589, .606 on the harder one: no pair reverses, and what changes is that the harder population compresses every score toward chance, so a threshold that fell in a clear gap now falls inside a compressed continuum. The $\Delta\rho$ contrast then subtracts a no-context baseline whose standard error is about .064, and baselines of $-.036$ for Qwen3-8B and $+.025$ for Qwen3-14B are enough to put one interval above zero and the other .004 below it, with the two intervals overlapping by 78%. We therefore read the harder population as showing that the ordering replicates while the significance threshold is not locatable to a checkpoint, rather than as a reversal. Section C.8 should be read alongside the seven-for-seven agreement, which is evidence that the probe reads what the forecasts express on the episodes both were measured on and not evidence for a scale law. The positives span three model generations and two vendors, and every positive shows the same shape: the decoded regime follows the cue at $k = 0$ and the truth at $k = 4$. Qwen3-4B and Qwen3-8B do reject at $k = 4$ (.685, $p = .002$ and .610, $p = .04$), which is expected rather than contrary: at $k = 4$ the displayed target cases exhibit the regime whatever the label says, and the mean-pooled input embeddings rise to .585 and .621 accordingly, in Qwen3-8B's case above the probe itself. The induction claim is a $k = 0$ claim throughout. The sealed test is 88 episodes per model, well above the 16 the roster began with but not a large sample, and the permutation nulls are computed rather than assumed.

At 16 episodes, the same arbitrary-symbol procedure returns $k = 0$ regime AUC .312 with $p = .90$. Its $[.203, .797]$ null is too broad to resolve a signal of the magnitude observed at 88 episodes.

C.7.7 Causal Test: Patching the City C Label Representation

Decodability is not mediation. To test whether the representation the probe reads is the one the forecast uses, we intervene on it directly. Under arbitrary labels, an episode's correct and swapped prompts are positionally aligned through the City C label and differ at exactly two label-token positions. We replace the residual stream at those two positions with the states the opposite arm produces, in both directions, and regenerate the forecast greedily. The endpoint is the episode-level mean of the regime-aligned correct-into-swapped shift and the sign-reversed swapped-into-correct shift, tested one-sided against an episode sign-flip null.

The analysis uses three patch sites and two controls. The primary site is a three-layer window centred on the development-selected relational layer; patching at the embedding output only, and at every causally effective layer, are the comparison sites; the unpatched arm and a self-patch that writes each state back over itself are the controls. Layer, window and ridge were selected on development episodes with the sealed test unread.

Table 34: **Causal effect of patching the two City C symbol subtokens, Qwen3-14B.** Shift is the regime-aligned symmetric patch mean; positive means the forecast moved toward the relationship the donor label names. The 16- and 88-episode analyses select their windows independently on their own development sets. The null interval is the 88-episode sign-flip null.

		16 sealed, window 33–35			88 sealed, window 19–21			
Patch site	k	Shift	p	n	Shift	p	n	Null 95%
Selected window	k=0	−.033	.75	14	.333	<.0001	82	[−.177, .178]
	k=4	−.007	.75	15	−.035	.8256	87	[−.073, .073]
Embedding only	k=0	.236	.15	14	.319	.0002	84	[−.182, .182]
	k=4	.115	.23	15	−.017	.6419	87	[−.087, .087]
All layers	k=0	.236	.15	14	.319	.0002	84	[−.182, .182]
	k=4	.115	.23	15	−.017	.6419	87	[−.087, .087]

At $k = 0$ the intervention moves the forecast: patching the selected window shifts the regime-aligned prediction by .333 against a null interval of $[-.177, .178]$ ($p < .0001$, 82 complete episodes), and patching the embedding alone gives .319. All eight stability checks pass, including $347/347$ exact self-patch matches.

Two readings of the table matter more than the significance. First, the embedding-only and all-layer rows are identical, and necessarily so: the two prefixes differ only at the patched label positions, so overwriting their embeddings and letting the model recompute reproduces what overwriting every later state does. They are one intervention reported at two depths, not two converging results. What distinguishes the mid-window row is that it reaches the same magnitude while touching only three layers late in the network.

Second, the effect is confined to $k = 0$. With four City C cases every patch site returns a null ($p = .83$ and $p = .64$), so the induced label no longer controls the forecast once direct target evidence is available. That is the revision claim of Section C.5 in causal form, and it matches the representational result: at $k = 4$ the inverted-arm probe recovers the true regime rather than the cue-assigned one.

At 16 episodes, the primary patch estimate is $-.033$ at $p = .75$ and the sign-flip null spans $[-.387, .387]$. Its 64-episode development set selects layers 33–35, whereas the 160-episode development set selects 19–21 at a higher cross-validation score (.903 against .806).

Permutation implementation. The test enumerates all 2^n sign assignments for 16 episodes; exhaustive enumeration is infeasible for 88 ($2^{88} \approx 3 \times 10^{26}$). Enumeration is retained for $n \leq 20$; above it the same null is sampled with 200,000 draws under a fixed seed, giving a Monte Carlo standard error of .0005 at $p = .05$. The estimand, endpoint, and direction are identical under both implementations.

Scope. This is one checkpoint under arbitrary labels. It shows that the label representation Qwen3-14B computes is causally responsible for its zero-shot regime assignment, and that direct evidence displaces it. It does not establish the same for the hosted deployments, the semantic cue, or other architectures.

C.7.8 Reproducibility and Scope

The checkpoints are pinned to commits `cdbee75f17c01a7cc42f958dc650907174af0554` (Qwen3-4B), `b968826d9c46dd6066d109eabc6255188de91218` (Qwen3-8B), `0e9e39f249a16976918f6564b8830bc894c89659` (Llama-3.1-8B), `40c069824f4251a91eefaf281ebe4c544efd3e18` (Qwen3-14B), `9216db5781bf21249d130ec9da846c4624c16137` (Qwen3-32B), and `495f39366efef23836d0cfae4fbe635880d2be31` (Qwen2.5-72B). The activation hashes are `e191385963d639aa5b8574a9abbeb91b3519aed934ac491f863655631b26dbbc` (Qwen3-4B) and `7f65aeec7a91e404cc60068618fedf25c679abde5d48c2b8477f736670fbc54f` (Llama-3.1-8B); all six hashes and tensor dimensions are enforced by the fail-closed generator. Frozen tasks, activation tensors, extraction manifests, generated behavior, selected predictions, complete layer curves, and probe results are under `exp2_v2/biased_news/data/coin_city_stable_relationship_claude_n250_v4/mechanistic_probe/`. The analysis and asset generator are `exp2_v2/biased_news/mechanistic_probe/analyze_probe.py` and `paper/generate_coin_city_mechanistic_comparison.py`. The causal run is `mechanistic_probe/qwen3_14b_symbol_relational_v2/`, frozen by `freeze_symbol_relational_protocol.py` and executed by `extract_symbol_label_states.py`, `patch_symbol_label_activations.py` and `analyze_symbol_relational_probe.py`; every count in that pipeline is derived from the frozen task manifest and checked by `tests/test_symbol_relational_scaling.py`. `paper/generate_coin_city_causal_patch_table.py` fails closed on the run's study name, sealed-test size and stability gate. The power-matched runs are `mechanistic_probe/qwen3_14b_probe_v2/`, `mechanistic_probe/qwen3_14b_symbol_probe_v2/`, `mechanistic_probe/qwen3_8b_symbol_probe_v2/`, `mechanistic_probe/qwen3_4b_symbol_probe_v2/`, `mechanistic_probe/llama3_1_70b_symbol_probe_v2/`, `mechanistic_probe/qwen2_5_72b_symbol_probe_v2/`, `mechanistic_probe/qwen2_5_32b_symbol_probe_v2/` (checkpoint `5ede1c97bbab6ce5cda5812749b4c0bdf79b18dd`) and `mechanistic_probe/llama3_1_8b_symbol_probe_v2/` (checkpoint `0e9e39f249a16976918f6564b8830bc894c89659`), built by `exp2_v2/biased_news/mechanistic_probe/build_probe_tasks.py` and `build_arbitrary_symbol_probe_tasks.py` with 62 and 31 episodes per factorial cell; `paper/generate_coin_city_probe_power_table.py` regenerates their table and validates the sealed-test size and task hash. Each run's activation tensor is regenerable from its frozen task file and pinned checkpoint; the extraction manifests record the tensor hashes rather than the tensors. Cross-vendor probe evidence here is the Llama 3.1-8B null rather than a positive. The diagnostic covers six checkpoints spanning Qwen and Llama at 16 sealed episodes per arm, plus the matched 88-episode semantic and arbitrary-symbol Qwen3-14B pair. For the semantic cue it is correlational and does not establish that the decoded direction mediates the forecast; under arbitrary labels Section C.7.7 provides a causal intervention for one checkpoint. Both complement the paired behavioral interventions above.

C.8 Robustness Across Generator Populations and Deployments

C.8.1 Generator-population robustness

Fresh seeds 940000–940249 target strong/weak means of $.75/.40$ with slope SD $.10$, jointly narrowing the regime gap and widening within-regime dispersion. The realized target slopes are $.752 \pm .100$ and $.404 \pm .105$. Draws were neither truncated nor reordered: two reference pairs reverse their nominal latent ordering and remain in the analysis. Case noise, factorial balance, prompts, prefix set, local checkpoint settings, and episode-randomized symbol mapping are unchanged.

All ten open checkpoints were run on the three matched arms, producing 37,500 response records; 37,355 contain a numeric forecast, and unresolved formats remain missing. Table 35 compares the $k = 0$ symbol-minus-no-context contrast across generator populations. Bold intervals exclude zero; negative ΔMAE would favor symbol context. Because this population changes separation and dispersion together, it tests the joint perturbation rather than identifying either parameter's effect separately.

Table 35: Arbitrary-symbol recovery on the original and harder generator populations. Intervals are 95% episode-bootstrap intervals with 5,000 draws; n is the harder-population complete-case count.

Model	n	Original $\Delta\rho$ [95% CI]	Harder $\Delta\rho$ [95% CI]	Symbol acc.	Harder ΔMAE [95% CI]
Qwen3-4B	250	$+.095$ [$-.060$, $.230$]	$+.050$ [$-.079$, $.197$]	.528	$+.052$ [$-.056$, $.161$]
Qwen3-8B	249	$+.023$ [$-.128$, $.171$]	$+.231$ [$.074$, $.370$]	.554	$-.113$ [$-.269$, $.036$]
Qwen3-14B	246	$+.398$ [$.253$, $.541$]	$+.155$ [$-.004$, $.303$]	.589	$+.361$ [$.076$, $.644$]
Qwen3-32B	249	$+.302$ [$.177$, $.433$]	$+.193$ [$.063$, $.325$]	.606	$+.848$ [$.539$, 1.168]
Qwen2.5-7B	250	$+.264$ [$.112$, $.406$]	$+.149$ [$.009$, $.279$]	.540	$+.303$ [$.000$, $.627$]
Qwen2.5-14B	250	$+.236$ [$.107$, $.356$]	$+.133$ [$.005$, $.254$]	.584	$+.365$ [$.115$, $.609$]
Qwen2.5-32B	250	$+.372$ [$.226$, $.502$]	$+.247$ [$.109$, $.383$]	.616	$+.882$ [$.590$, 1.186]
Qwen2.5-72B	250	$+.381$ [$.250$, $.508$]	$+.336$ [$.204$, $.463$]	.604	$+1.012$ [$.717$, 1.316]
Llama 3.1-8B	229	$+.045$ [$-.123$, $.250$]	$-.077$ [$-.216$, $.079$]	.524	$+.597$ [$.229$, $.952$]
Llama 3.1-70B	249	$+.424$ [$.314$, $.543$]	$+.242$ [$.119$, $.369$]	.558	$+.856$ [$.465$, 1.242]

Which Qwen3 checkpoints clear significance changes, but their ordering does not. Qwen3-8B moves from an original null to $\Delta\rho = .231$ $[.074, .370]$ while Qwen3-14B falls to $.155$ $[-.004, .303]$; Qwen3-4B remains null and Qwen3-32B remains positive. Regime accuracy in the symbol arm, which subtracts no baseline, orders the four identically in both populations—$.512$, $.516$, $.660$, $.656$ originally against $.528$, $.554$, $.589$, $.606$ here—so no pair reverses. What the harder population removes is separation: it compresses every score toward chance, turning a $.14$ gap between 8B and 14B into $.035$. The $\Delta\rho$ contrast then subtracts a no-context baseline carrying a standard error of about $.064$, and baselines of $-.036$ for Qwen3-8B against $+.025$ for Qwen3-14B suffice to place one interval above zero and the other $.004$ below it, with the two overlapping by 78%. We read this as a detection threshold that is no longer locatable to a checkpoint on the compressed population, not as a reversal.

All four Qwen2.5 intervals remain positive, from $.149$ $[.009, .279]$ at 7B to $.336$ $[.204, .463]$ at 72B, and both Llama checkpoints keep their classification, 8B null in each population and 70B positive in each. Seven of ten intervals therefore exclude zero in each population and eight of ten checkpoints keep their classification, the Qwen3 8B/14B pair being the two that do not. Mapping recovery survives the substantially less separable population; the checkpoint at which it becomes detectable does not, and neither does a scale reading across families, since Qwen2.5-14B sits below Qwen2.5-7B in both. Part of the shrinkage is mechanical: the regime explains less of the slope here, capping ρ at .87 against .996. The observed $\Delta\rho$ falls further than that alone predicts. Numerical use stays distinct: Qwen2.5-72B's discrimination improves while symbol context raises forecast MAE by 1.012 $[.717, 1.316]$.

C.8.2 One independent hosted repeat

GPT-5.6 was replayed once on the 250 $k = 0$ prompts in each no-context, semantic-context, and arbitrary-symbol arm. The deployment, low reasoning effort, provider-native sampling, prompt hashes, parser, and one-attempt policy were unchanged; all 750 repeat responses parsed.

Table 36: Independent GPT-5.6 $k = 0$ repeat. Paired intervals resample episodes; mean $|\Delta\hat{y}|$ is the per-task absolute change in forecast.

	Forecast MAE			$\rho(\hat{g}, g_C)$			Mean $\lvert\Delta\hat{y}\rvert$
Arm	Original	Repeat	Repeat − original [95% CI]	Original	Repeat	Repeat − original [95% CI]	
No C context	2.518	2.451	$-.068$ $[-.130, -.012]$	-.041	-.013	$+.029$ $[-.008, .066]$	.166
Semantic context	1.596	1.613	$+.016$ $[-.051, .082]$	.678	.678	$+.000$ $[-.032, .037]$	.193
Arbitrary symbol	1.744	1.708	$-.036$ $[-.104, .023]$	.655	.648	$-.006$ $[-.016, .002]$	.167

Per-task mean absolute forecast changes are .166, .193, and .167 points across the three arms. The arbitrary-symbol-minus-no-context correlation effect is .696 originally and .661 on repeat; their difference is $-.035$ $[-.075, .003]$. The corresponding forecast-MAE benefit is $-.774$ originally and $-.743$ on repeat, with difference $+.031$ $[-.046, .104]$. This one repeat supports stability of the paired conclusions for this named snapshot, not a general estimate of provider or model variance.

C.9 Missingness, Scope, and Artifact Map

C.9.1 Missingness and record recovery

Claude Opus 5 has two unparseable primary task cells: one correct-context response at $k = 1$ and one no-context response at $k = 4$. Both exhausted the output budget during extended thinking and returned no answer. GPT-5.6 has one no-context failure at $k = 1$, where the provider returned a model-side HTTP 500 before any response. None was rerun. Complete-case pairing therefore gives $n = 249$ for the corresponding deployment-prefix rows and $n = 250$ elsewhere. These three failures constitute 0.013% of the 22,500 primary task cells.

Claude Opus 4.8, DeepSeek-V4-Pro, Kimi K3, and Gemini 3.6 Flash have 3,750 parseable primary responses each after the one offline Claude key-whitespace normalization described above. That normalization changed neither the forecast value nor the raw model answer and made no additional API call.

All six misleading-context analytic files contain exactly 1,250 unique parseable tasks. Concurrent writes in the Kimi collection produced 11 duplicate records and 14 malformed intermediate lines. Construction of the analytic file retained the earliest response for each task and identified 11 absent task IDs; those 11 tasks were collected under the unchanged Kimi configuration. Thus 7,500 is the number of final unique analytic records for the misleading arm rather than the number of intermediate records or provider-level attempts.

The expanded arbitrary-symbol evaluation contains fifteen response-record triplets, or 56,250 records. Within every model, each arm contains 1,250 unique task IDs matched to the frozen task-file prompt hash. Original outputs are preserved. When a model placed an explicit arithmetic expression rather than a number in `predicted_poll`, a no-call reparse accepts only numeric constants, parentheses, unary signs, and the four arithmetic operators, and requires the result to lie in the primary parser's 0–100 range. For Llama 3.1-8B, this reparse leaves 1,234/1,241/1,232 parseable no-context, semantic-context, and symbol records, respectively. Its matched complete-case n is 242 at $k = 0$ and ranges from 238 to 246 across the five prefixes. Qwen3-32B retains one unresolved symbol response, leaving 1,249 parseable symbol records; every other model parses all 1,250 in every arm, and every model except Llama 3.1-8B has $n = 250$ at $k = 0$. The final table reports these complete-case denominators, and unresolved outputs remain missing rather than being re-asked.

C.9.2 Interpretive scope

The aggregate behavior is consistent with context-conditioned response-regime assignment and integration of direct City C evidence in this controlled synthetic setting. A label-conditioned analogical-regression benchmark—map the label to a reference, estimate that reference's slope, transfer it to City C, and revise with City C's own cases—is sufficient to reproduce the pattern but does not identify the model's internal algorithm. In the misleading arm, reversing only the City C label reverses forecast assignment while all numeric content remains fixed.

The primary binary semantic cue is aligned by construction with the higher- or lower-response family and has a fixed meaning across episodes. The arbitrary-symbol control removes that meaning by reversing the `KIV`/`ZOR` mapping across episodes; it therefore tests whether the semantic-arm result can arise solely from a pre-existing national-greater-than-local prior. The expanded 15-model comparison shows detectable symbol-based discrimination for Qwen3-14B and Qwen3-32B, all four tested Qwen2.5 checkpoints, Llama 3.1-70B, and all five hosted systems. In this original population, intervals span zero for Qwen3-4B, Qwen3-8B, and Llama 3.1-8B. Within Qwen3 the mapping is recovered only from 14B upward, and within Llama only at 70B, providing within-family contrasts on the original population; Qwen2.5-7B's positive result alongside Qwen3-8B's interval spanning zero, and Qwen3-32B's smaller point estimate than Qwen3-14B's, show that the transition is neither monotone in nor determined by parameter count alone. Four quantitative analyses bear on whether a model induces the arbitrary label-to-regime mapping, and they are computed from different quantities: the symbol-minus-no-context correlation contrast on forecasts; the partial correlation between the implied slope and the cue-named reference's own fitted slope (Section C.6); held-out linear decoding of hidden states; and causal patching of those states (Section C.7.7). Where more than one was run on a checkpoint they agree, including Llama 3.1-8B, where every analysis run is null. Agreement among measures that could have disagreed is the reason to prefer the induction account over a lexical or pooling one. It does not extend the claim to the hosted deployments, whose internals are unavailable, and only one checkpoint carries the causal test. A fifth, descriptive audit asks what models themselves say in their one-sentence rationales.

Deterministic lexical rules were developed on a hash-selected 50-episode subset and then applied without revision to the remaining 200 episodes. Across all three comparison arms, 56,057 of 56,250 response records contain a rationale. At $k = 0$ in the arbitrary-symbol arm, 2,992 of 3,000 held-out records contain one; the coder identifies a selected City A/B reference in 1,832, and every codable statement names the reference that actually shares City C's episode-local symbol. In the inverted semantic arm, all 1,200 held-out records contain a rationale: all 1,089 codable reference statements name the reference selected by the false cue, and all 1,186 codable national/local statements name the false regime. These outputs are behavioral self-reports, not privileged evidence of computation; in particular, the coder abstains whenever a statement does not make an unambiguous reference or regime claim. The conditional pattern also appears in the three checkpoints with null behavioral symbol contrasts, so it establishes declared cue interpretation rather than successful numerical use.

Because discrimination and forecast MAE can move in opposite directions, mapping induction is not equivalent to calibrated numerical use. Ambiguous, continuous, and adversarial cues remain untested. All prompts also tell models that within-city response is stable and explain the matched-pair structure.

The query target is a latent conditional mean with no newly sampled observation noise. This isolates response recovery but is not equivalent to calibration on a future noisy poll. The six primary deployments differ in provider, reasoning mode, output budget, and service tier. The symbol module mixes local vLLM checkpoints with hosted calls and includes the documented cross-resource Opus 4.8 arm. Cross-model comparisons are therefore descriptive rather than configuration-matched capability rankings. The cue-inversion module is a matched stress test over the same task identities and numeric inputs.

The primary and harder designs are two reproducible seeded populations in the same synthetic world. The second jointly narrows the regime gap and widens slope dispersion, but two settings do not characterize generator-parameter uncertainty or isolate those changes. Bootstrap intervals resample episodes within a population and are pointwise; they do not incorporate provider drift or correct for the many model, arm, and prefix comparisons. The evaluated populations also do not characterize factorial variation in generator parameters, ambiguous cues, or calibration against noisy held-out outcomes.

C.9.3 Reproducibility and artifact map

The frozen data root is `exp2_v2/biased_news/data/coin_city_stable_relationship_claude_n250_v4/`, and the experiment-code root is `exp2_v2/biased_news/`. Paths below are relative to those roots unless a repository-level path is shown.

Design and preflight. Under the data root, `design/manifest.json` and `design/validation.json` freeze and validate the primary design; `design/tasks_abc_wrong_context.manifest.json` records the misleading arm, and `design/tasks_abc_symbol_context.manifest.json` records the balanced, episode-randomized symbol control.

Generator and task builders. Under the code root, the generator is `engine/coin_city_stable_relationship_claude_n250.py`; the builders are `eval/build_coin_city_stable_relationship_claude_n250.py` and `eval/build_coin_city_wrong_context_tasks.py`. The symbol implementation is `engine/coin_city_symbol_context_arm.py`, with task builder `eval/build_coin_city_symbol_context_tasks.py`.

Execution and parsing. The manifest-locked Opus 4.8 wrapper is `eval/run_coin_city_stable_relationship_claude_n250.py`; shared append-only execution and parsing are in `eval/run_frozen_task_shard.py`. Model-specific runners and `eval/run_coin_city_wrong_context.py` preserve configurations and response records under the data root. The symbol runner is `eval/run_coin_city_symbol_context.py`; its contemporaneous comparison arms are under `responses/symbol_control_matched_20260813/`. The open-model runner is `eval/run_coin_city_symbol_context_open_model.py`; the module-specific Qwen and Llama responses, hosted symbol arms, launchers, and run manifests are under `local_results/symbol_context_model_comparison_20260824/` relative to the code root.

Primary analysis. `analysis/analyze_coin_city_stable_relationship_multimodel.py` under the code root produces `analysis/multimodel_results.json` under the data root, the authority for the primary-arm metrics and paired contrasts.

Arbitrary-symbol analysis. `analysis/analyze_coin_city_symbol_context.py` supplies the common estimator; `local_results/symbol_context_model_comparison_20260824/build_final_comparison.py` writes the paper authority `analysis/symbol_context_model_comparison_20260824.json`. Focused design checks are in `tests/test_coin_city_symbol_context.py`; `paper/generate_coin_city_symbol_context_tables.py` generates both appendix tables from the final comparison.

Reference-selection analysis. `analysis/analyze_coin_city_reference_selection.py` under the code root recomputes the partial correlations of Section C.6 from the frozen responses and writes `analysis/reference_selection_results.json` under the data root; it makes no model call. Checks are in `tests/test_coin_city_reference_selection.py`, including one that requires the partial correlation and the symbol-arm $\Delta\rho$ contrast to name the same systems. `paper/generate_coin_city_reference_selection_tables.py` writes both appendix tables.

Probe-target ceilings. `analysis/compute_bayes_ceilings.py` under the code root derives the Bayes-optimal ceiling for each probe target in Table 32 and the forecast-MAE floors drawn in Figure 5, importing the generator constants from the engine module so they cannot drift, and writes `analysis/bayes_ceilings.json` under the data root. It reads no response file and makes no model call, and fails closed if its closed form and its simulation of the generator's own row logic disagree. Checks are in `tests/test_bayes_ceilings.py`.

Misleading arm and paper outputs. The repository-level script `paper/generate_coin_city_split_figures.py` recomputes misleading-arm metrics from the final response files and scoring key. `paper/generate_coin_city_appendix_tables.py` generates the primary tables and pairwise contrasts.

Generator and deployment robustness. `engine/coin_city_robustness_population.py` and `eval/build_coin_city_robustness_population.py` build the fresh-seed population; its design, responses, scheduler ledgers, and analysis are under `data/coin_city_population_075_040_sd010_v1/`. `analysis/analyze_coin_city_rationales.py` writes `analysis/rationale_audit_v1.json` under the primary data root. `eval/run_coin_city_gpt56_k0_repeat.py` and `analysis/analyze_coin_city_gpt56_k0_repeat.py` isolate and score the hosted repeat. `paper/generate_coin_city_robustness_tables.py` renders both appendix tables from the saved analysis artifacts.

D Experiment 3: Controlled Relationship Transfer

As the controlled transfer step, Experiment 3 asks whether outcome-based training learns to select and apply an episode-matched relationship beyond its training setting. The primary Qwen3-8B endpoint jointly changes the domain and response mechanism; Qwen3-4B and Llama-3.1-8B provide scale and architecture comparisons.

D.1 Design and Evaluation

D.1.1 Data-generating processes

Training uses only Structure A, the direct relationship from Experiment 2. On the displayed scale of domain d,

$$y_{t+1} = b_d + g_j x_t + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma_d^2). \tag{6}$$

A pulse acts in the next period and has no residual effect under zero subsequent input. The unseen evaluation mechanism, Structure B, is

$$m_{t+1} = \rho m_t + g_j x_t, \tag{7}$$

$$y_{t+1} - b_d = \phi(y_t - b_d) + \lambda m_{t+1} + \epsilon_t. \tag{8}$$

This is a linear-Gaussian latent-state process in the sense of Kalman [1960]: it introduces a persistent mediator and outcome carry-over, so a single pulse has both immediate and delayed effects. Model prompts disclose neither equation, parameter values, internal variable, nor structure label.

Coin City uses $b = 50$, range $[0, 100]$, driver pool $\{-12, -8, -4, 0, 4, 8, 12\}$, and $\sigma = 1.6$. Held-out Coin Harbor uses $b = 180$, range $[40, 320]$, driver pool $\{-24, -16, -8, 0, 8, 16, 24\}$, and $\sigma = 4.5$, together with new terminology, units, and qualitative descriptions. Evaluation gains follow $g \sim \mathrm{Uniform}(.22, .92)$. Structure B independently samples $\rho \sim \mathrm{Uniform}(.48, .74)$, $\phi \sim \mathrm{Uniform}(.18, .42)$, and $\lambda \sim \mathrm{Uniform}(.68, .94)$. A reference and its matched target share these parameters but have independent inputs and noise.

Table 37: **Exact Coin City–Coin Harbor domain contrast.** Both surfaces instantiate the same Structure A and B simulators and task template; Coin Harbor changes the semantic setting, time unit, numeric scale, interventions, noise, and pathway descriptions.

Field	Coin City (train and test)	Coin Harbor (test only)
Driver → outcome	Net campaign news → candidate support	Net shipping bulletin → cargo-flow index
Time unit	Week	Tide cycle
Resting level / support	50 / $[0, 100]$	180 / $[40, 320]$
Observed-input pool	$\{-12, -8, -4, 0, 4, 8, 12\}$	$\{-24, -16, -8, 0, 8, 16, 24\}$
Evaluation shocks	$\{-12, -6, 0, 6, 12\}$	$\{-24, -12, 0, 12, 24\}$
Observation-noise SD	1.6 support points	4.5 cargo-index points
Structure A description	Synchronized citywide alerts; reactions occur mostly in the same week	Synchronized fleetwide radio; responses occur mostly in the same tide cycle
Structure B description	Neighborhood conversations and local organizations; reactions build gradually and persist	Dockside crews and harbor groups; responses build gradually and persist

D.1.2 Training episodes and matched control

Each training prompt contains two 14-case Coin City reference systems, both governed by Equation 6. One draws $g \in [.18, .46]$ and the other $g \in [.68, .98]$; their order is randomized. The target shares exactly one reference gain, selected in balanced fashion, and shows $k \in \{0, 2, 4, 8\}$ independently generated cases. Its national/local background description identifies the matching weak or strong response as in Experiment 2. There are 1,200 rows at each k, or 4,800 per training run.

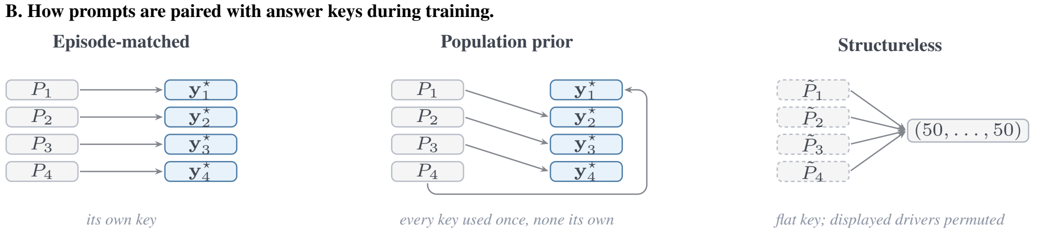
The episode-matched and population-prior datasets have byte-identical prompts. Episode-matched reward uses the target's noise-free forecasts. For the control, targets are deranged within each k stratum with a fixed permutation: no row keeps its own target, while the complete multiset of target vectors is exactly preserved. Consequently the control can learn the output format and population distribution but cannot improve reward by using that row's references or target history. The Qwen3-4B structureless diagnostic additionally scrambles displayed drivers and rewards a flat intervention response. It isolates task-format learning and is excluded from the episode-matched-versus-prior estimand.

Table 38: **One training episode, and what each policy is rewarded against.** Panel A is the prompt for episode `train:1:k2`, abridged from the rendered text. Panel B is the prompt–key pairing used during training, drawn for four episodes at one evidence depth: P_i is episode i's prompt and $\mathbf{y}_i^\star$ its ten-answer key. Episode-matched and population-prior training use byte-identical prompts and the same multiset of keys; only the pairing differs. The control shifts cyclically across all 1,200 episodes at a depth, so its key is a real key drawn from the same distribution but never this prompt's, and no reward can be earned by reading this row's references. Base is untrained and receives no supervision.

A. Shared prompt. Coin City; resting level 50.0, bounded to $[0, 100]$; no equation or coefficient is shown.

REFERENCE 1	"…immediate national alerts … react **strongly**"	14 wk: $-8.0 \to 43.8$, $-12.0 \to 40.9$, …, $0.0 \to 51.6$
REFERENCE 2	"…local civic briefings … react **weakly**"	14 wk: $+8.0 \to 52.6$, $+4.0 \to 51.3$, …, $-12.0 \to 46.9$
TARGET (k=2)	"…immediate national alerts … react **strongly**"	$-8.0 \to 42.3$, $+4.0 \to 53.9$
QUERY	ten counterfactuals A–J: shock $\in \{-12, -6, 0, 6, 12\}$ × horizon $\in \{1, 3\}$; return `{"forecasts":[A,...,J]}`	

B. How prompts are paired with answer keys during training.



D.1.3 Evaluation interface and qualitative hint

Every evaluation prompt contains a complete 14-period Structure A reference, a complete 14-period Structure B reference, and a target governed by one of them. For Coin City the two reference descriptions are:

Residents receive campaign developments through synchronized citywide alerts. Reactions occur mostly during the same week as the news.

Campaign developments usually spread through neighborhood conversations and local organizations. Reactions may build gradually and persist after the original news has faded.

Coin Harbor uses the matched fleetwide-radio versus dockside-network descriptions. The target receives the correct description, no description, or the other pathway's description. The three prompt variants reuse the exact reference and target trajectories, scenario grid, and gold forecasts; the manifest records a common numeric hash for every cue triplet.

Each task requests ten counterfactuals: shocks $\{-2u_d, -u_d, 0, u_d, 2u_d\}$ at horizons one and three, with $u_d = 6$ in Coin City and $u_d = 12$ in Coin Harbor. The shock occurs in the next period and all later inputs are zero. The response vector consists of the eight nonzero-shock minus zero-shock contrasts. The full endpoint contains

$$2 \text{ domains} \times 2 \text{ structures} \times 3 \text{ hints} \times 4\ k \text{ values} \times 30 \text{ episodes} = 1,440$$

paired prompts. Training and evaluation seed bands begin at 81,000,000 and 91,000,000, respectively.

Exact task template and prompt variants. The following is the task string emitted by the prompt generator before the model-specific chat wrapper is applied. Bracketed uppercase fields below denote row-specific substitutions; they are not shown to the model. Line wrapping is typographic, while the blank-line-separated sections and all fixed wording are part of the prompt.

```
Forecasting task in [DOMAIN].

The resting level of [OUTCOME] is [BASELINE]. Values are bounded to [LOWER, UPPER].

The reference systems are complete examples.  The target system may share a stable
response pattern with one reference.  Infer from the information shown; no response
equation or coefficient is provided.

REFERENCE SYSTEM 1
Background:  [REFERENCE 1 CONTEXT]
[REFERENCE 1 TABLE: period, driver, and 14 observed outcomes]

REFERENCE SYSTEM 2
Background:  [REFERENCE 2 CONTEXT]
[REFERENCE 2 TABLE: period, driver, and 14 observed outcomes]

TARGET SYSTEM
[TARGET BACKGROUND, WHEN PRESENT]
[TARGET TABLE WITH k OBSERVATIONS, OR THE EXACT NO-OBSERVATION SENTENCE]

COUNTERFACTUAL FORECASTS
For each scenario, apply the stated [DRIVER] in the next [PERIOD] and then set it to 0
in later periods.  Forecast [OUTCOME] at the stated horizon.

A: shock=[VALUE]; horizon=1  ...  E: shock=[VALUE]; horizon=1
F: shock=[VALUE]; horizon=3  ...  J: shock=[VALUE]; horizon=3

Return only strict JSON with exactly ten finite numbers in scenario order:

{"forecasts":[A,B,C,D,E,F,G,H,I,J]}
```

The renderer realizes the variants as follows. Training always includes the correct strong/weak background and uses the same input string for episode-matched and population-prior training; only the hidden reward target differs. At evaluation, `correct` inserts the target pathway's description, `none` omits the background line entirely, and `misleading` inserts the other pathway's description. Every cue triplet otherwise has identical tables, interventions, and targets. If $k = 0$, the target block contains the exact sentence *No target-system observations are available yet.*; otherwise it contains the same table format as the references with the first k rows. The Qwen3-4B `structureless` diagnostic retains this shell but swaps every displayed $-12.0 \leftrightarrow 12.0$ and $-8.0 \leftrightarrow 8.0$ token through a collision-safe substitution. Untrained bases receive the same held-out prompts as trained models. Thus cue, reward-arm, structureless, and base comparisons do not introduce an unreported instruction change.

D.1.4 Estimands, roster, and validation criteria

The reported factorial crosses Qwen3-4B, Qwen3-8B, and Llama-3.1-8B, two reward arms, and three training seeds, for 18 runs. Three Qwen3-4B structureless runs and three untrained endpoints provide diagnostic comparisons. The primary estimand is population-prior minus episode-matched response MAE on Structure B in Coin Harbor with the correct hint, averaging the four k values equally. Positive values favor episode-matched training. The same estimator defines factorial contrasts for Structure A/Coin Harbor (domain transfer), Structure B/Coin City (structural transfer), and the in-distribution cell. Paired absent-minus-correct and misleading-minus-correct contrasts measure hint use. Their paired $k = 8$ minus $k = 0$ difference tests whether observations override a misleading prior; negative values indicate a shrinking misleading-hint penalty.

Every reported round-300 endpoint uses five draws per prompt at temperature .7; temperature-zero checkpoints serve only as online training monitors.

Each episode's score is the mean of its five per-draw response MAEs, so "averaging draws" averages errors and not forecasts. The hierarchical bootstrap resamples the three training seeds at the top level and paired episodes within each selected seed. Parse failures receive the full observable-range error penalty rather than being discarded. That penalty is the clip width—280 points in Coin Harbor, roughly fifty times a typical error—so it carries large leverage: a single unparsed draw of 150 raises the Qwen3-8B base joint-cell MAE from 5.26 to 5.72, which is most of that arm's reported gap to matched training. Both trained arms parse completely in that cell, so the primary estimand is unaffected; the base comparison is not. The launch ledger contains all 24 endpoints and 172,800 stochastic rows; 172,732 parse (99.96%), and every endpoint has at least 99.82% coverage.

Sensitivity of the intervals to three training seeds. The hierarchical interval resamples three top-level clusters, and a nonparametric cluster bootstrap at $G = 3$ is known to understate between-cluster uncertainty [Cameron et al., 2008]. We therefore report a deliberately conservative alternative alongside it: a Student-t interval on the three seed-level means with two degrees of freedom, which uses the same paired episode differences but charges the full small-sample penalty for estimating a variance from three numbers. Under that alternative none of the twelve transfer-cell intervals excludes zero, the primary Qwen3-8B contrast becoming $.251$ $[-.259, .762]$ and the Qwen3-4B comparator $.319$ $[-.120, .758]$. Nine of the twelve intervals that exclude zero under the hierarchical bootstrap therefore do not survive the t alternative, and we do not claim that any single interval in this experiment is robust to the choice of small-cluster procedure.

What does survive is the sign pattern, which requires no variance estimate. Across the four transfer cells and three training seeds there are twelve seed-level contrasts per model. All twelve favor episode-matched training for Qwen3-8B and all twelve for Qwen3-4B, so all 24 Qwen seed-by-cell contrasts point the same way; these reuse training seeds across cells and are not independent replications. The corresponding count for Llama-3.1-8B is three of twelve. We take the two-Qwen sign consistency, rather than any interval, as the principal evidence for the episode-matching effect, and we read the interval estimates as descriptive of magnitude.

Before training, the dataset passed 17 fail-closed checks: exact row counts and factorial balance; byte-identical episode-matched/population-prior prompts and held-out datasets; exact within-k target marginals with no fixed reward links; absence of Structure B from training prompts and metadata; cue-triplet numeric identity; train/evaluation identifier disjointness; hidden-equation leak scans; strict gold round trips; and oracle separation from the blind pathway mixture. Mean A–B response separation is 2.65 points in Coin City and 4.80 in Coin Harbor. Across both tokenizers, the longest prompt is 1,071 tokens against the configured 2,304-token limit.

We estimate population-prior-minus-episode-matched intervals in all four transfer cells, paired intervals for both cue penalties at every k, and the paired $k = 8$ minus $k = 0$ change in misleading-minus-correct error to evidence override. The protocol archive records versioned input, analysis, and output hashes. Full optimization and compute parameters are in Section F.

D.1.5 Stochastic scale and architecture comparators

Qwen3-4B scale replication. The Qwen3-4B comparison includes three seeds in each of the episode-matched, population-prior, and structureless arms. Each is evaluated on the same 1,440 tasks at round 300 with five stochastic draws per task; the untrained base uses the same endpoint. Parse rates are 100% for every trained arm and 99.97% for the stochastic base.

Joint-cell MAE is 6.15, 6.76, 4.81, and 4.49 for base, structureless, population-prior, and episode-matched training, respectively. The prior-minus-episode-matched difference is $.319$ $[.134, .489]$; episode-matched training has the lowest point estimate in all four cells, and all four intervals exclude zero. The Qwen3-8B main estimate is likewise positive, $.251$ $[.058, .470]$, giving the same interval-excluding-zero result at both tested Qwen sizes.

Averaged over evidence depths, omitting the hint raises episode-matched response MAE by $.151$ $[.079, .235]$, and a misleading hint raises it by $.313$ $[.228, .415]$. The misleading-hint penalty is smaller at $k = 8$ than at $k = 0$, but the evidence-override contrast $-.116$ $[-.309, .100]$ includes zero.

Llama-3.1-8B architecture comparator. In the joint-transfer cell the prior-minus-episode-matched training contrast is positive, $.420$ $[-.115, .958]$, but its interval spans zero. The three seed-level effects are $+.762$, $+.622$, and $-.125$, so this result does not establish a robust cross-architecture training advantage.

The joint cell is also the most favorable of the four Llama cells, and reporting it alone would misrepresent the factorial, so we give all four. The contrasts are $-.351$ $[-.680, -.130]$ in the in-distribution cell, $-.167$ $[-.862, .306]$ under structure-only shift, $-.581$ $[-.868, -.243]$ under domain-only shift, and $+.420$ $[-.115, .958]$ under joint shift. In the two Structure A cells the population-prior arm is therefore better than the episode-matched arm by an interval that excludes zero, and only three of the twelve Llama seed $\times$ cell contrasts favor episode-matched training. On this architecture the factorial runs against the hypothesis in the cells whose response mechanism was seen during training, and recovers a positive point estimate only where both shifts apply. We report this as a failure of the episode-matching effect to replicate on Llama-3.1-8B rather than as a directionally-consistent-but-underpowered result, and the manuscript's transfer claim rests on the two Qwen models.

Llama nevertheless uses the qualitative pathway cue at the stochastic endpoint: absent-minus-correct MAE is $.219$ $[.123, .313]$ and misleading-minus-correct is $.210$ $[.092, .320]$. Its misleading-hint penalty shrinks from $k = 0$ to $k = 8$ by $-.621$ $[-.957, -.323]$. All three cue-related intervals exclude zero. These contrasts indicate sensitivity to the supplied mechanism cue and reduced misleading-cue effects as observations accumulate.

Table 39: **Stochastic Coin City appendix comparators in the joint-transfer cell.** The Qwen3-8B primary result remains in the main body. Entries are response MAE under the correct hint after averaging five draws per task; lower is better. Positive prior-minus-episode-matched contrasts favor episode-matched training. Intervals use the paired seed-first bootstrap with a fixed cell-specific bootstrap seed.

Model	Base	Prior	Episode-matched	Δ [95% CI]
Qwen3-4B	6.15	4.81	4.49	0.32 [0.13,0.49]
Llama-3.1-8B	9.43	4.69	4.27	0.42 [-0.12,0.96]

E Experiment 4: Historical-Market Training and Held-Out Evaluation

Experiment 4 asks whether base-relative gains extend from the controlled setting to events and normalized question families unseen during training. It uses proper-score training, a point-in-time cutoff, and family-disjoint held-out data.

E.1 Sample Construction and Leakage Controls

Source archive and sample. The source export[2] has 1,788,703 rows and SHA-256 `5697f767...f638e7d` (the full digest is stored in the manifest). The builder scans the entire

[2] 🤗 Full Polymarket archive. Link anonymized for review.

file and fails on checksum drift. The calendar splits contain 1,736 train, 512 development, and 1,024 test markets, with YES rates .311, .275, and .301. Train contains 991 distinct events and 1,456 normalized question templates; development contains 292 and 453; test contains 493 and 866. The selected archive rows have no official series identifiers, so event and template components supply the operative family groups.

Eligibility and point-in-time censoring. We admit literal Yes/No markets resolved by an extreme final token price. A boundary-aware domain rule selects political, governmental, macroeconomic, and geopolitical questions while explicitly rejecting culture, awards, sports, esports, and crypto categories. This rule is fixed before model training. One task is built 30 days before the earliest archived terminal timestamp. At least four distinct UTC dates must have genuine pre-cutoff YES candles, the latest candle must be no more than seven days stale, and its price must remain in $[.02, .98]$. We keep at most the latest 14 dates. Empty, carry-forward, and post-cutoff candles are rejected. The prompt omits the outcome, final prices and volume, market status, realized terminal time, and the tags used for domain selection.

The archive records timestamped price history but not revisions to static question or resolution-rules text. We therefore treat those strings as time-invariant. This is a residual point-in-time limitation: the audit verifies that no timestamped or outcome field crosses the cutoff, but cannot determine whether an archived rule changed between the forecasting cutoff and resolution.

Family-disjoint temporal split. Before applying sample caps, every development row sharing an event, available series, or normalized template with train is removed; the same rule removes test rows matching train or development. Numeric values and month names are normalized in template keys. The final intersection is zero for each pair of splits. Sampling within a split is a stable hash of task identifier and protocol version, so row order in the export cannot change membership.

E.2 Training and Held-Out Evaluation

Matched update budget. Training comprises 1,736 unique markets arranged in a deterministic, balanced schedule of 4,800 presentations, matching Experiment 3's 300-round update budget. Every market appears twice and 1,328 appear a third time. Schedule order and task files are checksum-verified. With batch size 16, this gives exactly 300 rollout-and-update rounds (2,400 AdamW steps). The manifests and tables distinguish unique markets from presentations. The longest Qwen3-4B wrapped prompt contains 1,012 tokens against the 1,792-token training limit. Data, schedule, reward, optimizer, update count, seeds, and scoring rule are held constant across models; each uses its native chat template. Shared optimizer, batching, and runtime details appear in Section F.

Table 40: Experiment 4 training configuration. Held-out data are absent from training and evaluated once per model after all three training seeds.

Quantity	Value
Model roster / adapter	Qwen3-4B-Instruct-2507, Qwen3-8B, and Llama-3.1-8B-Instruct; LoRA rank 32, $\alpha = 64$
Algorithm	Dr. GRPO; $\beta = 0$; AdamW, lr 10^{-6}
Unique train / presentations	1,736 / 4,800 balanced repetitions
Rounds / seeds	300; $42, 43, 44$
Rollout	16 prompts, 8 samples each; temperature 1.3
Output	one JSON probability; 128 new-token limit
Reward	$1 - (\hat{p} - y)^2$ only
Chat formatting	Qwen3-4B uses its frozen ChatML wrapper; Qwen3-8B uses its non-thinking template; Llama uses its native instruct template.
Development	512 tasks; temperature-zero evaluation every 25 rounds
Held-out evaluation	1,024 tasks; temperature zero; one evaluation per model

E.3 Held-Out Results

Decision rule. The primary learned contrast is the mean loss over three trained seeds minus base loss. Invalid JSON receives Brier loss one. Intervals use 5,000 bootstrap resamples of connected event/template components.

All Qwen3-8B test forecasts parse. Its three-seed trained mean obtains Brier .11642 versus .12867 for the base, a difference of $-.01225$ with 95% interval $[-.01948, -.00602]$. Every training seed improves on the base. The interval excludes zero, supporting transfer to held-out events for Qwen3-8B.

Table 41: **Qwen3-8B on the 1,024-market held-out set.** The mean-row interval resamples connected event/template families; lower is better.

Forecaster	Brier ↓	Δ base	Log loss ↓
Untrained Qwen3-8B	.1287	—	.5003
Trained, seed 42	.1164	-.0123	.4058
Trained, seed 43	.1168	-.0119	.4069
Trained, seed 44	.1161	-.0126	.4146
Trained mean	.1164	-.0123 [-.01948,-.00602]	.4091

Table 42: **Qwen3-4B and Llama held-out comparisons.** The Qwen3-8B result is immediately above. Δ columns are trained mean minus comparator, so negative values favor training. Parse is the minimum coverage across base and trained endpoints; Llama's 12 unparsed base outputs receive Brier loss one.

Model	Base	Seed 42	Seed 43	Seed 44	Trained mean	Δ base [95% CI]	Δ market [95% CI]	Δ Platt [95% CI]	Parse
Qwen3-4B	.1316	.1142	.1140	.1194	.1159	-.0157 [-.02528,-.00667]	.0022 [-.00264,.00721]	.0036 [-.00125,.00874]	1.000
Llama-3.1-8B	.1520	.1151	.1149	.1151	.1150	-.0370 [-.05376,-.02258]	.0013 [-.00004,.00371]	.0027 [.00067,.00539]	.988

Qwen3-4B replicates the base-relative improvement: its trained mean is $.11589$ versus $.13160$ for base, a difference of $-.01572$ $[-.02528, -.00667]$. Llama-8B improves from $.15203$ to $.11502$, a difference of $-.03701$ $[-.05376, -.02258]$; all three trained Llama endpoints parse.

The latest pre-cutoff market price obtains Brier $.11369$, and logistic recalibration of its logit fit only on the training split obtains $.11232$. Neither comparator informs language-model training. The trained-minus-market interval spans zero for every model. Qwen3-4B is also statistically indistinguishable from the calibrated price, but Qwen3-8B and Llama-8B have higher error by $+.00409$ $[+.00098, +.00785]$ and $+.00269$ $[+.00067, +.00539]$, respectively. Thus the complete roster shows base-relative improvement rather than superiority to market-based forecasts.

E.4 Scale and Architecture Comparison

The scale comparison uses all 318 eligible 2026-window tasks whose identifiers and event, series, and normalized question-template families are absent from the training, development, and 1,024-market evaluation sets. The holdout contains 227 event families and 288 normalized templates; selection does not use outcomes or market probabilities.

Qwen3 checkpoints at 4B, 8B, and 14B use three seeds, a common reward, a 4,800-presentation schedule, and the same development rule. Each untrained base and trained adapter receives five non-thinking draws per task at temperature .7, top-p .8, and top-k 20. All 19,080 Qwen draws satisfy the strict JSON contract.

Table 43: **Exact values behind the main-text scale figure.** Lower Brier is better. Trained means average seeds 42–44. Missing five-draw task groups receive Brier loss one.

Checkpoint	Base	Trained mean	Trained − base	Trained parse
Qwen3-1.7B	0.2253	0.1248	-0.1005	100.0%
Qwen3-4B-Instruct-2507	0.1522	0.1366	-0.0156	100.0%
Qwen3-8B	0.1369	0.1286	-0.0083	100.0%
Qwen3-14B	0.1281	0.1270	-0.0012	100.0%
Llama-3.2-3B	0.2189	0.1367	-0.0822	100.0%
Llama-3.1-8B	0.1642	0.1264	-0.0377	100.0%

The trained means are .13664, .12859, and .12695 at 4B, 8B, and 14B; the corresponding bases are .15224, .13686, and .12812. The 14B-minus-8B trained contrast is $-.00164$ $[-.00332, -.00024]$, and the 14B-minus-4B contrast is $-.00969$ $[-.01869, -.00148]$. The 14B trained-minus-base contrast is $-.00117$ $[-.00298, +.00016]$.

On this holdout, contemporaneous Polymarket prices have Brier .12655. Trained 14B minus market is $+.00040$ $[-.00028, +.00139]$, so the interval crosses zero. Thus 14B has lower trained error than 8B while its base-relative and market-relative differences remain unresolved.

Llama-3.1-8B uses the same holdout and five-draw endpoint. All 4,770 trained draws parse; nine of the 1,590 base draws do not, causing seven base task groups to receive Brier loss one. Trained seed Briers are .12638, .12629, and .12662, for a mean of .12643 versus .16417 for base. Trained minus base is $-.03774$ $[-.05860, -.01946]$; trained minus Polymarket is $-.00012$ $[-.00046, +.00012]$. The latter interval crosses zero.

Qwen3-1.7B and Llama-3.2-3B preserve the same training/development data, three seeds, 4,800-presentation schedule, five stochastic draws, and holdout. Qwen3-1.7B seed Briers are .12413, .12610, and .12415 (mean .12479) versus .22531 for base; trained minus Polymarket is $-.00175$ $[-.00644, +.00265]$. Llama-3.2-3B seed Briers are .13197, .13666, and .14149 (mean .13671); trained minus Polymarket is $+.01016$ $[+.00121, +.02060]$. The pinned public BF16 Llama mirror is recorded because the gated upstream repository was unavailable to the workspace account. Its unconstrained base initially produced only 4/318 complete five-draw groups at the 128-token cap; an output-cap-only rerun reached 21/318. A disclosed, holdout-informed syntax recovery therefore repeated the full base endpoint with identical prompts and sampling settings while constraining output to a JSON object containing one numeric probability. All 1,590 recovered draws parse. Base Brier is .21887, and trained minus base is $-.08216$ $[-.11062, -.05367]$. Because the syntax constraint was introduced after observing the failures and differs from the trained endpoints, this base-relative contrast is descriptive. Both trained models also parse every draw.

Hosted frontier references. The API panel evaluates the same 318-task holdout. It is categorical rather than a parameter-scale curve: provider parameter counts are unavailable or not comparable, and the APIs do not expose identical decoding controls. DeepSeek-V4-Pro uses the common temperature .7/top-p .8 stochastic request. GPT-5.6-Sol uses provider-native low reasoning without user-set temperature or top-p; Claude Opus 4.8 likewise uses provider-native sampling. Each endpoint receives five independent calls per task and uses the same strict parser, five-draw mean, Brier scoring, and family-clustered 5,000-repetition interval procedure as the local panel.

DeepSeek parses all 1,590 draws and has Brier .12660, differing from the crowd by $+.00005$ $[-.00105, +.00102]$. Claude parses all draws and has Brier .12927, a difference of $+.00272$ $[-.00049, +.00614]$. GPT receives all 1,590 responses but six are malformed; two task groups therefore receive Brier loss one, and task-level coverage is 99.37%. Its Brier is .13335, a difference of $+.00680$ $[+.00016, +.01635]$.

Kimi uses a 1,024-token output budget. Thirteen transport failures are retried without replacing the 1,577 successful responses; all 1,590 final draws parse. Its Brier is .12682, differing from the crowd by $+.00027$ $[-.00046, +.00109]$. Hosted results provide performance context, not evidence for a common scale law or an identical-decoder comparison.

E.5 Qwen3-4B Evidence-Updating Evaluation

This evaluation applies the Experiment 1 update prompt and all nine fixed packets for each of its 100 markets (three pro-H_1, three anti-H_1, and three orthogonal). The packet and initial-context SHA-256 digests are `6741be0a...8796d36` and `c550cf05...0c098`. Every condition receives the same canonical first structured Qwen 2.5-7B forecast for each market, so this paired analysis isolates conditional revision rather than web retrieval or initial event-model construction. Search and thinking are disabled; decoding is greedy with a 1,500-token output cap. The longest prompt is 1,480 tokens against a 6,144-token context limit. These contrasts characterize revision behavior and are separate from the held-out forecasting evaluation.

The paired evaluation produces all 900 packet-conditioned revisions for the base and each final adapter, with 100% parse coverage and ICS equal to one. Base EHC/HFC is .909 on 591 eligible directional updates. Seeds 42, 43, and 44 obtain .917, .922, and .918 (592, 588, and 591 eligible updates), for a three-seed mean of $.919$. The paired market-bootstrap trained-minus-base EHC difference is $+.0101$ with 95% interval $[+.0004, +.0207]$; HFC has the same estimate and interval $[+.0009, +.0210]$. Training reduces mean absolute directional revision by $.52$ percentage points $[-.70, -.36]$ and orthogonal revision by .22 points $[-.38, -.06]$; the directional-minus-orthogonal gap narrows by .30 points $[-.52, -.09]$. The sensitivity ratio changes from $4.64\times$ to $4.83\times$ ($+.19$ $[-.07, +.48]$). These components show improved directional consistency alongside smaller revisions overall; they do not indicate an increase in absolute packet selectivity.

Table 44: **Paired evidence-updating contrasts.** Movement rows are mean absolute probability revisions in percentage points; intervals resample the 100 markets.

Measure	Trained − base	95% interval
Directional consistency (EHC)	$+.0101$	$[+.0004, +.0207]$
Headline-forecast consistency (HFC)	$+.0101$	$[+.0009, +.0210]$
Directional movement	$-.52$ pp	$[-.70, -.36]$
Orthogonal movement	$-.22$ pp	$[-.38, -.06]$
Directional − orthogonal movement	$-.30$ pp	$[-.52, -.09]$
Sensitivity ratio	$+.19\times$	$[-.07, +.48]$

F Experiments 3–4: Shared Training Configuration and Runtime

Tables 45–47 record the effective OAT/Dr. GRPO [Liu et al., 2024] arguments for the reported Coin City and Polymarket studies, including framework defaults omitted from the command line. Execution manifests hash the data, trainer, launcher, evaluator, prompt renderer, and each cell's environment.

Step accounting. A "step" in the figures and earlier tables is one rollout-and-update round rather than one AdamW call. Each full run consumes 16 prompts per round and samples eight responses per prompt (128 trajectories). With one learner GPU, per-device microbatch one, one PPO epoch, and gradient accumulation 16, OAT performs eight AdamW parameter updates per round. Thus 4,800 prompt presentations give 300 reported rounds and 2,400 AdamW updates. The command line supplied a .03 warmup ratio, but OAT selected Transformers' `constant` scheduler, which does not use the warmup argument. The effective learning rate was therefore 10^{-6} from the first optimizer update. We use "round" below to avoid counting the eight within-round parameter steps as independent data exposures.

Table 45: **Shared Dr. GRPO implementation parameters.** These values apply to the Coin City and Polymarket studies; the study-specific tables below record exceptions.

Quantity	Effective value
Trainable parameters	Frozen base model; LoRA rank 32, $\alpha = 64$, dropout 0, no bias; targets `q_proj`, `k_proj`, `v_proj`, `o_proj`, `down_proj`, `up_proj`, and `gate_proj`; no quantization.
Algorithm	OAT Dr. GRPO (`critic_type=drgrpo`); eight-response group advantage $r_i - \bar{r}$ with no group-standard-deviation division; policy loss is token-summed with the generation limit as a fixed normalizer (no response-length normalization).
Policy objective	PPO ratio clipping 0.2; truncated importance-sampling cap 2.0; one PPO epoch; KL coefficient $\beta = 0$; no critic/value model; reward whitening, advantage whitening, and REINFORCE mode all off.
Optimizer	PyTorch AdamW [Loshchilov and Hutter, 2019] (substituted for DeepSpeed fused/CPU Adam); learning rate 10^{-6}; $\beta_1 = .9$, $\beta_2 = .95$; weight decay 0; maximum gradient norm 1.0.
Schedule	Constant 10^{-6} learning rate from the first optimizer update; the supplied .03 warmup ratio is ignored by Transformers' `constant` scheduler; one prompt epoch; 4,800 prompt presentations $=$ 300 rollout-and-update rounds $=$ 2,400 AdamW steps.
Batching	16 prompts per round; 8 sampled responses per prompt; 128 trajectories; training microbatch 1; gradient accumulation 16; one learner GPU.
Training sampling	Temperature 1.3; top-$p = 1$; top-$k = -1$; one independently sampled completion at a time, repeated eight times per prompt.
Truncation and stopping	Model EOS only (no external stop string); a completion ending at the length limit receives reward zero. Missing or invalid required JSON also receives reward zero.
Precision and memory	bfloat16; FlashAttention 2; gradient checkpointing; DeepSpeed ZeRO-2 [Rajbhandari et al., 2020]; reference-model offload; vLLM prefix caching [Kwon et al., 2023]; dropout disabled.
Seeds and shuffling	Training seeds 42, 43, and 44. Each dataset is deterministically shuffled once, then OAT shuffles prompt batches independently under the training seed.
Checkpointing and logging	Evaluation includes step zero and the final round; intermediate evaluation at the setting-specific frequency; one final LoRA adapter retained; metrics logged each round; Weights & Biases disabled.

F.1 Coin City A-to-B Factorial

Table 46 records the shared configuration for Experiment 3. The main analysis uses five-draw stochastic round-300 evaluations from the three Qwen3-8B episode-matched and three population-prior seeds. The appendix reports the Qwen3-4B stochastic replication, its three-seed structureless diagnostic, and the Llama-8B architecture comparator. The table supplements the shared optimizer details in Table 45; where the two tables differ, this setting-specific table governs. The execution manifest stores the environment; the protocol archive hashes the preflight manifest, data files, generator, prompt renderer, scorer, and trainer entry point.

Table 46: **Evaluated configuration for Coin City hidden-structure transfer.**

Quantity	**Effective value**
Model roster	`Qwen/Qwen3-4B-Instruct-2507`, `Qwen/Qwen3-8B`, and `meta-llama/Llama-3.1-8B-Instruct`.
Chat formatting	Qwen3-4B uses the explicit ChatML wrapper; Qwen3-8B and Llama-3.1-8B use their tokenizer templates, with Qwen thinking disabled. All include an assistant-generation prefix.
Evaluated grid	3 models $\times$ 2 reward arms (episode-matched reward, stored under the legacy artifact key `causal`, and `population_prior`) $\times$ 3 seeds $(42, 43, 44) = 18$ full training runs.
Diagnostics	Qwen3-4B `structureless` at the same three seeds, plus one untrained endpoint for each model. These six diagnostics are excluded from the primary episode-matched-versus-prior estimand.
Training data	Coin City and Structure A only; 4,800 prompts, balanced across $k \in \{0, 2, 4, 8\}$ and weak/strong target match. Two reference trajectories of length 14 and one target trajectory of maximum length 8.
Comparator target construction	Episode-matched reward uses the episode truth. Population-prior reward targets are a fixed no-self-match derangement within each k; its target-vector multiset is exactly identical to episode-matched training. Prompt strings are byte-identical across these two arms.
Held-out data	2 domains $\times 2$ target structures $\times 3$ hint arms $\times 4$ evidence depths $\times 30$ episode seeds $= 1,440$ rows. The complete held-out dataset is byte-identical across all training arms.
Mechanism ranges	Evaluation $g \in [.22, .92]$; Structure B $\rho \in [.48, .74]$, $\phi \in [.18, .42]$, and $\lambda \in [.68, .94]$. Training reference gains are $[.18, .46]$ and $[.68, .98]$.
Domain ranges	Coin City: baseline 50, support $[0, 100]$, noise 1.6, driver pool $\{-12, -8, -4, 0, 4, 8, 12\}$. Coin Harbor: baseline 180, support $[40, 320]$, noise 4.5, driver pool $\{-24, -16, -8, 0, 8, 16, 24\}$.
Interventions	Five shocks at horizons one and three; Coin City $\{-12, -6, 0, 6, 12\}$, Coin Harbor $\{-24, -12, 0, 12, 24\}$. Ten joint forecasts produce eight nonzero-minus-zero response contrasts.
Reward	$.4\, r_{\text{level}} + .6\, r_{\text{response}}$. Each component averages $\max(0, 1 - \lvert e\rvert/S)$; level tolerance is 10% of the domain's observable range (10 Coin City points; 28 Coin Harbor units), and response tolerance is half one shock unit (3 and 6, respectively). Invalid or length-truncated output receives zero online reward.
Output contract	Exactly one JSON object with sole key `forecasts` and exactly ten finite JSON numbers. The same strict parser is used online and offline; no prose, extra/duplicate key, Boolean, non-finite value, or wrong-length list is accepted.
Structured decoding	Syntax-only XGrammar GBNF in training, online evaluation, endpoint evaluation, and base evaluation. It fixes object shape and arity only; numeric values and their relationship to targets remain unconstrained.
Optimization	Frozen base plus LoRA rank 32, $\alpha = 64$; OAT Dr. GRPO; PyTorch AdamW learning rate 10^{-6} with an effective constant schedule and no warmup; $\beta = 0$; one PPO epoch; gradient norm 1; ZeRO-2 and reference offload.
Batch and update accounting	16 prompts per round; eight rollouts per prompt; 128 trajectories; microbatch 1; gradient accumulation 16; 4,800 presentations = 300 rollout-and-update rounds = 2,400 AdamW steps.
Training generation	Temperature 1.3, top-$p = 1$, top-$k = -1$, model EOS only; 192 new-token limit; prompt limit 2,304; total model context 3,072; vLLM prefix caching enabled.
Precision / memory	BF16, FlashAttention 2, gradient checkpointing, dropout off, no quantization; vLLM GPU fraction .38; expandable-segments CUDA allocator.
Online evaluation	All 1,440 rows at temperature 0, one draw, top-$p = 1$, 192-token limit, batch 120, at step zero and every 50 rounds including round 300.
Reported endpoint evaluation	Five draws per prompt at temperature .7 and top-$p = .95$; trained decode seed is training seed $+20{,}260{,}818$ and the base seed is 20,260,819. Temperature-zero checkpoints are retained only as online monitoring artifacts and are not reported as endpoints.
Offline scoring	Level MAE, response MAE, response error normalized by the blind 50/50 pathway mixture, response correlation, strict parse coverage, and nearest behavioral pathway. A parse failure receives the full domain range as both level and response error.
Variance and estimands	Five-draw outputs first average within episode. Hierarchical intervals resample three training seeds, then paired episodes. Primary: prior-minus-episode-matched response MAE for correct-hint Structure B in Coin Harbor, equal over k. Domain, structure, joint, hint, and hint-by-k contrasts form the factorial analysis.
Preflight validation	A 10-round run for every model and evaluated arm must save an adapter, produce the full temperature-zero monitoring and five-draw stochastic evaluation sets, achieve at least .95 strict parse rate under each decoder, and exhibit nonconstant finite reward. The 17-check data preflight must also pass before estimating the factorial.
Compute	Training and trained-checkpoint evaluation use two NVIDIA A6000 48-GB GPUs (one actor, one learner), 8 CPU cores, and 56–100 GB allocated host memory. Untrained base evaluation uses one A6000, 8 CPU cores, and 24 GB host memory. All runs load models from the same pinned local cache.

Table 47: **Runtime environment for the reported training runs.** Package versions come from the job execution environment.

Component	**Version / configuration**
Python / CUDA build	Python 3.10.20; PyTorch 2.6.0 built for CUDA 12.4.
Training stack	OAT 0.2.4 at Git revision `869706620a35ec5304fa80a0c74f0c566cc2bc57`; DeepSpeed 0.16.8; PEFT 0.15.2.
Generation stack	Transformers 4.51.3; vLLM 0.8.4; FlashAttention 2.7.4.post1.
Data stack	Datasets 2.16.1; Accelerate 1.14.0.
Allocator / logging	`PYTORCH_CUDA_ALLOC_CONF=expandable_segments:True`; Weights & Biases disabled; stdout/stderr and per-round metrics retained with each run.

G Experiments 3–4: Reproducibility and Artifact Map

The artifact roots `exp3_training_transfer/coin_city_structural/` and `exp3_training_transfer/polymarket/scripts/` contain the generators, prompt builders, trainers, evaluators, and analysis code for the Coin City and historical-market studies. Execution ledgers record effective hyperparameters, Git revisions, adapter identities, and SHA-256 hashes of data and executable protocol files; result renderers verify these fields before aggregation. For Coin City, `make_qwen3_8b_endpoint_outputs.py` validates the seven five-draw files (50,400 scored draws) that support Figure 9. `make_paper_outputs.py` validates the complete 24-endpoint roster and all 48 recorded endpoint files before writing the machine-readable result and the stochastic appendix comparator table. Temperature-zero files are checked as execution provenance but are excluded from manuscript estimates. For the historical study, `analyze_exp1_reapplication.py` validates four matched 900-record files and computes the paired-market direction, sensitivity, and movement-component contrasts. The Polymarket manifests preserve sample membership, prompt and adapter hashes, parse coverage, and bootstrap settings.